

RESEARCH ARTICLE

A Comprehensive Survey on Machine and Deep Learning for Optical Communications

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ABSTRACT The increasing complexity of optical communication systems and networks necessitates advanced methodologies for extracting valuable insights from vast and heterogeneous datasets. Machine learning (ML) and deep learning (DL) have emerged as pivotal tools in this domain, revolutionizing data analysis and enabling automated self-configuration in optical communication systems. Their adoption is fueled by the growing intricacy of systems and links, driven by numerous adjustable and interdependent parameters. This complexity is particularly evident in areas such as coherent transceivers, advanced digital signal processing, optical performance monitoring, cross-layer network optimizations, and nonlinearity compensation. While ML and DL offer immense potential, their application in optical communications is still in its early stages, with significant opportunities remaining unexplored. Many algorithms have yet to be fully deployed in practical settings, underscoring the emerging nature of this research area. This paper presents a comprehensive survey of ML and DL applications across optical fiber communication (OFC), optical wireless communication (OWC), and optical communication networking (OCN), highlighting the challenges, current advancements, and future potential of these approaches. To address the identified gaps, this survey evaluates and compares ML and DL algorithms in terms of their performance, complexity, objectives, input data, metrics, and applications in optical communication. Specific emphasis is placed on identifying how these algorithms enhance system performance and optimization. Furthermore, the advantages and limitations of existing methods are analyzed, offering a clear perspective on the role of ML and DL in this domain. The survey also includes updated visual representations and domain-specific examples to elucidate the practical applications of ML and DL in OFC, OWC, and OCN. It concludes by discussing the open challenges, proposing potential solutions, and identifying promising future research directions to foster advancements in intelligent optical communication systems.

INDEX TERMS Machine learning, deep learning, optical fiber communication, optical communication networking, optical wireless communication.

I. INTRODUCTION

Artificial intelligence (AI) has revolutionized various fields by enabling machines to perform tasks traditionally requiring human intelligence, such as knowledge storage, problem-solving, and adaptive learning. Within optical

communications, AI—particularly machine learning (ML) and deep learning (DL)—has emerged as a groundbreaking force, offering innovative solutions to address the increasing computational complexity of modern communication systems. ML focuses on algorithms that learn from data to make predictions or decisions, while DL, inspired by neural networks, extends ML's capabilities by achieving high accuracy in complex and data-intensive scenarios [1], [2], [3].

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TABLE 1. List of acronyms.

Abbreviation	Definition
AI	Artificial Intelligence
ANN	Artificial Neural Network
BER	Bit Error Rate
BLP	Blocking Probability
BPNN	Backpropagation Neural Network
BR-MFI	Bit Rate-Modulation Format Identification
CD	Chromatic Dispersion
CNN	Convolutional Neural Network
CSI	Channel State Information
DBP	Digital Backpropagation
DDO-OFDM	Direct Detection Optical OFDM
DDPG	Deep Deterministic Policy Gradient
DL	Deep Learning
DNN	Deep Neural Network
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
EM	Expectation Maximization
EON	Elastic Optical Network
EVM	Error Vector Magnitude
FEC	Forward Error Correction
FLOPs	Floating Point Operations
FSO	Free-Space Optical
GRU	Gated Recurrent Unit
ICA	Independent Component Analysis
kNN	k-Nearest Neighbors
KRR	Kernel Ridge Regression
LMS	Least Mean Square
LoS	Line-of-Sight
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MANETs	Mobile Ad-Hoc Networks
MDM	Mode Division Multiplexing
MILP	Mixed-Integer Linear Programming
MIMO	Multiple-Input Multiple-Output
ML	Machine Learning
MLP	Multi-Layer Perceptron
MMSE	Minimum Mean Square Error
MSE	Mean Squared Error
MUI	Multi-User Interference
NDA	Non-Data-Aided
NLE	Nonlinear Equalization
NLPN	Nonlinear Phase Noise
OCDMA	Optical Code Division Multiple Access
OCN	Optical Communication Networks
OFC	Optical Fiber Communication
OFDM	Orthogonal Frequency Division Multiplexing
OOK-NRZ	On-Off Keying Non-Return-to-Zero
OPM	Optical Performance Monitoring
OSNR	Optical Signal-to-Noise Ratio
PAM	Pulse Amplitude Modulation
PCA	Principal Component Analysis
PINNs	Physics-Informed Neural Networks
PM	Polarization-Multiplexed
PMD	Polarization Mode Dispersion
QAM	Quadrature Amplitude Modulation
QoS	Quality of Service
QoT	Quality of Transmission
QPSK	Quadrature Phase Shift Keying
RBFNN	Radial Basis Function Neural Network
RL	Reinforcement Learning
RMSA	Routing, Modulation, and Spectrum Assignment
RNN	Recurrent Neural Network
RSSI	Received Signal Strength Indicator
RSA	Routing and Spectrum Allocation
RWA	Routing and Wavelength Allocation
SDM	Spatial Division Multiplexing
SDNs	Software-Defined Networks
SNR	Signal-to-Noise Ratio
SL	Supervised Learning
SVM	Support Vector Machine
USL	Unsupervised Learning
VLC	Visible Light Communication
VLP	Visible Light Positioning
Volterra-DFE	Decision Feedback Equalizer
Volterra-FFE	Volterra-Based Feedforward Equalizer
WDM	Wavelength Division Multiplexing

The growing demand for efficient and resilient optical communication systems has accelerated the adoption of ML

and DL techniques. These methods have demonstrated their potential in diverse applications, from the physical layer to optical communication networking (OCN), and across various media such as optical fiber communication (OFC) and optical wireless communication (OWC). Despite these advancements, the full potential of ML and DL in optical communications remains underexplored, highlighting the importance of surveys that consolidate recent developments and identify opportunities for further research [4], [5], [6], [7].

In the physical layer, ML and DL algorithms have been effectively utilized to address several challenges. These include enhanced modeling, where complex relationships between system parameters are captured to develop accurate and reliable models for performance evaluation and system design. Additionally, adaptive signal processing optimizes tasks like noise reduction, error correction, and adaptive modulation in real-time, improving quality of service. Nonlinearity equalization helps mitigate signal distortions caused by nonlinear optical effects, ensuring robust and high-quality signal transmission. ML and DL are also applied in resource allocation to efficiently manage network resources, such as bandwidth and power, thereby enhancing scalability and efficiency. Moreover, constellation shaping optimizes signal patterns, improving transmission efficiency and error resilience [8], [9], [10].

In OCN, ML and DL are employed for various purposes. Network optimization is a key area, improving network routing, resource allocation, and traffic management to achieve more efficient and resilient communication systems. Fault detection and mitigation techniques are used to proactively identify and resolve issues such as fiber cuts or optical impairments, maintaining network reliability. Resource management optimizes power consumption and spectrum usage, ensuring system sustainability and efficiency. ML and DL also enable optical performance monitoring (OPM), providing real-time monitoring of network performance parameters to detect and diagnose problems. Furthermore, quality of transmission (QoT) estimation evaluates network performance metrics, ensuring data integrity and reliability.

ML algorithms used in optical communication are generally classified into three categories. Supervised learning (SL) algorithms learn from labeled data to make predictions and classifications. Common techniques in this category include support vector machines (SVMs) [11], artificial neural networks (ANNs) [12], k-nearest neighbors (kNNs) [13], ensemble learning algorithms [14], and linear models [15], [16]. Unsupervised learning (USL) algorithms analyze unlabeled data to identify patterns or clusters. Examples include hierarchical clustering [17], K-means [18], expectation maximization (EM) [19], principal component analysis (PCA) [20], and independent component analysis (ICA) [21]. Reinforcement learning (RL) algorithms, on the other hand, interact with an environment to learn optimal decision-making strategies, such as policy-based RL and value-based RL [22].

DL, as a subset of ML, uses artificial neural networks with multiple layers to extract complex patterns from large datasets. DL applications in optical communications include deep neural networks (DNNs) [23], which generalize neural networks by employing multiple layers to analyze complex data relationships. Recurrent neural networks (RNNs) [24] are designed for sequential data and are particularly useful for optical signal processing and traffic prediction in dynamic networks. Convolutional neural networks (CNNs) [25] excel at processing spatial data, such as optical channel images or network topology visualizations. Lastly, deep RL (DRL) combines RL and DL [26] to facilitate intelligent decision-making in complex and dynamic optical network environments.

A. RELATED WORKS

ML and DL have become pioneering tools across various fields, including optical communications. Numerous surveys have examined the advancements and applications of ML/DL in this domain, providing valuable insights into their potential and identifying opportunities for future exploration (Table 2). In [27], the authors investigate the use of ML for failure management in optical networks, highlighting its ability to transform fault detection practices and enhance network resilience. Similarly, ML techniques have been applied for routing optimization in software-defined networks (SDNs). The survey in [28] categorizes ML approaches for routing optimization into three main groups: SL, USL, and RL. The work in [29] provides a comprehensive review of routing and resource allocation (RRA) challenges in optical networks. It discusses critical problems such as routing and wavelength allocation (RWA) in wavelength division multiplexing (WDM), routing and spectrum allocation (RSA) in elastic optical networks (EONs), and routing, core, and resource allocation (RCSA) in spatial division multiplexing (SDM) systems. This survey also explores ML's role in addressing issues like QoT estimation, traffic prediction, and crosstalk mitigation. The potential of ML to optimize optical network performance is explored in [30]. This work focuses on algorithm selection, data and model management, and ML integration into network control systems. Four practical case studies demonstrate ML's versatility in addressing real-world challenges such as predictive maintenance, virtual network topology management, capacity optimization, and optical spectral analysis. The applications of ML in intelligent optical networks are further reviewed in [31]. The authors classify ML use cases into two broad categories: network control and resource management (e.g., RWA, bandwidth allocation, traffic management, self-healing networks) and optical network monitoring and survivability (e.g., fault detection, performance monitoring, network planning, and intrusion detection).

A broader review of AI applications in optical communications is provided in [32], encompassing both optical transmission and network management. This work

highlights AI's potential to improve signal processing, optimize resource allocation, and enhance network resilience. In a related study, [33] delves into the motivations for ML adoption in optical networks, emphasizing its capacity to address challenges such as nonlinear equalization, constellation shaping, adaptive modulation, and fault detection. Mathematical foundations of ML techniques, including SL, USL, and RL, are explored in [34]. This survey also examines ML-based solutions for channel estimation, signal detection, and equalization to tackle physical layer issues such as optical nonlinearities and interference. ML's application in SDNs, particularly for traffic classification, routing optimization, quality of service (QoS) prediction, and security, is analyzed in [35]. The integration of DL in optical communications, focusing on DNNs, RNNs, and DRL, is highlighted in [36]. This work emphasizes the innovative potential of DL in tasks like optical signal processing and network traffic prediction. A focused review of ML for coherent optical-orthogonal frequency division multiplexing (OFDM) systems is presented in [37]. This survey systematically analyzes ML-based approaches, including linear prediction, Volterra series, and neural networks, and compares them with conventional digital signal processing solutions. QoT prediction techniques utilizing random forests, neural networks, and SVMs are discussed in [38]. Finally, [39] reviews ML applications in OPM. This includes tasks such as signal quality monitoring, impairment detection, fault localization, and Bit Rate-Modulation Format Identification (BR-MFI), demonstrating ML's effectiveness in enhancing OPM capabilities and enabling proactive network management.

These studies collectively underscore the significant advancements and diverse applications of ML/DL in optical communications. However, further work is needed to explore the integration of these technologies across emerging domains, such as OWC, and to address challenges related to scalability, efficiency, and deployment in real-world networks.

B. MOTIVATIONS, NOVELTIES, AND CONTRIBUTIONS

1) MOTIVATIONS

The field of AI in optical communications is progressing at an unprecedented rate, driven by the continuous advancements in ML and DL techniques. However, despite numerous studies exploring these technologies, a critical analysis of existing literature reveals that many AI-driven methodologies remain underexplored. Additionally, the rapid emergence of new techniques, such as end-to-end learning for optical systems and RL in optical networking, highlights the necessity of an up-to-date, structured, and comprehensive survey.

Existing surveys often focus on specific subsets of AI applications, typically within a single domain such as OFC or OWC, without providing a holistic perspective across multiple optical communication paradigms. Moreover, prior works primarily categorize research based on application areas,

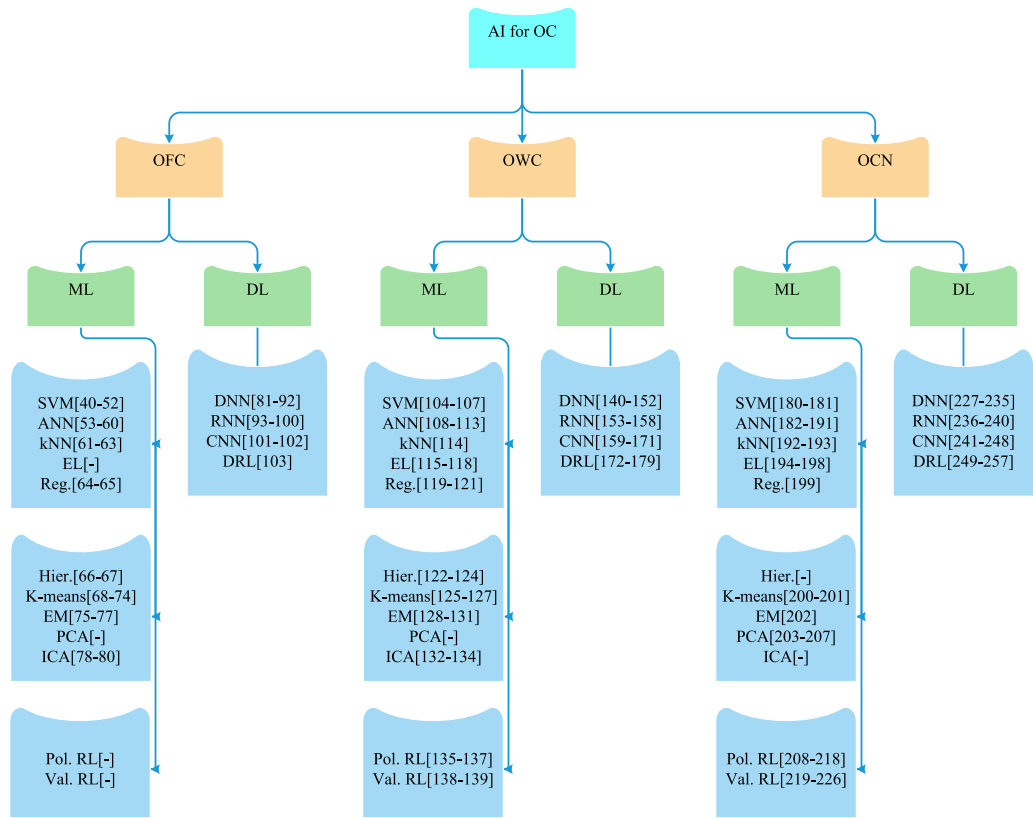


FIGURE 1. Schematic diagram categorizing different AI methods applied in OFC, OWC, and OCN.

TABLE 2. Summary of DRL applications in OFC, OWC, and OCN.

Ref	Technique	Media	Year	Number of Refs	Highlights
[27]	ML	OCN	2019	73	Review of ML for failure management in optical networks
[28]	ML	OCN	2021	201	Survey on ML for routing optimization in SDN
[29]	ML	OCN	2020	110	Overview of ML for routing and resource allocation in optical networks
[30]	ML	OCN	2018	43	ML to improve performance and efficiency of optical networks
[31]	ML	OCN	2020	132	Survey of ML applications in intelligent optical networks
[32]	ML	OCN	2018	148	Review of AI for enhancing optical communication systems
[33]	ML	OCN	2018	125	Overview of ML in optical communications and networking
[34]	ML	OFC	2019	86	Mathematical foundations of ML for optical communications and networking
[35]	ML	OCN	2018	278	Review of ML in SDN: traffic classification, routing, QoS, security
[36]	DL	OCN	2021	42	Contributions of DL to optical communications
[37]	ML	OCN	2018	48	ML for coherent optical-OFDM with a focus on practical aspects
[38]	ML	OCN	2022	108	Applications of ML in QoT prediction for optical networks
[39]	ML	OCN	2022	79	Review of OPM techniques with ML algorithms
[This work]	ML, DL	OFC, OCN, OWC	2025	286	Survey on ML and DL applications in OFC, OWC, and OCN

offering limited insights into the fundamental algorithmic methodologies. This lack of a methodological focus makes it difficult for researchers to understand how AI techniques address the core challenges in optical communication.

Given this dynamic landscape, the need for a well-structured survey that synthesizes recent advancements, identifies research gaps, and differentiates emerging trends has become increasingly evident. Such surveys are crucial for enabling researchers to make informed decisions,

remain updated on cutting-edge developments, and contribute effectively to the evolving field of AI-driven optical communications.

2) NOVELTIES AND CONTRIBUTIONS

While existing surveys provide valuable insights into AI applications in optical communications, they often have several limitations, including a lack of discussion on end-to-end learning approaches and RL in optical networks.

Furthermore, few studies critically differentiate their contributions from prior works, limiting their practical value. This survey addresses these gaps and offers a unique perspective, with key contributions summarized as follows:

- **Comprehensive and Up-to-Date Survey:** This work provides a structured and detailed review of AI applications in OFC, OWC, and OCN, integrating recent advancements in end-to-end learning and RL-based methods.
- **Methodology-Centric Approach:** Unlike prior surveys that primarily focus on application domains, this paper adopts an algorithm-centric perspective, offering a deeper understanding of how ML and DL methodologies address optical communication challenges.
- **Incorporation of RL and End-to-End Learning:** The survey includes a discussion of RL for dynamic network optimization and end-to-end learning for optical systems, ensuring the coverage of state-of-the-art techniques that have recently gained traction.
- **Quantitative and Comparative Analysis:** The paper presents a comprehensive evaluation of ML and DL approaches, including their advantages, limitations, and performance gains compared to conventional algorithms.
- **Summarized Tables for Research Insights:** Detailed tables categorizing reviewed studies based on methodology, objectives, performance metrics, and computational complexity are provided, facilitating knowledge sharing and informed decision-making.
- **Identification of Research Gaps and Future Directions:** Key findings are summarized, existing challenges are critically analyzed, and future research directions are proposed, ensuring that the survey remains a valuable resource for both new and experienced researchers in the field.

Additionally, this survey critically differentiates itself from prior literature by engaging with previous surveys, highlighting their contributions and limitations, and positioning this work as a complementary and forward-looking analysis. By synthesizing advancements in AI-driven optical communication and incorporating emerging methodologies such as RL and end-to-end learning, this survey enhances the understanding of AI's significant potential in this domain. The insights provided herein will serve as a valuable reference for researchers, enabling them to navigate the rapidly evolving landscape of AI applications in optical communication effectively.

C. PAPER ORGANIZATION

This review systematically explores the applications of ML and DL in optical communications through a structured framework, as illustrated in Fig. 1. The core focus lies on three key domains: OFC, OWC, and OCN each addressed in dedicated sections (Sections II, III, and IV). Within each domain, ML and DL approaches are analyzed separately in subsections (A and B), enabling an effective comparison. The

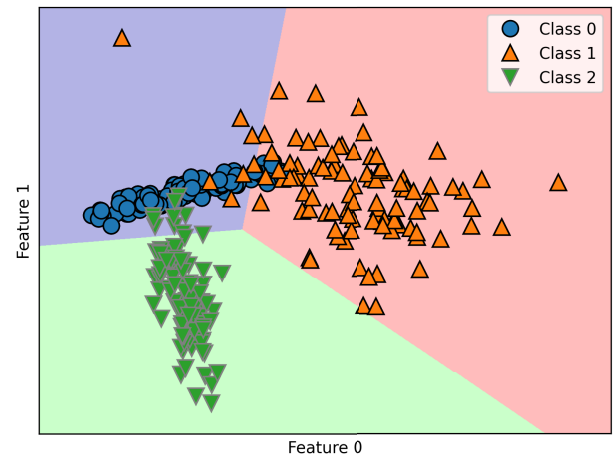


FIGURE 2. Descriptions of SVM.

ML section examines three primary categories in an order; SL (including SVM, ANN, kNN, ensemble learning, and linear models), USL (including hierarchical clustering, K-means, EM, PCA, and ICA models), and RL (including policy-based RL and value-based models). Similarly, the DL section highlights four major architectures: DNNs, RNNs, CNNs, and DRL, detailing their unique applications and advantages in optical communications. Each subsection begins with a visual representation of the ML or DL paradigm, followed by a comprehensive review of its implementations in OFC, OWC, and OCN. This approach provides a clear understanding of how various ML and DL techniques address challenges in optical communications. Section V discusses key insights, identifies challenges, and proposes future research directions, while Section VI concludes the review, offering a holistic perspective on this rapidly evolving field.

II. OPTICAL FIBER COMMUNICATION

A. MACHINE LEARNING

1) SUPPORT VECTOR MACHINE ALGORITHM

Figure 2 illustrates a SVM classifier's decision boundaries for a multi-class classification problem with three classes (Class 0, Class 1, and Class 2) using two features (Feature 0 and Feature 1). The SVM aims to find optimal hyperplanes that maximize the margin between different classes in the feature space. The decision boundaries, represented by the colored regions, are formed by these hyperplanes, which are linear in this case, dividing the space into distinct class regions. New data points are classified based on which side of the hyperplane they fall into. The support vectors, which are the data points closest to the decision boundary, play a crucial role in defining the margin and thus the classifier's performance. The plot demonstrates how SVM effectively separates classes even when they are not perfectly linearly separable by projecting them into a higher-dimensional space (implicitly through kernel trick)

TABLE 3. Summary of ML applications in OFC.

Ref	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[45]	Learn the relation between received and transmitted symbols	k-means, Bit-based SVM	BER	1.2-1.8 dB improvement in sensitivity, lower complexity than conventional linear detection
[52]	Learn the relation between received and transmitted symbols	HD detector, SVM detector	Identification accuracy	BER of 10^{-6} at 18 dB SNR, 10x lower complexity than conventional detectors
[48]	Compensate for VCSEL nonlinearity	SVM-RBF, SVM-Linear, SVM-Poly	Sensitivity, Specificity, Accuracy, F1-score	BER reduced by up to 10x, 50% lower complexity than conventional equalizers
[49]	Compensate for fiber nonlinearities	Volterra, Fast Newton-based SVM NLE	BER, Computational complexity	2 dB extended optimum launched power, 6x lower computational load than SVM
[53]	Compensate for fiber nonlinearities	Adaptive filter, ANN-based equalizer	BER, Spectral efficiency	2 dB Q-factor improvement compared to Volterra, lower complexity than Volterra
[54]	Equalize received signal compensating nonlinearities	RBF-NN, Volterra	BER	4 dB Q-factor improvement compared to previous equalizers, complexity independent of nonlinear terms
[58]	Classify received signal points into bit classes	ANN, Volterra, Linear equalizer	BER, Q-factor	1.5 dB OSNR improvement compared to LMS, more complex than LMS equalizer
[59]	Classify received signal points into bit classes	ANN, Volterra, DFE	BER, SER, MSE, NMSE, EVM	3 dB Q-factor improvement vs linear equalizer, 50% lower complexity than conventional linear equalizer
[60]	Predict optimal gain and noise figure of EDFAs	Proportional-integral-derivative control, RL	MSE, PAPR, BER	2 dB improvement in noise figure and 3 dB in gain flatness, real-time computation even for large networks
[62]	Mitigate impairments caused by nonlinearities in system link	Volterra equalizer, kNN-based detector	BER, Q-factor, Eye diagram	0.7 dB Q-factor improvement, real-time implementation with lower complexity
[63]	Compensate linear and nonlinear impairments	Non-Data-Aided-kNN, Linear equalizer, Volterra, DFE	BER, Q-factor, Eye diagram	0.5 dB BER improvement in 800-km SMF, low complexity
[64]	Reduce nonlinearity-induced distortion by 3 dB	Sparse Identification for Nonlinear Optics, Orthogonal Matching Pursuit	BER, Q-factor, Computational complexity	Achieves 3 dB reduction in Kerr nonlinearity compared to conventional Volterra series-based compensation
[199]	Predict fault location with high accuracy	Linear regression, Perceptron neural network	MAE, RMSE, R-squared	Provides 0.06m MSE, 0.08m MAE; faster than conventional statistical models for fault localization
[67]	Improve Q-factor in nonlinear equalization	Hierarchical clustering, Fuzzy-logic C-means, K-means, SVM, ANN	BER, SNR, NMSE	Improves Q-factor by 2.5 dB over Volterra-based equalization; fuzzy-logic C-means has lower complexity
[68]	Mitigate time-varying distortions in Nyquist-WDM	K-means, LSTM	MSE, BER, Q-factor	0.8 dB OSNR improvement at BER of 10^{-2} ; real-time applicability with lower computational cost than Volterra series
[69]	Cluster signal points based on density	Linear equalizer, K-means	BER, Q-factor	2.158 dB Q-factor improvement at 500 km; complexity $O(N^2)$, lower than Volterra $O(N^3)$
[70]	Classify received signal points into bit classes	RF, SVM, ANN	BER, Q-factor	0.4 dB Q-factor improvement over conventional decision thresholding; minimal computational resources required
[72]	Improve SNR in digitized radio-over-fiber	PCM, K-means	PAR, BER, Spectral efficiency	9 dB SNR improvement over conventional PCM; computationally efficient clustering
[75]	Estimate phase noise parameters efficiently	RF, SVM, XGB	MSE, RMSE, Correlation coefficient	20 dB MSE improvement in phase noise estimation; 10x faster than Wiener filtering
[78]	Uninformed equalization of received signals	ICA, CMA	BER, Q-factor	More robust to noise and nonlinearities than CMA; lower complexity
[79]	Separate received signal into polarization states	CMA, ICA	BER, Q-factor	Achieves same BER as CMA but with 3 dB lower SNR; increased memory complexity
[80]	Equalization without training symbols	ICA, Maximum likelihood	BER, MSE	Up to 3 dB reduction in BER; lower complexity than training-symbol-based equalizers

and finding a linear separating hyperplane in that higher space, which corresponds to a non-linear boundary in the original feature space.

SVMs have gained significant traction in OFC due to their adaptability, robustness, and ability to improve system performance. They provide efficient solutions for mitigating impairments such as nonlinear phase noise (NLPN), laser phase noise, and in-phase/quadrature modulator imperfections, ensuring signal integrity and minimizing error rates [40], [41], [42], [43]. Furthermore, SVMs address challenges associated with modulator nonlinearity and the Kerr effect, which limits transmission capacity, while also suppressing amplified spontaneous emission noise, thereby enhancing overall signal quality. Their applicability extends to both linear and nonlinear transmission

channels, making them indispensable in long-distance optical communication [44].

SVMs operate by mapping input data into a high-dimensional space using kernel functions, enabling effective classification and regression analysis. The decision boundary is determined by maximizing the margin between different classes, which is particularly useful in detecting signals in complex optical environments. For instance, in M-ary modulation schemes like 16-Quadrature Amplitude Modulation (QAM), 64-QAM, 16-APSK, and 32-APSK, SVMs employ techniques such as $\log_2(M)$ hyperplanes [45] or multi-layer binary classification [46], significantly improving classification accuracy compared to traditional detectors. Experimental studies indicate that SVM-based detection achieves a 1.2-1.8 dB improvement in sensitivity while

maintaining lower complexity than conventional linear detection [45].

In fault diagnosis, SVMs transform classification tasks into nonlinear optimization problems, allowing precise fault identification and localization. This capability facilitates timely maintenance interventions and enhances service reliability [47]. Their performance in identifying fiber faults and optimizing fault recovery is superior to conventional threshold-based detection methods.

SVMs also play a critical role in nonlinear equalization (NLE). Unlike conventional Volterra-based equalizers, which struggle with computational complexity, SVM-based NLEs achieve at least a 3.5 dB improvement at 60 Gbps and enable error-free transmission at 40 Gbps [48]. By learning the relationship between received and transmitted symbols, SVM-based equalizers effectively compensate for fiber nonlinearities, significantly extending the optimal launched power by up to 2 dB [49]. This is especially relevant for high-speed transmission, where nonlinear distortions limit system performance. Compared to traditional equalizers, SVM-NLEs exhibit up to 50% lower computational complexity, making them a viable solution for practical deployment.

In high-density modulation scenarios, SVMs are often combined with the constant modulus algorithm (CMA) to enhance Carrier-less Amplitude Phase modulation performance. This hybrid approach enables direct demapping of rotated constellations without additional correction, reducing BERs compared to conventional methodologies [50], [51], [52]. Experimental validation has shown that SVM-based detectors can achieve a bit error rate (BER) of 1×10^{-6} at 18 dB signal to noise ratio (SNR) with ten times lower complexity than traditional detectors [52].

In multi-channel WDM and Optical OFDM systems, SVM-based NLEs have demonstrated superior performance in mitigating fiber nonlinearity, stochastic noise interactions, and inter-channel nonlinear crosstalk. Specifically, in multi-channel Quadrature Phase Shift Keying (QPSK) systems, SVM-NLEs outperform digital back-propagation (DBP), achieving a 2 dB improvement in optimal launch power for 40 Gbps 16-QAM coherent optical-OFDM transmission over 2000 km [49]. Despite their effectiveness, SVMs face computational limitations, particularly in large-scale implementations, where alternative approaches like Fast Newton-based SVMs are explored to reduce computational load by a factor of six compared to traditional SVMs [49].

In conclusion, SVMs have emerged as a highly effective tool in OFC, addressing key challenges such as signal detection, fault diagnosis, and nonlinear equalization. Their ability to learn complex relationships, adapt to diverse signal environments, and provide computational efficiency makes them integral to the development of next-generation optical communication systems. Nevertheless, challenges such as computational scalability and kernel selection require further research to optimize their practical deployment in ultra-high-speed networks.

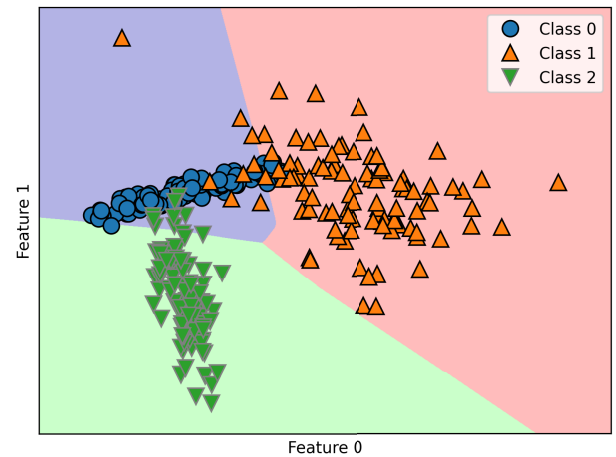


FIGURE 3. Schematic of an ANN.

2) ARTIFICIAL NEURAL NETWORK ALGORITHM

Figure 3 is a visualization of the output of a trained neural network with two input features (Feature 0 and Feature 1) and three output classes (Class 0, Class 1, and Class 2). The colored regions would then represent the decision boundaries learned by the network after training, where each region corresponds to the area in the feature space where the network predicts a specific class with the highest probability. The data points represent the training samples used to adjust the network's weights and biases, enabling it to learn a complex, non-linear mapping from the input features to the output classes. The network effectively partitions the feature space into distinct regions, demonstrating its ability to classify data based on learned patterns. The complexity of the decision boundaries suggests that the network likely employs multiple layers with non-linear activation functions to capture the underlying structure of the data.

ANNs have emerged as powerful tools for enhancing OFC systems, particularly in tasks such as NLE, gain adjustment, and signal constellation shaping. Their ability to model complex, nonlinear relationships between input and output data makes them well-suited for these tasks. ANNs, by mimicking the way biological neurons process information, can adaptively learn from data, enabling them to provide efficient solutions to signal distortion, noise, and nonlinearity, all of which are common in optical communication systems.

The successful application of ANNs in OFC systems requires careful tuning of their hyperparameters, which control network architecture and the learning process. Hyperparameters, such as the number of layers, the number of neurons in each layer, and the learning rate, can be adjusted either manually or automatically using optimization techniques. In [53] and [54], hyperparameter tuning is emphasized as crucial for precise fitting of the network to the training data, ensuring optimal performance.

Two prominent ANN architectures used in OFC are Radial Basis Function Neural Networks (RBFNNs) and Multi-Layer Perceptrons (MLPs). RBFNNs are particularly effective for linearly separable data and are well-suited for problems where the relationship between inputs and outputs can be approximated by smooth functions [55]. On the other hand, MLPs, which are deep feedforward networks, excel at handling nonlinearly separable data and have been extensively used for complex signal processing tasks such as NLE [56]. In [57], a detailed study of ANN hyperparameter tuning in OFC provides valuable insights into how these hyperparameters can be optimized to achieve better performance in various OFC applications.

One of the key applications of ANNs in OFC is NLE. ANNs can significantly improve the Q-factor of optical communication systems by compensating for nonlinear distortions, which are typically caused by fiber nonlinearity and cross-phase modulation. For instance, [53] reports a 2 dB improvement in Q-factor for an ANN-based NLE compared to Volterra-based NLE in a 40 Gbps 16-QAM coherent optical-OFDM system over 2000 km. The ability of ANNs to adapt to complex nonlinearities makes them a powerful tool in compensating for fiber nonlinearities, which is a major challenge in long-distance high-speed communication. Similarly, [55] demonstrates that RBFNN-based NLE improves the Q-factor by 8.7 dB in a 16-QAM coherent optical-OFDM system, enhancing both data rate and transmission reach (1200 km).

ANN-based NLE techniques also offer enhanced non-linearity tolerance by mitigating interference such as cross-phase modulation. For example, [58] shows that an ANN-based NLE doubles the transmission distance (320 km) compared to the least mean square (LMS) algorithm, with a 0.7 dB improvement in optical signal-to-noise ratio (OSNR). Further, [59] demonstrates that ANN-based NLE outperforms linear equalization and Volterra-based NLE, providing a 3 dB and 1 dB Q-factor improvement, respectively, at a transmission distance of 1000 km in an 80 Gbps 16-QAM coherent optical-OFDM system. These results highlight the superiority of ANNs over conventional methods in terms of performance, especially in terms of handling nonlinear distortions and improving transmission distance.

In addition to NLE, ANNs have shown promise in gain adjustment for optical amplifiers. Reference [60] proposes an ANN-based method for optimizing cascaded amplifier gains, ensuring that both input and output powers remain within predefined limits while minimizing noise figure and maintaining gain flatness. This approach results in a noise figure of 30.06 dB and a gain flatness of 5.26 dB for a 6-amplifier system, with minimal error (less than 0.1 dB), which demonstrates the capability of ANNs in improving amplifier performance while reducing operational costs and complexity.

However, while ANN-based techniques offer significant improvements, there are also practical challenges associated

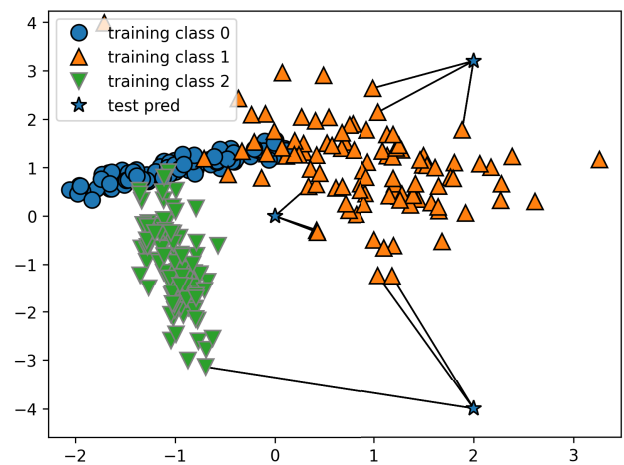


FIGURE 4. Visualization of kNN classification procedure.

with their use. The complexity of ANN models can be significantly higher compared to traditional methods such as linear equalizers, and training these models requires large amounts of data and computational power. Furthermore, the performance of ANN-based methods can be sensitive to the quality of the training data and the selection of hyperparameters. In comparison, conventional methods such as Volterra-based equalization or the LMS algorithm are computationally simpler but often fail to provide the same level of performance, particularly in systems with strong nonlinearities.

Overall, the use of ANNs in OFC systems presents a promising avenue for improving transmission, enhancing spectral efficiency, and achieving more robust and reliable communication. As research continues, ANN-based techniques are expected to play an increasingly important role in optimizing optical communication systems, although challenges related to complexity and training remain key areas for future improvement.

3) K-NEAREST NEIGHBORS ALGORITHM

Figure 4 depicts a kNN classification scenario where three training classes (Class 0, Class 1, and Class 2) are represented by circles, triangles, and inverted triangles, respectively, in a two-dimensional feature space. The “test pred” points, denoted by stars, represent unseen data points whose class labels are to be predicted. The kNN algorithm, in essence, classifies these test points based on the majority class among their ‘k’ nearest neighbors in the training data. The black lines connecting each test point to its nearest neighbors visually illustrate the distance metric used (likely Euclidean distance in this case) to determine proximity. The class prediction for each test point would be determined by the dominant class label within its ‘k’ closest training points, where ‘k’ is a user-defined parameter. This method is a

non-parametric, instance-based learning approach that relies on local information rather than building an explicit model.

The kNN algorithm is a widely used non-parametric method in optical communication systems for detecting and mitigating impairments, especially in the presence of circularly symmetric non-Gaussian noise such as NLPN, I/Q imbalance, and amplified spontaneous emission noise. In an optical phase conjugation system, specifically for 112 Gbps 16-QAM transmission, the kNN algorithm demonstrates significant performance improvement over conventional methods. It effectively enhances system performance over 800-km dispersion-managed and 1280-km dispersion-shifted-fiber links [61].

One of the challenges with the basic kNN algorithm is its sensitivity to the distribution of training data. The kNN method typically requires the calculation of distances between input data and its neighbors to make decisions, which can be computationally intensive, especially with non-uniformly distributed data. To address this limitation, Wang et al. [62] proposed incorporating distance-related weights for surrounding points. This enhancement boosts the kNN algorithm's robustness by improving its tolerance to linewidth variations and nonlinearity, showing improvements of 180 kHz in linewidth tolerance and 1.7, 1.0, and 0.4 dBm in nonlinear tolerance for dispersion-neglected, managed, and unmanaged links, respectively. This is achieved in comparison to traditional maximum likelihood post-compensation approaches, providing a clear benefit in terms of performance for optical communication systems operating in challenging environments.

However, the complexity of kNN increases with the size of the dataset and the choice of the k value, the number of neighbors used in classification. To alleviate this, Zhang et al. [63] introduced a Non-Data-Aided-kNN technique that focuses on using the density parameter of test data to identify noiseless data centers. This allows the algorithm to operate with low complexity, as only a small number of reference points are required for classification. Experimental results demonstrate the effectiveness of this approach, with a 0.5 dB improvement in BER in an 800-km single-mode fiber (SMF) 16-QAM system and a 2 dB improvement in an 80-km SMF 64-QAM system. The Non-Data-Aided-kNN method remains robust to noise and converges quickly, making it an efficient solution for mitigating both linear and nonlinear impairments in optical fiber networks.

Despite its advantages, the kNN algorithm still has some inherent limitations. The method's performance heavily depends on the quality and quantity of training data, and the optimal selection of the k parameter. A larger k value can lead to smoother decision boundaries but may also increase computational complexity, particularly for large datasets. Additionally, kNN's reliance on distance-based calculations makes it less efficient in high-dimensional spaces, which is a common challenge in modern optical communication systems.

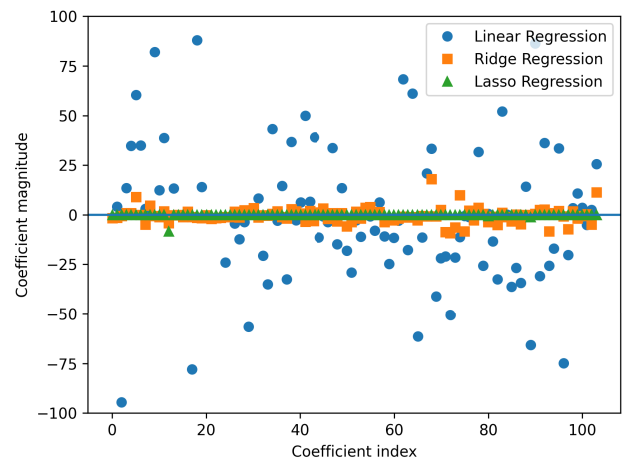


FIGURE 5. An example of linear models.

In comparison to conventional methods like the Volterra equalizer, kNN-based detectors provide real-time implementation with lower complexity, particularly in nonlinear equalization tasks. However, the performance improvement in kNN is context-dependent, and the method's suitability should be evaluated based on the specific system configuration and impairment characteristics. For instance, while kNN offers a clear advantage in compensating for nonlinearities and mitigating impairments like NLPN, traditional methods like Volterra might still outperform kNN in cases involving more complex interference patterns or where highly accurate compensation is required. Furthermore, kNN algorithms can be computationally expensive in larger optical systems due to the need for distance calculations across large datasets.

Overall, kNN serves as an effective tool for mitigating impairments in optical communication systems, especially when combined with improvements such as distance-weighted neighbors and the non-data-aided approach. Its low complexity, especially in nonlinear compensation, makes it a promising candidate for real-time applications, although its dependence on training data and parameter selection should be carefully managed to optimize performance.

4) LINEAR MODELS

Figure 5 presents a comparative analysis of coefficient magnitudes from three linear regression models: Linear Regression, Ridge Regression, and Lasso Regression. Linear Regression, represented by blue circles, exhibits a wide dispersion of coefficient values, indicating that it does not inherently impose constraints on the magnitude of coefficients. Ridge Regression (orange squares) demonstrates a noticeable shrinkage effect, where coefficient magnitudes are generally smaller compared to Linear Regression, suggesting the regularization term (L2 norm) in Ridge is effective in reducing coefficient size. Lasso Regression (green triangles)

shows an even more pronounced shrinkage, with many coefficients driven to exactly zero, signifying its ability to perform feature selection through the L1 norm regularization. This visual comparison highlights how different regularization techniques in linear models impact coefficient magnitudes and, consequently, model complexity and feature importance.

Regression algorithms, including linear regression, Lasso regression, and ridge regression, are widely used in optical communication systems for tasks such as signal prediction, parameter estimation, and impairment compensation. These techniques model the relationship between input variables and output signals by finding the best-fitting function that minimizes the difference between observed and predicted values. In optical communication, regression algorithms are primarily employed for mitigating the effects of nonlinearities, such as inter-channel and intra-channel Kerr nonlinearity, which are common challenges in systems with high-speed data transmission.

Among the regression techniques, Lasso (Least Absolute Shrinkage and Selection Operator) regression has gained significant attention, particularly for compensating nonlinearities in SDM-based transmission systems. Lasso regression incorporates a penalty term in the cost function, which encourages sparsity in the model by driving some coefficients to zero. This sparsity-promoting property makes Lasso particularly effective for identifying the most relevant perturbation terms in nonlinear optical systems while minimizing computational complexity. By reducing the number of active terms in the model, Lasso regression effectively simplifies the overall computational load, making it well-suited for real-time implementation in high-speed optical communication systems, such as those involving SDM.

Several studies have demonstrated Lasso regression's ability to mitigate nonlinearities and enhance the performance of SDM-based transmission systems. For instance, Sorokina et al. [64], [65] showed that Lasso regression could reduce nonlinearity-induced signal distortion by approximately 3 dB, improving signal integrity over long transmission distances. This capability makes Lasso regression particularly useful in SDM systems, where the increased number of channels and the complex interactions between them exacerbate nonlinear distortion. The reduction in nonlinearity-induced distortion directly translates into improvements in key performance metrics, such as BER and quality factor (Q-factor), which are crucial for maintaining reliable high-speed communication.

Despite its effectiveness, Lasso regression has certain limitations. The main challenge lies in the selection of the regularization parameter, which controls the strength of the penalty term. A poor choice of this parameter can lead to either overfitting (when the penalty is too weak) or underfitting (when the penalty is too strong), resulting in suboptimal performance. Furthermore, Lasso regression assumes that the relationships between input features and output variables are linear, which may not always hold

true in highly nonlinear optical systems. While Lasso can approximate nonlinearities effectively in many cases, it may struggle with more complex, non-linear relationships that require more advanced models.

In terms of computational complexity, Lasso regression is generally considered efficient due to its sparse nature. This makes it faster and less resource-intensive compared to other regression methods like ridge regression, which does not promote sparsity and may result in higher computational overhead. The low complexity of Lasso also makes it an attractive choice for real-time applications, where fast processing is crucial.

Lasso regression's performance can be further enhanced when combined with other techniques, such as the Sparse Identification for Nonlinear Optics method and Orthogonal Matching Pursuit (OMP), which help in identifying sparse nonlinear perturbations in optical systems. These hybrid approaches are especially useful for high-capacity SDM transmission, where they facilitate efficient nonlinearity mitigation over long transmission distances. As SDM technology continues to evolve, the role of Lasso regression in optimizing high-speed optical communication systems is expected to expand, offering a scalable and computationally efficient solution for nonlinear compensation.

However, a key comparison with conventional methods reveals that while Lasso regression excels in low-complexity, real-time applications, traditional techniques such as Volterra series and polynomial models may offer more accurate nonlinear compensation at the cost of increased computational complexity. These traditional methods are often better suited for systems where high precision is required in mitigating complex, multi-dimensional nonlinearities, such as those present in dense WDM systems or when multi-band and multi-core fibers are used. In contrast, Lasso regression's simplicity and effectiveness make it an ideal choice for applications where computational resources are limited, and real-time performance is essential.

In conclusion, Lasso regression provides a valuable tool for compensating nonlinearities in optical communication systems, particularly in the context of SDM transmission. Its low complexity, real-time implementation capability, and ability to reduce nonlinearity-induced signal distortion make it a promising technique for high-speed optical networks. However, its limitations, such as sensitivity to regularization parameter selection and its assumption of linear relationships, suggest that it may not always be the best solution for highly nonlinear systems. Combining Lasso regression with other advanced techniques could offer a more comprehensive solution, balancing the need for computational efficiency and high performance in next-generation optical communication systems.

5) HIERARCHICAL CLUSTERING ALGORITHM

Figure 6 depicts the progression of a hierarchical clustering algorithm over multiple steps, illustrating the iterative grouping of data points into clusters. Starting with an initial

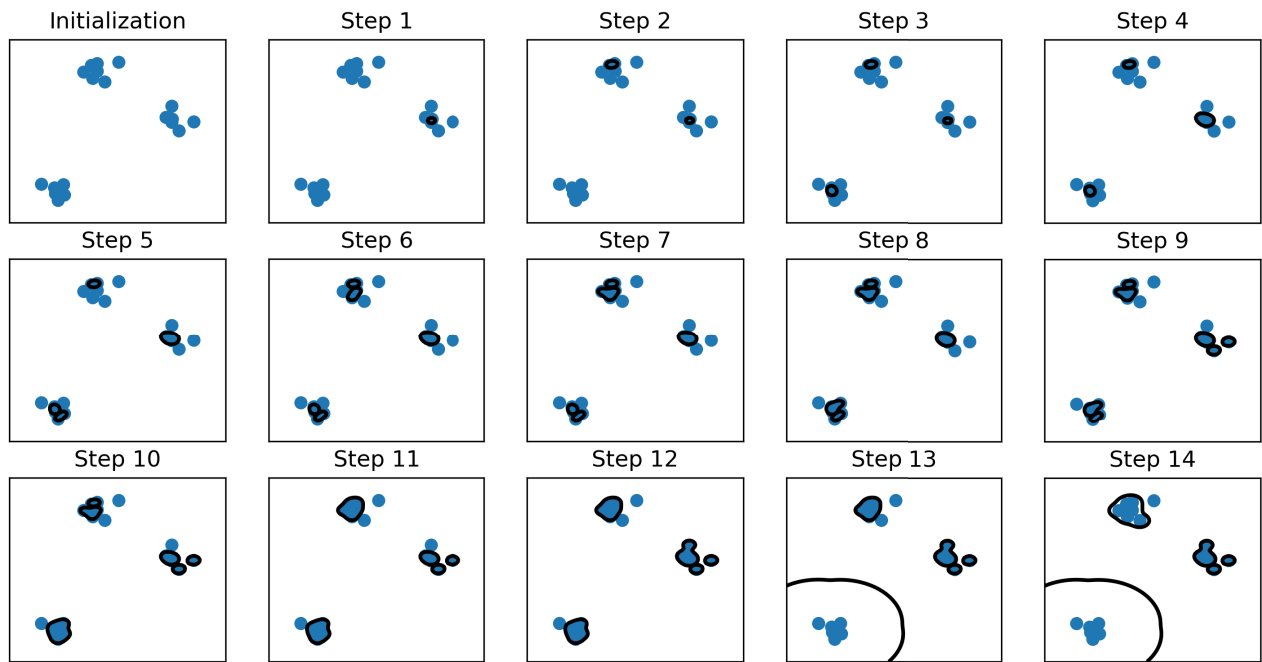


FIGURE 6. Demonstration of hierarchical clustering.

state where each data point is considered its own cluster, the algorithm progressively merges the closest clusters based on a distance metric, visualized by the expanding black contours. In the early steps, nearby points are grouped together, and as the process continues, larger, more encompassing clusters emerge. The increasing size of the black contours reflects the agglomeration of data points, demonstrating the hierarchical nature of the clustering process. The final steps show the formation of distinct, well-separated clusters, highlighting the algorithm's ability to reveal underlying data structures without requiring a pre-defined number of clusters. This sequential visualization effectively demonstrates the bottom-up, agglomerative approach of hierarchical clustering, where smaller clusters are iteratively merged to form a hierarchical representation of the data.

Hierarchical clustering is a powerful USL algorithm that organizes similar data points into a hierarchical structure. It is commonly used in optical communication systems for tasks like signal classification, modulation recognition, and nonlinear impairment compensation. The algorithm operates by successively merging (agglomerative) or splitting (divisive) data clusters based on a distance metric, such as Euclidean distance or correlation, which measures the similarity between data points. The result is a tree-like structure, known as a dendrogram, which visually represents the grouping of data points at different levels of similarity.

In optical communication, hierarchical clustering has been applied to NLE, where it helps group received signal points into clusters that exhibit similar nonlinear distortion

characteristics. For instance, in [66], hierarchical clustering was compared to Fuzzy-Logic C-means clustering for NLE, with [67] showing that Fuzzy-Logic C-means outperforms hierarchical clustering in terms of NLE performance. The key advantage of Fuzzy-Logic C-means is its ability to adaptively determine the optimal clustering based on fuzzy membership values, allowing it to handle varying degrees of nonlinearity more effectively.

When applied to low-level modulation formats like BPSK and QPSK, Fuzzy-Logic C-means excels in determining the optimal launched power, providing more robust compensation across a wide range of operating conditions. On the other hand, hierarchical clustering demonstrates greater effectiveness at higher powers, suggesting that it may be better suited for environments with less variation in signal strength. However, while hierarchical clustering is effective for clustering signal points based on power, its performance can degrade under conditions of significant power variation, where Fuzzy-Logic C-means remains more consistent.

A notable benefit of Fuzzy-Logic C-means over hierarchical clustering is its lower complexity in comparison to other NLE techniques, such as K-means, fast-Newton SVM, ANN, and Volterra-based NLE. In particular, Fuzzy-Logic C-means demonstrates substantial improvements in signal quality for BPSK and QPSK modulation formats, achieving Q-factor increases of 2.5 dB and 0.4 dB, respectively. This makes it highly attractive for real-time applications where computational efficiency is crucial, and it can significantly improve system performance without introducing high processing overhead.

Hierarchical clustering, while simpler in structure and easier to implement, may not be as flexible in capturing complex nonlinearities compared to more advanced techniques. For instance, while it is effective at clustering similar data points, its rigid clustering scheme may not always capture the dynamic nature of nonlinear distortions that change over time, especially in high-speed optical communication systems. This limitation can hinder its performance in systems with rapidly changing channel conditions, where Fuzzy-Logic C-means can dynamically adapt to the varying degrees of nonlinearity.

Despite these limitations, hierarchical clustering remains a useful tool for optical communication systems where the complexity of the channel is relatively simple, and the computational resources are constrained. It offers a straightforward solution to clustering signal data based on predefined similarity criteria and is less computationally intensive than more sophisticated techniques like SVM or ANN. In such systems, hierarchical clustering may still outperform conventional methods that do not explicitly account for the inherent nonlinearities in optical transmission, such as traditional equalizers based on linear filtering techniques.

In comparison to conventional methods, the performance of ML-based clustering algorithms, such as hierarchical clustering and Fuzzy-Logic C-means, shows clear advantages in handling complex signal distortion and mitigating nonlinear effects. Conventional approaches like Volterra-based NLE or linear equalizers are often limited by their inability to handle high-dimensional, nonlinearly separable data effectively. In contrast, ML-based methods can provide superior performance, especially in systems where nonlinearities dominate the signal processing tasks. However, these advantages come at the cost of increased computational complexity, which is a significant factor when considering the practical implementation of these techniques in real-world optical networks.

In conclusion, hierarchical clustering offers a valuable technique for clustering signal points based on similarity, particularly in applications with simpler nonlinearities and lower computational resource demands. However, Fuzzy-Logic C-means clustering, due to its adaptability and ability to handle complex nonlinearities, provides a more robust and efficient solution in many optical communication systems. As the complexity of optical networks increases, ML-based clustering methods, such as Fuzzy-Logic C-means, will likely play a more dominant role, enabling higher performance and more efficient real-time operation in next-generation optical communication systems.

6) K-MEANS CLUSTERING ALGORITHM

Figure 7 illustrates the iterative process of K-means clustering, demonstrating how data points are grouped into clusters based on their proximity to cluster centroids. Starting with an “Initialization” step where initial centroids are randomly assigned, the algorithm proceeds to “Assign

Points” where each data point is assigned to the nearest centroid, forming initial clusters. The “Recompute Centers” step then recalculates the centroids as the mean of the points within each cluster. This assignment and recomputation cycle repeats (“Reassign Points” and “Recompute Centers”) until convergence, where cluster assignments stabilize. The “Final Cluster Assignment” shows the resulting clusters, with each point belonging to a specific cluster, and the “Decision Boundaries” depict the regions in the feature space where points are assigned to each cluster, effectively visualizing the partitioning achieved by the K-means algorithm. This process highlights K-means’ ability to partition data into K distinct, non-overlapping subsets based on minimizing the within-cluster sum of squares.

K-means clustering is a versatile ML algorithm widely used for mitigating signal distortion and enhancing performance in optical communication. Its ability to identify patterns and group similar data points makes it effective for addressing fiber nonlinearity, optimizing transmission, and enabling efficient resource management. In [68], time windowing within the K-means algorithm was used to track signal variations in multi-carrier 16-QAM Nyquist-WDM systems, compensating for distortions from I/Q imbalance, NLPN, and inter-channel interference, resulting in reduced OSNR requirements and improved BER performance. A modified version of K-means was introduced in [69] for 24.72 Gbps DQPSK coherent optical-OFDM transmission, outperforming linear equalization and improving Q-factor performance over 500 km fiber links. To address fiber non-Gaussian nonlinearities in dispersion-managed and unmanaged fibers, [70] employed a Parzen window-based detector with K-means clustering, achieving Q-factor improvements up to 0.4 dB. K-means has also been applied in spectrally efficient digitized radio over fiber transmission, where it improved SNR and enabled 30 Gbps transmission with error vector magnitudes (EVMs) of 8% and 1% for 4 and 7 quantization bits, respectively [71], [72]. In optical phase-modulated radio over fiber systems, [73] and [74] demonstrated K-means clustering for RF phase recovery. Reference [73] achieved demodulation of three-channel 8-PSK subcarrier multiplexed signals at 5 GHz over 40 km, while [74] showed efficient demodulation of three-channel WDM signals at 6 GHz RF carrier frequency over 78.8 km.

These examples highlight K-means clustering’s versatility in optical communication systems, making it a valuable tool for mitigating distortions, optimizing transmission, and improving system performance. As optical communication technology advances, K-means clustering will likely play an increasingly important role in enhancing system reliability and efficiency.

7) EXPECTATION MAXIMIZATION CLUSTERING ALGORITHM

Figure 8 illustrates the iterative process of Expectation-Maximization clustering, a probabilistic approach to

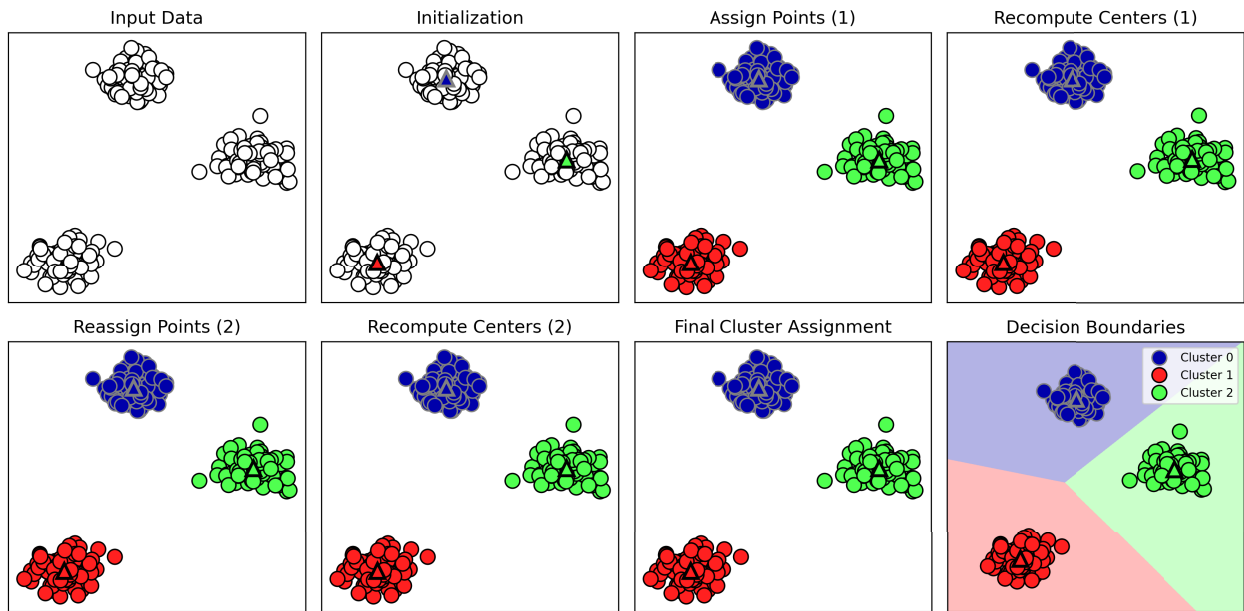


FIGURE 7. Example of K-means clustering.

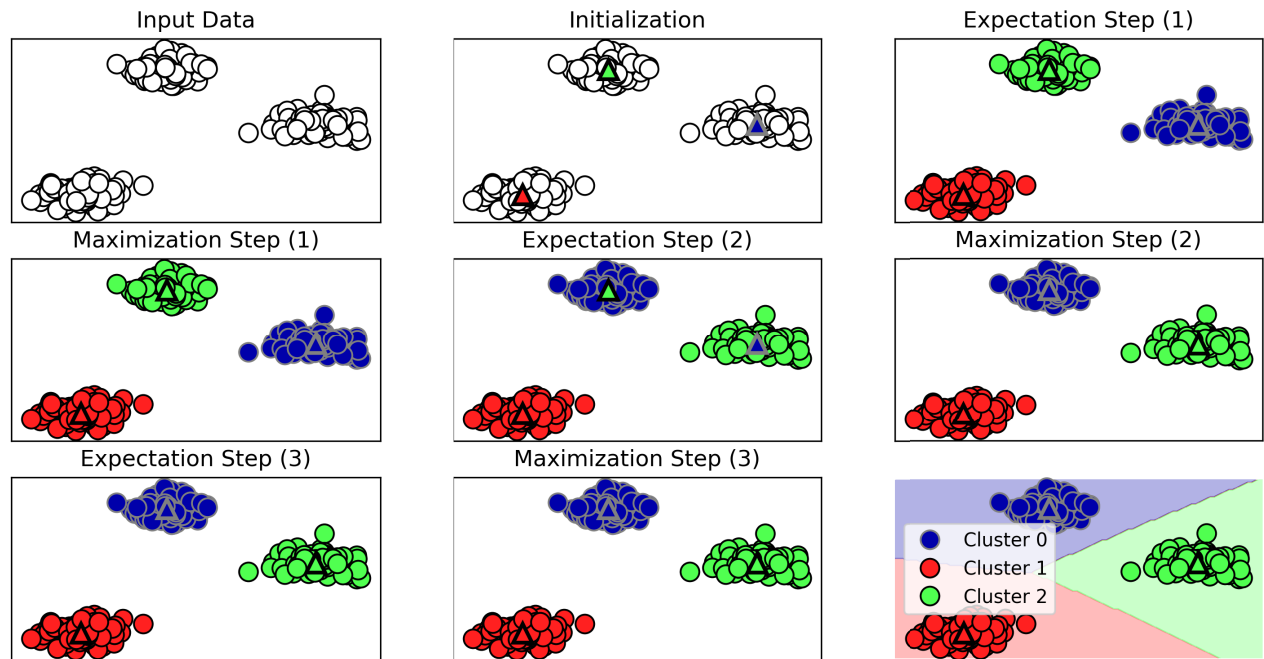


FIGURE 8. Example of EM clustering.

assigning data points to clusters. The process starts with “Initialization,” where initial cluster parameters (e.g., means and covariances) are estimated. The algorithm then alternates between two steps: the “Expectation Step,” where data points are assigned probabilities of belonging to each cluster based on the current parameters, and the “Maximization Step,” where the cluster parameters are updated to maximize the likelihood of the observed data given the cluster

assignments. This iterative refinement of cluster assignments and parameter estimations continues through multiple steps, as shown in Figure, until convergence. The “Final Cluster Assignment” and the “Decision Boundaries” demonstrate the resulting cluster assignments and the probabilistic regions in the feature space, respectively, reflecting the algorithm’s ability to handle clusters with varying shapes and sizes by modeling data as a mixture of probability distributions. This

process highlights EM's ability to handle soft assignments and model complex cluster structures compared to simpler methods like K-means.

EM clustering is an effective technique that is widely used for mitigating laser amplitude and phase noise in optical communication systems. EM clustering is based on Bayesian analysis, which assumes that the observed data (such as amplitude and phase signals) are generated by a probabilistic model. The algorithm iterates between two main steps: the expectation step (E-step), where it estimates the probabilities of data points belonging to a particular cluster (based on the current parameters), and the maximization step (M-step), where it updates the model parameters to maximize the likelihood of the observed data. This iterative process continues until convergence, yielding optimized parameters that best fit the data.

In the context of optical communication, EM clustering is applied to evaluate the likelihood that an incoming signal represents amplitude or phase noise. The advantage of EM clustering lies in its ability to incorporate prior knowledge (such as the distribution of laser noise) and estimate the noise parameters with high accuracy. The algorithm improves the detection and compensation of laser amplitude and phase noise, which are critical impairments in high-speed optical communication systems.

Several studies have demonstrated the effectiveness of EM clustering in mitigating laser noise. For example, in [75] and [76], simulations and experiments showed that EM clustering significantly improves the launch power tolerance by 1.5 dB and reduces the OSNR requirements by 0.3 dB, thereby enhancing the transmission efficiency. This improvement indicates that EM clustering helps in better signal detection by more accurately modeling and compensating for noise in the system. Furthermore, [77] integrated EM clustering with a soft decision error-control code, which iteratively mitigated NLPN and further boosted system performance, demonstrating the potential of combining ML with error-correction techniques to achieve even better noise mitigation.

One of the key benefits of EM clustering is its computational efficiency. According to [75], the algorithm achieves a 20 dB improvement in Mean Squared Error (MSE) for phase noise estimation compared to conventional methods, with a tenfold increase in speed. This makes EM clustering an attractive option for real-time applications in optical communication, where both accuracy and computational efficiency are critical. Additionally, EM clustering can be easily integrated with other ML techniques, such as Random Forests (RF), SVM, and eXtreme Gradient Boosting (XGB), to further enhance its performance in more complex scenarios.

Despite its advantages, there are some limitations to EM clustering. One challenge is that the algorithm's performance is highly sensitive to the initialization of its parameters, particularly the initial estimates of the noise model. Poor initialization can lead to suboptimal performance or even

convergence to a local optimum. To mitigate this issue, various methods, such as multiple initializations or more sophisticated optimization techniques, can be used, but they may introduce additional computational complexity. Furthermore, while EM clustering is powerful in noise mitigation, its performance may degrade in highly nonlinear or non-Gaussian noise environments where more advanced techniques, such as DL, may provide better results.

In comparison to conventional methods, such as RF or SVM, EM clustering has a clear advantage in its ability to model the underlying probabilistic structure of noise and perform iterative optimization to enhance noise estimation. Traditional methods often rely on fixed, predefined models that may not adapt well to varying channel conditions or changing noise characteristics. EM clustering, on the other hand, can dynamically adjust to the observed data, providing more accurate estimates of noise parameters. However, conventional methods, like SVM, tend to perform better in scenarios where the decision boundaries between signal and noise are more complex, especially in high-dimensional spaces. In these cases, ML techniques like SVM and XGB may offer superior performance by leveraging their ability to model non-linear decision boundaries.

In summary, EM clustering provides a robust method for compensating laser amplitude and phase noise in optical communication systems, offering improvements in signal quality, launch power tolerance, and OSNR. While it has some limitations, such as sensitivity to parameter initialization, its computational efficiency and adaptability make it a valuable tool for modern high-speed optical networks. As optical communication systems evolve, it is likely that EM clustering will continue to play a significant role, especially when combined with other ML techniques, to address the growing challenges of noise mitigation in increasingly complex and dynamic communication environments.

8) INDEPENDENT COMPONENT ANALYSIS ALGORITHM

ICA (Figure 9) is a powerful and versatile technique that has shown great promise in uninformed equalization and phase recovery for coherent optical communication systems. ICA works by exploiting the statistical independence of the underlying source signals, which allows it to effectively decompose a received signal into its independent components. This decomposition is particularly useful in mitigating impairments such as PMD and polarization-dependent loss (PDL), which are critical challenges in optical systems.

ICA operates by separating mixed signals into statistically independent components without the need for prior knowledge of the signals, unlike other methods such as the conventional CMA. This makes ICA particularly suitable for uninformed equalization tasks, where the receiver does not have access to known training symbols. In [78], ICA was applied to uninformed equalization and phase recovery, achieving comparable performance to CMA equalization in QPSK modulation. However, ICA demonstrated superior performance in higher-order modulation formats such as

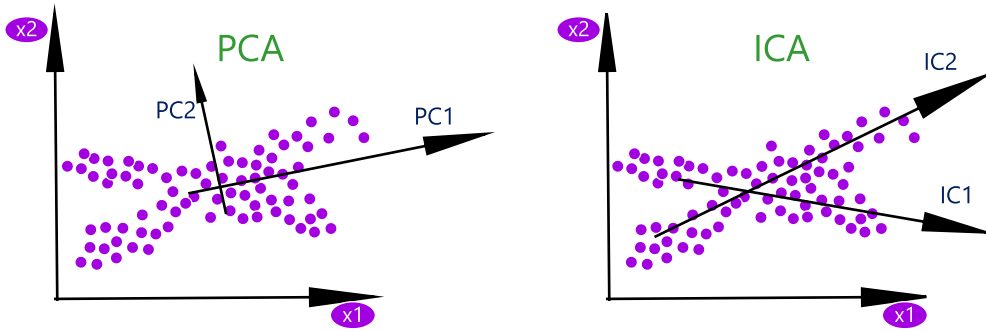


FIGURE 9. Example of data fitted using PCA and ICA.

16-QAM, particularly in compensating for the effects of PMD and PDL. This highlights ICA's advantage in systems using high-order modulation, where signal impairments can have a more significant impact on performance.

Furthermore, [79] showcased ICA's effectiveness in addressing polarization demultiplexing, where it outperformed CMA in handling the singular convergence problem. This problem is particularly challenging in conventional equalization algorithms, where the signal can become trapped in suboptimal solutions. ICA, however, was able to maintain robust polarization tracking performance, providing a more stable solution, especially under varying transmission conditions. Despite this, it is important to note that ICA-based methods can be more complex and require more memory than CMA-based methods, as demonstrated in [79], where ICA achieved the same BER as conventional methods but with a 3 dB lower SNR. This highlights a trade-off between performance and computational complexity.

In addition, [80] extended ICA for channel equalization in Direct Detection Optical OFDM (DDO-OFDM) and Polarization-Multiplexed (PM) coherent optical-OFDM systems. The study showed that ICA-based equalizers, when used without training symbols, achieved performance on par with training symbol-based methods, significantly improving system adaptability. This feature makes ICA a strong candidate for future optical communication systems, where minimizing the reliance on training symbols can reduce overhead and increase system flexibility.

One of the key advantages of ICA over traditional methods, such as CMA, is its robustness to noise and nonlinearities. For example, ICA has been shown to be more resilient to noise and nonlinear distortions compared to CMA, which makes it particularly beneficial in challenging transmission conditions. In simulations, ICA demonstrated a more robust performance with less complexity, achieving better results in high-order modulation schemes like 16-QAM while maintaining simplicity in implementation.

However, ICA does have some limitations. The algorithm's complexity increases with the number of independent components that need to be extracted, and it can require more memory compared to CMA, especially in large-scale

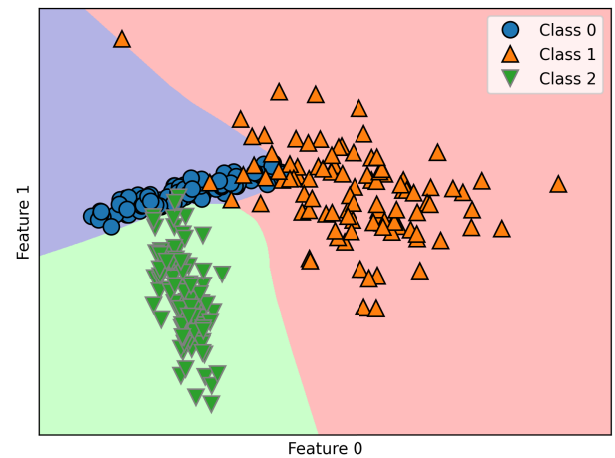


FIGURE 10. Schematic of DNN.

systems. This can be a significant consideration in practical implementations where computational resources are limited. Moreover, ICA's performance is sensitive to the mixing model, which assumes the signals are linearly mixed. In more complex, non-linear environments, ICA might not perform as well as other methods like Maximum Likelihood or DL-based approaches, which can model non-linearities more effectively.

When comparing ICA to other conventional methods, such as CMA, the trade-offs between computational complexity and performance must be carefully considered. While CMA is a simpler algorithm and widely used for uninformed equalization, its performance can degrade in the presence of PMD and PDL, especially for higher-order modulation formats. ICA, on the other hand, provides a more powerful solution for these impairments, but it comes at the cost of increased complexity. This makes ICA a better choice for systems where the benefits of improved performance outweigh the computational cost.

In conclusion, ICA is a highly effective algorithm for mitigating signal impairments in optical communication systems, particularly in uninformed equalization, phase recovery,

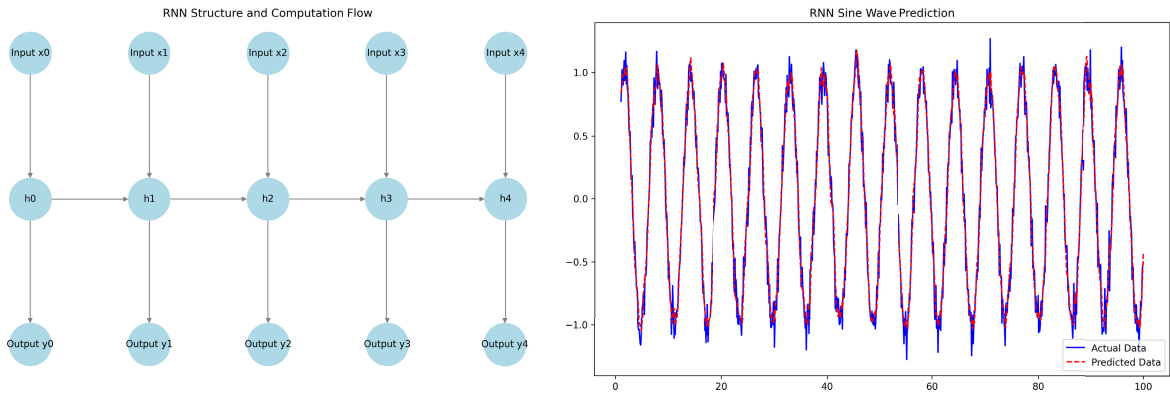


FIGURE 11. Schematic of RNN.

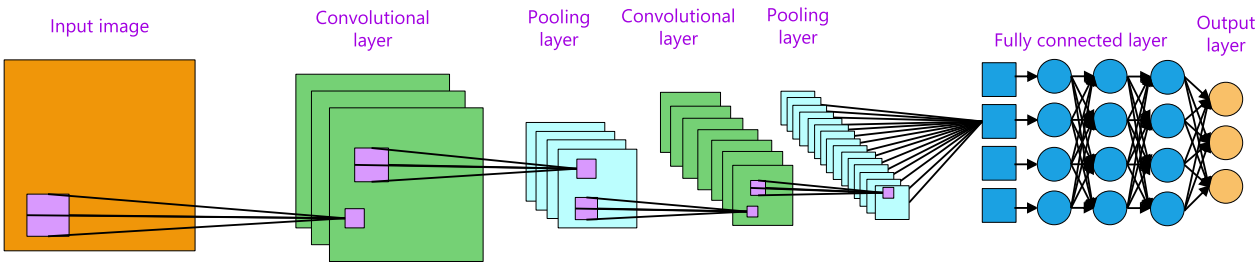


FIGURE 12. Structure of CNN algorithm.

TABLE 4. Summary of DL applications in OFC.

Ref	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[83]	Learn inverse of nonlinear Schrödinger equation	DNN, DBP, FFNN, CNN, RNN	BER, SER, Q-factor, Complexity	2 dB SNR penalty reduction, 1000x faster than DBP at 3200 km (vs. DBP)
[84]	Detect transmitted signal in MDM optical system	Zero forcing, Semi-definite relaxation, DNN	BER, Q-factor, MAE, RMSE, MSE	0.5 dB BER improvement, Complexity $O(N^2)$, lower than zero forcing
[85]	Recover original data	MLP, CNN, RNN	BER, SER, NMSE, Capacity, Spectral efficiency, Complexity	BER below 6.7% HD-FEC threshold beyond 40 km, Lower complexity than conventional methods
[86]	Learn mapping between Raman gain and pump parameters	DNN, SVM, DTs	MSE, Latency, Complexity	99% accuracy in Raman gain profile prediction, Less complex than traditional methods
[87]	Generate optimal constellation shapes for FMF channel	QAM, DNN-based GCS	BER	0.13 bit/4D nonlinearity gain, Reduced complexity compared to constellation shaping
[93]	Learn temporal correlations for equalization	LSTM, GRU, Bi-LSTM	BER, Q-factor	20% BER reduction, 50% less complexity than conventional equalizers
[96]	Learn channel-symbol relationship	FNN, RNN, CNN, CNN-RNN	MSE, MAE, BER	34% fewer FLOPs than RNN, Suitable for real-time applications
[97]	Compensate linear and nonlinear impairments	FFE, FNN/RNN, Cascade FNN/RNN	BER, NMSE, EVM	100x BER reduction, 60% complexity reduction vs non-cascade FNN/RNN
[98]	Equalize received symbols from optical link	FNN, RNN	MSE, BER, NMSE, Q-factor, EVM	90% fewer epochs, 62.5% fewer training symbols, Low complexity
[100]	Predict received signal from transmitted signal	FFNN, Bidirectional RNN	BER, Spectral efficiency	10 dB BER reduction, 50% less complexity in transceiver
[101]	Reconstruct modal coefficients of optical fiber output	Gerchberg-Saxton, CNN, Genetic	MSE, Correlation coefficient	97.0% intensity pattern similarity, 100x faster than IGP
[102]	Compensate for nonlinear impairments in OAM-MDM IM/DD	CNN, Volterra, Linear equalizer	BER, Q-factor, Spectral efficiency, Complexity	100x BER reduction for two-mode groups, $O(N^2)$ complexity, lower than Volterra's $O(N^3)$
[103]	Optimize Volterra equalizer structure for any given channel	Auto-Volterra, Greedy search, Volterra-pruning	BER, Equalizer complexity	50% BER reduction vs greedy search, 80% less complexity than greedy search

and polarization demultiplexing. Its ability to separate independent signal components without requiring training symbols makes it an attractive choice for adaptive and flexible

communication systems. However, its higher complexity and memory requirements may limit its practicality in some systems, especially when compared to simpler methods like

CMA. As research progresses, ICA is expected to play a significant role in future optical transmission systems, particularly in environments where high-order modulation schemes and polarization-related impairments are prevalent.

B. DEEP LEARNING

1) DEEP NEURAL NETWORK ALGORITHM

Figure 10 is the output of a trained deep neural network with two input features (Feature 0 and Feature 1) and three output classes (Class 0, Class 1, and Class 2). The colored regions represent the decision boundaries learned by the network, indicating the regions in the feature space where the network predicts a specific class with the highest probability. The non-linear and complex shapes of these boundaries suggest that the network likely has multiple hidden layers with non-linear activation functions, enabling it to learn intricate relationships between the input features and the output classes. The data points represent the training samples used to adjust the network's weights and biases through backpropagation, allowing it to effectively partition the feature space into distinct regions for classification. The ability to capture such complex decision boundaries highlights the power of deep neural networks in handling non-linearly separable data, a task that simpler models might struggle with.

DNNs are powerful ML models capable of learning complex input-output mappings. These networks are particularly useful for modeling intricate relationships in optical communication systems, where traditional methods may struggle. One of the primary challenges in using DNNs is selecting optimal hyperparameters, with grid search being the most commonly used method for this task. However, grid search can be computationally expensive, which has led to the development of more efficient manual search techniques, as discussed in [57].

DNNs have demonstrated exceptional performance in various optical communication tasks, particularly in equalization. They are adept at learning to compensate for both linear and nonlinear fiber impairments, which are often challenging for conventional methods. DNNs can address both deterministic and stochastic nonlinearities, as shown in [81] and [82]. For example, in nonlinear interference mitigation, DNN-based DBP provides significant complexity reductions compared to traditional DBP methods, as demonstrated in a 32×100 km fiber-optic link [83]. This reduction in complexity makes DNN-based approaches particularly appealing for long-distance transmission.

Moreover, in Mode Division Multiplexing (MDM) systems, DNNs have outperformed conventional Multiple-Input Multiple-Output (MIMO) detectors, such as Zero-Forcing and Semi-Definite Relaxation Row-by-Row methods [84]. DNNs implemented in these systems, where multiple modes of light are transmitted simultaneously, offer more robust detection and improved BER, showcasing their ability to handle complex, high-dimensional data more effectively than traditional techniques.

DNNs have also shown substantial promise in optical modulator modeling. For instance, at 42 Gbps over distances exceeding 40 km, DNN-based systems achieved a BER below the 6.7% hard-decision Forward Error Correction (FEC) threshold. This performance is significantly better than conventional Intensity Modulation/Direct Detection (IM/DD) schemes, including PAM-2 and PAM-4 [85]. Furthermore, DNNs have been successfully applied to predict Raman gain profiles, achieving a maximum error of just 0.6 dB. This level of accuracy is crucial for the accurate modeling of optical amplifiers, which are key components in long-distance optical communication systems [86].

A notable development in DNN applications is the end-to-end training of both transmitter and receiver components, which helps avoid suboptimal solutions that arise when DNNs are used only at the receiver side. By optimizing the entire transceiver chain, end-to-end DNNs improve performance in tasks such as autoencoding, constellation shaping, and power adjustment [87], [88], [89], [90], [91], [92]. These advancements are critical for improving the overall efficiency of optical systems. For example, end-to-end DNN-based constellation shaping has shown up to 0.13 bits/4D of nonlinearity gain, outperforming traditional methods and enabling more efficient optical signal transmission [90], [91].

The use of DNNs for autoencoding algorithms has also advanced probabilistic, geometric, and joint constellation shaping in MDM transmission, yielding superior results compared to conventional methods [87], [88], [89]. These innovations highlight the transformative potential of DNNs in improving the performance and efficiency of optical communication systems. In particular, DNNs enhance the system's ability to mitigate nonlinear distortions, a critical factor for maintaining signal integrity over long transmission distances.

However, despite their impressive performance, DNNs come with their own set of challenges. The main challenge lies in the computational cost of training deep networks, especially when large amounts of data are involved. The complexity of DNNs grows with the number of layers and parameters, which can be a limitation in real-time applications. Additionally, while DNNs can offer significant performance improvements, their opaque nature makes them less interpretable compared to traditional methods, which can be a disadvantage when understanding the decision-making process is crucial.

Comparing DNN-based approaches to conventional methods like DBP, Zero-Forcing, or traditional modulation schemes, DNNs generally offer better performance in terms of BER and system efficiency, especially when nonlinearities and impairments are present. However, the trade-off often lies in their increased computational complexity. While DNN-based methods can be several orders of magnitude faster than traditional methods, as demonstrated in [83], they may still require substantial computational resources, particularly for real-time processing in large-scale systems.

In conclusion, DNNs represent a highly promising solution for tackling various challenges in optical communication systems. They provide robust solutions for nonlinear interference mitigation, optical modulator modeling, and system optimization, with the potential to revolutionize optical communication technologies. However, their computational demands and complexity remain important considerations, particularly for systems that require low-latency processing or operate with limited computational resources. As research continues, DNNs are expected to play a pivotal role in advancing optical communication networks, especially with innovations in end-to-end training and constellation shaping.

2) RECURRENT NEURAL NETWORK ALGORITHM

Figure 11 illustrates a RNN designed for sequential data processing, specifically demonstrated in a sine wave prediction task. The left panel shows the unrolled structure of the RNN, where each input at a given time step is processed through a hidden state which is influenced by the previous hidden state enabling the network to maintain memory of past inputs. This recurrent connection allows the RNN to learn temporal dependencies within the data. The right panel demonstrates the network's performance in predicting a sine wave, where the "Predicted Data" closely aligns with the "Actual Data," showcasing the RNN's ability to model and forecast sequential patterns. The network's output at each time step is generated based on the current hidden state, effectively capturing the underlying dynamics of the sine wave. This example highlights the RNN's capability to process and predict time-series data by leveraging its inherent memory mechanism.

RNNs are a class of neural networks designed to handle sequential data by maintaining temporal dependencies in the network's hidden states. This capability makes RNNs particularly effective for nonlinear equalization in optical communication systems, where signal impairments such as chromatic dispersion and nonlinearities arise over time and distance. The ability to capture and exploit temporal correlations in signals is a key advantage of RNNs in optical systems. In [93], [94], and [95], RNNs were successfully applied as nonlinear equalizers in 56 Gbps 4-PAM MDM transmission systems, achieving significant performance gains, such as a 20% reduction in BER compared to conventional methods. Moreover, RNNs were able to reduce computational complexity by 50%, making them more efficient than traditional equalization techniques.

A hybrid approach that combines CNN with RNNs has also been explored to improve equalization performance. In [96], a CNN-RNN hybrid equalizer was developed for 100 Gbps 16-QAM transmission over 2000 km, where it demonstrated similar BER performance to traditional DBP equalization. The CNN-RNN model, however, achieved this with significantly fewer floating-point operations (FLOPs) compared to end-to-end MLP, CNN, RNN, and bidirectional RNN models, making it more suitable for real-time applications

where processing power is a limiting factor. This hybrid approach is particularly useful in long-distance systems where maintaining high throughput and low complexity is essential.

Further innovations in RNN-based equalization have included the use of cascade architectures. For instance, [97] proposed cascade DNN-RNN equalizers for 100 Gbps 4-PAM systems over a 15 km link, where the cascade model outperformed non-cascade configurations. This configuration achieved a remarkable BER reduction of up to 100x and reduced complexity by 60% compared to non-cascade models, highlighting the benefits of combining multiple neural network layers in the equalization process. The cascade structure helps improve the compensation of both linear and nonlinear impairments, which are common in optical fibers, thereby enhancing the system's overall performance.

RNN-based methods also demonstrate adaptability in dynamic network environments. One of the challenges faced by traditional equalizers is their dependence on fixed operating conditions, which can be problematic in real-world, dynamic optical networks. However, RNNs, especially when aided by transfer learning techniques, can overcome this limitation. As demonstrated in [98], RNN-based algorithms with transfer learning were able to reduce training epochs by 90% and training symbol requirements by 62.5%, facilitating the rapid adaptation of the model to new operating conditions without the need for extensive retraining. This is particularly advantageous in practical optical systems, where conditions such as channel noise or fiber nonlinearities can vary over time, and fast adaptation is necessary for maintaining optimal performance.

RNNs have also been employed in the context of autoencoding for optical communication systems. In [99], RNN-based autoencoders showed notable improvements in performance compared to DNNs in IM/DD systems. Furthermore, bidirectional RNNs (Bi-RNN) have been shown to provide even further enhancements in the modeling of dispersive nonlinear channels. In [100], Bi-RNN autoencoders demonstrated a 10 dB improvement in BER and 50% reduced complexity in transceiver design compared to traditional methods, making them an attractive option for high-speed optical communications. The Bi-RNN's bidirectional structure allows it to consider both past and future inputs, enhancing the model's ability to capture the full temporal dependencies in the communication channel.

Despite the promising results, RNNs do have limitations. Their application is often restricted by specific operating conditions, and the computational complexity of training deep RNN architectures can be a barrier, especially in real-time systems. However, as seen in [98], techniques such as transfer learning can mitigate these issues by reducing the amount of training data and time required, thus improving their scalability and adaptability. Additionally, although RNNs provide excellent performance in equalization tasks, they may still fall short in extremely high-dimensional or

noisy environments, where other DL architectures like CNNs or hybrid models could offer complementary advantages. For instance, CNNs excel in feature extraction from spatial data, which can be advantageous when handling high-dimensional optical signals, whereas RNNs shine in time-sequence-based tasks.

In conclusion, RNNs offer significant potential for improving nonlinear equalization, autoencoding, and overall signal processing in optical communication systems. They provide a unique ability to learn temporal dependencies, which are crucial for mitigating the effects of dispersion and nonlinearities in optical fibers. While they offer substantial performance improvements over conventional methods, challenges such as computational complexity and sensitivity to operating conditions must be addressed for practical deployment in large-scale systems. Nonetheless, RNN-based techniques, especially when combined with other DL architectures, represent a promising direction for the future of optical communications, particularly in real-time applications and dynamic environments.

3) CONVOLUTIONAL NEURAL NETWORK ALGORITHM

Figure 12 illustrates a simplified CNN architecture designed for image processing. The network begins with an “Input image” which serves as the raw data. This input then passes through a “Convolutional layer,” where feature extraction occurs through the application of filters or kernels. Following this, a “Pooling layer” reduces the spatial dimensions of the feature maps, contributing to computational efficiency and feature invariance. Another set of “Convolutional layer” and “Pooling layer” may follow to further refine feature extraction and downsampling. Finally, a “Fully connected layer” maps the learned features to specific output categories, and the “Output layer” produces the final classification or prediction. This architecture effectively captures spatial hierarchies in images, making CNNs well-suited for tasks like image recognition and computer vision.

CNNs have emerged as powerful tools in optical communication systems, particularly for applications in orbital angular momentum (OAM) communication. OAM utilizes the phase structure of light to encode information, and CNNs have demonstrated significant potential in improving the efficiency and accuracy of these systems. In [101], CNNs were employed to predict mode coefficients by analyzing interference patterns obtained from distribution histograms of near-field intensity and phase information. The CNN-based method showed impressive results, achieving 97.0% and 85.6% similarities between the reconstructed and given intensity and phase patterns, respectively. This indicates the high precision of CNNs in reconstructing the modal coefficients of optical fibers, a crucial task for enhancing signal quality and minimizing transmission errors. The approach was found to be 100 times faster than traditional iterative methods such as the Iterative Gerchberg-Saxton algorithm, which is commonly used for optical beam

reconstruction, demonstrating the computational efficiency of CNNs in OAM systems.

In addition to their application in mode reconstruction, CNNs have also been successfully utilized for nonlinearity equalization in OAM communication systems using IM/DD. In [102], CNNs were employed to compensate for nonlinear impairments that occur due to signal distortion in long-distance transmission. Experimental results showed that CNN-based equalization outperformed Volterra-based NLE by 3 dB and 1.5 dB for two-mode groups, respectively. This performance improvement highlights the CNN's ability to model and correct nonlinearities more effectively than traditional methods, significantly enhancing the overall signal quality. Additionally, the CNN-based method demonstrated a computational advantage, with a complexity of $O(N^2)$ compared to Volterra's $O(N^3)$, making CNNs more suitable for real-time applications where computational efficiency is critical.

CNNs are poised to play a crucial role in OAM communication systems, especially for tasks such as mode classification, nonlinearity compensation, and error correction. Their ability to learn spatial features from input data enables them to effectively handle complex tasks in optical communication, where traditional methods may struggle. The results from [101] and [102] demonstrate the significant improvements in both accuracy and efficiency offered by CNNs, making them a powerful tool for next-generation OAM-MDM IM/DD systems. As research in CNNs and OAM communication continues to progress, CNNs are expected to further enhance the performance and robustness of OAM-based systems, enabling more reliable and high-capacity optical communication networks.

4) DEEP REINFORCEMENT LEARNING ALGORITHMS

Figure 13(c) illustrates a simplified RL framework, particularly focused on DRL where the agent interacts with an environment to learn optimal policies. The process begins with “Initialize Environment,” where the agent is placed in a specific state. Next, the agent chooses an “Action” based on its current state and policy. The environment then provides a “Reward” and transitions the agent to a “New State.” The agent uses this feedback to “Update its knowledge,” typically by adjusting its policy to maximize future rewards. The “Check Termination Condition” step determines if the task is complete, and the process repeats until the termination condition is met. This iterative learning process allows the agent to discover the best sequence of actions to achieve its goal within the given environment, highlighting the power of DRL in solving complex decision-making problems.

DRL combines the power of RL with DL techniques, enabling agents to make decisions based on unstructured input data without the need for manual state-space engineering. This ability to learn directly from raw data makes DRL highly suitable for optimizing complex systems, such as those used in optical communications. Traditional Volterra-based NLEs are known for their effectiveness in

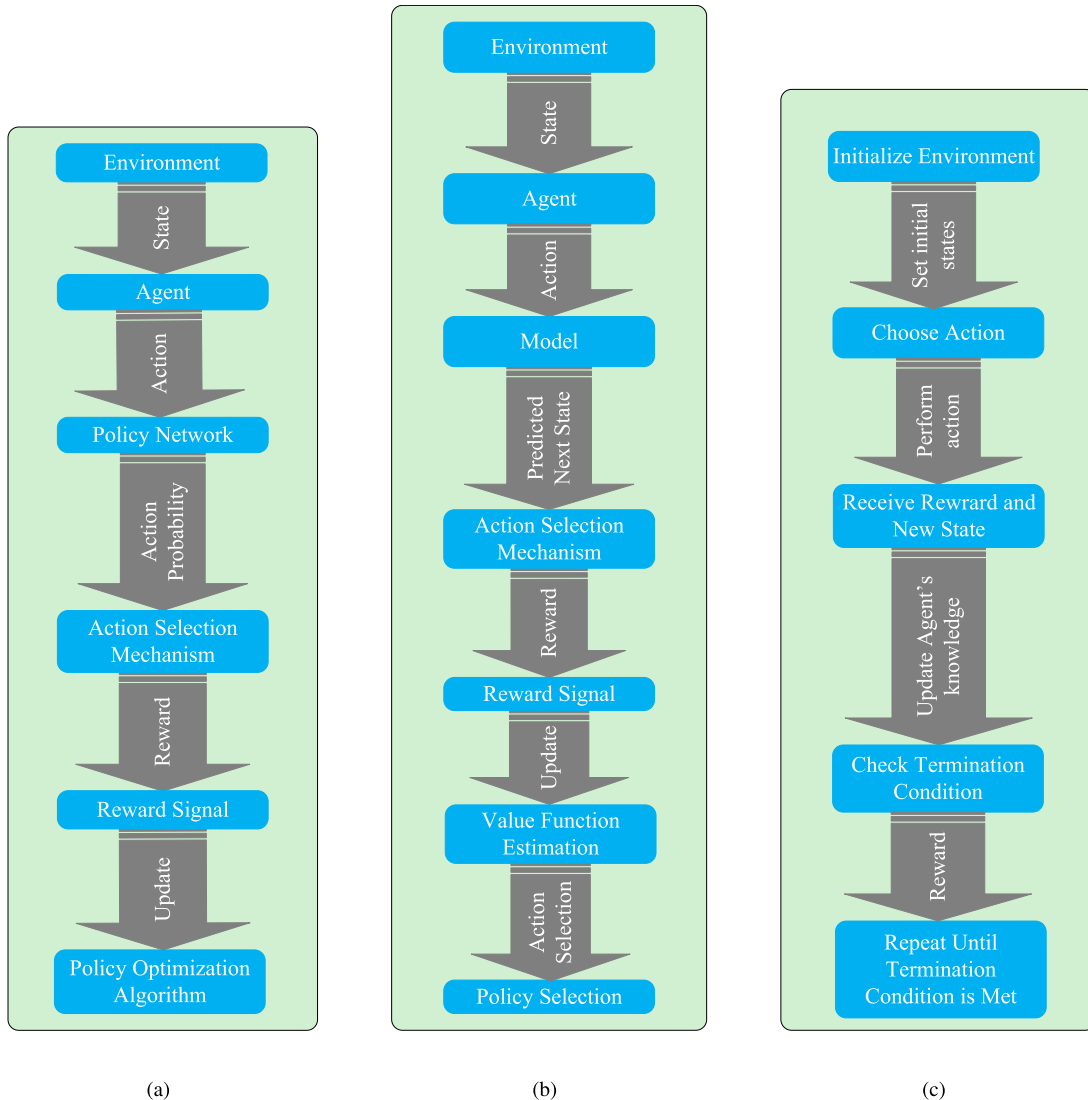


FIGURE 13. Concept of a) policy-based RL, b) value-based RL, and c) DRL algorithms.

compensating for nonlinear impairments, but they suffer from high computational complexity and inefficiencies when used in conventional uninformed optimization methods. In [103], DRL was introduced as a technique to optimize Volterra-based NLE structures by learning the functional relationship between optimization objectives and structural parameters. This approach eliminates the need for exhaustive search methods or heuristic algorithms, directly identifying the optimal structure for a given system. By using DRL, the Volterra equalizer structure can be optimized efficiently for any specific channel and signal conditions.

In an experimental setup for a 50 Gbps PAM-4 IM/DD system, DRL was applied alongside a greedy search method to optimize Volterra-based Feedforward Equalizer (Volterra-FFE), Decision Feedback Equalizer (Volterra-DFE), and Volterra-Pruning. The results showed that DRL outperformed

the greedy search by reducing computational complexity while maintaining or even improving performance with minimal or no BER loss. This experiment highlighted DRL's ability to significantly enhance the efficiency of Volterra equalization, making it a powerful tool for high-speed optical communication systems.

Additionally, DRL-based pruning techniques were shown to further enhance the Volterra equalization process by removing redundant taps, thereby significantly reducing complexity while maintaining the equalizer's performance. These advances underscore the potential of DRL in optimizing communication algorithms, making it a promising approach for improving the performance and efficiency of optical communication systems. The application of DRL to optimize Volterra-based equalizers represents a step toward more intelligent and adaptive systems capable of efficiently handling complex communication environments.

III. OPTICAL WIRELESS COMMUNICATION

A. MACHINE LEARNING

1) SUPPORT VECTOR MACHINE

SVM is a widely used supervised learning algorithm that constructs hyperplanes in a high-dimensional space to classify data points effectively. In optical communication systems, SVM has been employed for signal detection, phase offset mitigation, and equalization, significantly improving performance compared to conventional methods.

In [104], SVM was utilized for signal detection in visible light communication (VLC) systems, achieving a 35% higher data rate than traditional DD methods. The algorithm leveraged the received signal characteristics to improve classification accuracy, outperforming maximum likelihood estimation in challenging signal environments. Its computational complexity remains relatively low, making it a practical alternative to conventional detection schemes.

In the domain of free-space optical (FSO) communication, SVM was applied to enhance signal detection, demonstrating robust performance under high noise conditions [105]. The approach allowed reliable transmission under a 7% FEC threshold, ensuring improved system reliability in turbulent FSO links.

SVM has also been leveraged for phase offset mitigation in VLC and FSO systems. In [106], an SVM-based equalization technique was introduced for a 4-band Gaussian-shaped 8-QAM system integrated with fiber and VLC links. This method, applied after CMA equalization, improved the Q-factor by 11.5 dB, enabling stable transmission at 320 MHz and 960 Mbps with an optical power level of -2.5 dBm while maintaining BER performance under the 7% FEC threshold. Despite its cubic time complexity $O(n^3)$, the computational overhead was manageable compared to traditional maximum likelihood approaches.

Another notable application of SVM is in phase estimation for multiband carrierless amplitude and phase modulation VLC systems [107]. The proposed technique achieved a 60% BER reduction at an SNR of 10 dB, effectively compensating for phase distortions. The approach demonstrated lower computational complexity than the Viterbi algorithm while maintaining accuracy in phase estimation tasks, making it a viable alternative for real-time optical communication systems.

Compared to conventional methods such as direct decision and maximum likelihood estimation, SVM provides enhanced classification accuracy and robustness to noise. However, its primary limitation lies in its computational complexity, particularly when applied to large-scale systems. In contrast, heuristic methods like kNN and decision trees offer lower complexity but at the cost of reduced accuracy. Additionally, while least squares and maximum likelihood techniques remain popular for phase estimation, SVM presents a more efficient alternative with better BER performance.

Overall, SVM proves to be a powerful tool in optical communication, offering significant improvements in signal detection, phase estimation, and equalization. Its adaptability and classification accuracy make it a promising candidate for next-generation VLC and FSO systems, where reliability and data rate enhancements are critical.

2) ARTIFICIAL NEURAL NETWORK ALGORITHM

ANNs are effective ML models that excel in nonlinear mapping and pattern recognition, making them highly suitable for addressing impairments in FSO and VLC systems. These impairments include atmospheric turbulence in FSO and interference from artificial light sources in VLC.

In FSO systems, ANNs have been employed for adaptive wavefront correction, significantly improving system performance. In [108], an ANN-based wavefront correction approach for sensor-less adaptive optics enhanced the Strehl Ratio to over 0.9, surpassing conventional Zernike polynomial fitting and modal decomposition methods. This improvement leads to better energy coupling and reduced computational complexity, making ANN-based approaches 50% less complex than traditional model-based methods. Similarly, ANN-based channel state information (CSI) estimation in [109] demonstrated a high reliability rate, achieving a BER of 1.2×10^{-12} , outperforming conventional Kalman filters. These results highlight the ANN's capability to learn complex distortions in FSO links, providing superior robustness and adaptability.

For VLC systems, ANNs have proven instrumental in mitigating interference and enhancing signal classification. In [110], an ANN-based receiver, combined with discrete wavelet transform, achieved error-free data transmission at 150 Mbps with On-Off Keying Non-Return-to-Zero (OOK-NRZ) modulation. Compared to traditional matched filters and linear equalizers, the ANN-based approach demonstrated 50% lower complexity while maintaining superior performance in suppressing artificial light interference. Further advancements in ANN-based equalization techniques, such as the model in [111], enabled data rates up to 170 Mbps using a low-bandwidth LED transmitter and a PIN photodetector receiver. However, these methods often introduce higher computational complexity, which remains a practical challenge for real-time implementations.

ANNs have also been applied to MIMO VLC systems, improving data transmission rates and system resilience against LED nonlinearity and cross-LED interference. In [112], an ANN-based equalization approach enhanced an OOK-MIMO VLC system, increasing the achievable data rate from 200 kbps (without equalization) to 1.8 Mbps while maintaining a BER of 1×10^{-4} . Similarly, Extreme Learning Machine, a specialized ANN model, was introduced in [113] for MIMO VLC detection, achieving comparable BER performance (1×10^{-4}) with reduced computational complexity, requiring only $N \log N$ floating-point operations. Compared to LMS and Linear Minimum Mean Square Error estimators,

TABLE 5. Summary of ML applications in OWC.

Reference	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[104]	Detection	SVM, Direct decision, Maximum likelihood	Accuracy, Precision, Recall, F1-score	35% higher data rate than DD, low complexity increase over DD
[106]	4-band Gaussian-shaped 8-QAM equalization	kNN, DT	Accuracy, Precision, Recall, F1-score	Q-factor improved by 11.5 dB, Complexity $O(n^3)$
[107]	Phase estimation	Least squares, Maximum likelihood, SVM	BER, Computational complexity	60% BER reduction at 10 dB SNR, lower complexity than Viterbi
[108]	Learn wavefront distortion relationship	BP ANN, Zernike polynomial fitting, Modal decomposition	BER, Q-factor, Eye Diagram	Strehl ratio > 0.9, 50% less complexity than model-based method
[109]	CSI estimation	ANN, EKF, LMS	BER	BER of 1.2e-12, less complex than Kalman filter
[110]	Signal classification	Matched filter, Linear equalizer, Wavelet-neural network	BER, Q-factor, MSE	Error-free data at 150 Mb/s, 50% less complexity than conventional VLC receivers
[111]	Signal compensation	Wavelet transform, ANN	BER	Data rate of 150 Mb/s, high complexity
[112]	Estimate transmitted signal	Equalizer-based MIMO, ANN-based MIMO	BER, Complexity	BER of 1e-4, 50% complexity increase
[113]	Estimate transmitted signal	Linear Minimum Mean Square Error, Volterra, Extreme learning machine	BER, Spectral efficiency, Complexity	BER of 1e-4, requires $N \log N$ flops
[114]	Estimate distance between receiver and transmitter	kNN, Trilateration	Accuracy, Precision, Recall, F1-score	Positioning accuracy of 2.7 cm, Complexity $O(N)$
[116]	Predict RSSI from weather parameters	kNN, SVM, RF	Accuracy, Precision, Recall, F1-score, AUC	95% accuracy, lower complexity than traditional methods
[117]	Classify channel estimates	kNN, SVM, RF	Accuracy, Precision, Recall, F1-score	90% accuracy, better than k-NN and SVM, real-time object classification
[118]	Predict mobile user location	kNN, SVM, RF	Accuracy, Precision, Recall, F1-score	52.55% better positioning accuracy, 78.26% faster than conventional methods
[119]	Predict receiver position	Regression model, Trilateration	MAE, RMSE, R-squared	3 cm accuracy, 90% less training burden
[120]	2 cm positioning error	Ridge regression, kNN, SVM, RF	MAE, RMSE, R-squared	More efficient than RSS-based systems (baseline: RSS conventional methods)
[122]	99.9% accuracy in modulation format classification	K-means, Spectral clustering	Accuracy, Precision, Recall, F1-score	Lower complexity than traditional clustering methods (baseline: fuzzy c-means, hierarchical clustering)
[126]	BER of 0.28 at 5 dB SNR	K-means, Improved K-means	BER	50% less complex than standard K-means (baseline: standard K-means)
[127]	20% BER reduction	Mean phase estimation, K-means	BER	Linear complexity; improves over conventional phase estimation (baseline: conventional phase recovery)
[128]	BER reduced by two orders of magnitude	MMSE, EM-based ML	MSE, BER	10x faster channel estimation than conventional MMSE (baseline: MMSE)
[129]	<5% estimation error	EM-based ML, Newton-Raphson, GMM	MSE, RMSE, Complexity	Faster convergence than Newton-Raphson and GMM methods (baseline: Newton-Raphson, GMM)
[130]	BER of 10^{-3} at 10 dB	EM	BER	Complexity $O(N)$, suitable for real-time applications (baseline: traditional detection algorithms)
[131]	BER of 10^{-5} , better than LS	EM-VLC, LS, MMSE	BER, MSE, Complexity, Convergence rate	Higher complexity ($O(N^2)$) than LS but better BER (baseline: LS, MMSE)
[135]	75% reduction in mean positioning error	RL, kNN	MAE, RMSE, 2D Localization Error	Real-time implementation with lower error than traditional methods (baseline: RSS-based positioning)
[137]	26.8% improvement over trust region policy	Policy gradient, RL, Meta-learning	Served users, Convergence, Energy, Delay, Throughput	Reduced complexity vs. traditional RL (baseline: trust region RL)
[139]	90% acceptance ratio	VNE-EACH, vNE-USE	Throughput, Delay, Loss, Utilization, Energy	26% lower delay than traditional resource allocation (baseline: conventional VNE methods)
[222]	30% throughput improvement	RL with Knowledge Transfer	Throughput, Latency, Loss, Energy	Complexity $O(N)$, faster than traditional methods (baseline: conventional AP selection)
[223]	10% sum rate increase	QL, MILP	Throughput, Fairness, Complexity	Lower complexity than MILP (baseline: MILP)
[225]	25% better than fixed time-to-trigger	Deep QL, Double QL, Dueling QL	Throughput, Delay, Loss, Efficiency	More efficient than fixed TTT methods (baseline: fixed TTT strategies)

ANN-based methods offer enhanced nonlinear compensation and robustness to LED distortions at the cost of increased processing requirements.

Despite their advantages, ANN-based approaches exhibit certain limitations. The high computational complexity of ANN models can hinder their real-time deployment, particularly in resource-constrained embedded VLC/FSO systems. Moreover, while ANNs excel at learning complex channel impairments, they require extensive training datasets and fine-tuned architectures to generalize well across

diverse transmission conditions. Future research should focus on developing lightweight ANN architectures, such as pruning-based models or hybrid techniques combining ANN with conventional equalization, to optimize performance and complexity trade-offs in practical FSO and VLC applications.

3) K-NEAREST NEIGHBORS ALGORITHM

The kNN algorithm is a non-parametric ML technique widely used in classification and regression tasks. It operates by identifying the k closest training samples in a feature

space and making predictions based on a majority vote (for classification) or an average (for regression). kNN is particularly effective in indoor positioning applications using VLC due to its simplicity, adaptability, and ability to handle nonlinear relationships in signal variations.

Traditional VLC-based positioning methods, such as trilateration and fingerprinting with deterministic algorithms, often suffer from high sensitivity to ambient light interference and fluctuations in received signal strength indicator (RSSI) values due to environmental changes. In contrast, kNN-based positioning leverages a pre-collected dataset of known RSSI values and their corresponding locations. When a new signal measurement is obtained, kNN computes the distances to stored reference points and estimates the receiver's position based on the closest matches.

In [114], kNN was applied to a 3D VLC-based indoor positioning system and demonstrated a high positioning accuracy of 2.7 cm, significantly outperforming conventional methods. Unlike model-based approaches, which require complex mathematical modeling of the indoor environment, kNN provides a data-driven alternative that is computationally efficient, with a complexity of $O(N)$. The method's robustness against varying indoor conditions makes it suitable for real-time applications, including indoor navigation, asset tracking, and gesture recognition.

However, despite its advantages, kNN has some limitations. The algorithm's performance depends heavily on the quality and density of the training dataset. In large-scale environments, maintaining an extensive and up-to-date fingerprint database can be challenging. Additionally, the computational cost increases with the number of stored reference points, making real-time processing more demanding in large deployments. Compared to DL-based localization methods, kNN lacks the ability to generalize beyond the observed training data, limiting its effectiveness in highly dynamic environments.

Nevertheless, due to its simplicity, interpretability, and high accuracy in controlled indoor settings, kNN remains a promising candidate for VLC-based positioning. Future research may focus on hybrid approaches that integrate kNN with DL techniques or adaptive filtering methods to further enhance accuracy and reduce computational complexity.

4) ENSEMBLE LEARNING ALGORITHMS

Ensemble learning, a ML approach that combines multiple weak learners to achieve superior predictive performance, has shown significant promise in improving signal strength prediction and positioning accuracy in FSO and VLC systems. By aggregating the predictions of multiple models, ensemble methods enhance robustness, reduce overfitting, and improve generalization, making them particularly effective for complex and dynamic communication environments.

In [115], XGB was utilized to predict the RSSI in a hybrid FSO/RF system, enabling seamless switching between links during FSO blockages. The ability to accurately predict signal strength in such conditions ensures uninterrupted

communication, addressing one of the key challenges in FSO systems affected by atmospheric turbulence. In [116], the performance of Random Forest, AdaBoost, and Gradient Boosting classifiers was evaluated for RSSI prediction in FSO links. RF demonstrated the highest accuracy (100%), followed by AdaBoost (99.7%), while Gradient Boosting exhibited performance degradation over time, highlighting the importance of selecting appropriate ensemble methods based on application requirements.

Beyond signal prediction, ensemble learning has also been employed in VLC-based object detection and positioning. In [117], Random Forest was applied to classify objects based on CSI from VLC links, achieving a 90% object detection accuracy. This method surpassed traditional kNN and SVM approaches, demonstrating the superiority of ensemble learning in real-time classification tasks. Similarly, [118] explored a dual-function ML approach for VLC-based indoor positioning, where SVM and Random Forest achieved positioning accuracies of 8.6 cm and 10.2 cm, respectively. SVM also exhibited a 78% improvement in execution time, making it well-suited for applications requiring low latency. Comparatively, kNN achieved 13 cm accuracy with a rapid execution time of 5.6 ms, making it an optimal choice for speed-sensitive applications.

Despite their advantages, ensemble learning techniques present several challenges. The increased computational complexity associated with training and inference can be a limiting factor for resource-constrained systems. Additionally, while methods like Random Forest and AdaBoost are resilient to overfitting, their performance heavily depends on parameter tuning and dataset quality. Compared to DL-based approaches, ensemble methods may struggle with feature extraction in highly nonlinear and unstructured data scenarios. Nevertheless, their interpretability, high accuracy, and adaptability make them valuable for FSO and VLC systems.

As ML techniques continue to evolve, ensemble learning is expected to play an increasingly crucial role in enhancing communication reliability and positioning accuracy in challenging environments. Future research may focus on hybrid models that integrate DL with ensemble techniques to further improve performance while maintaining computational efficiency.

5) REGRESSION ALGORITHMS

Regression algorithms play a crucial role in VLC-based positioning systems by establishing mathematical relationships between received signal strength (RSS) or light intensity and the receiver's position. Traditional VLC positioning approaches, which segment an area into unit cells to reduce computational complexity, often suffer from high positioning errors due to limited granularity and sensitivity to environmental variations. To address these issues, ML-based regression techniques such as linear regression and kernel ridge regression (KRR) have been explored for improving positioning accuracy.

Linear regression, a fundamental statistical technique, models the relationship between dependent and independent variables using a linear equation. In [119], linear regression, combined with a light spatial sequence arrangement, significantly reduced positioning errors by 90.71%, achieving a mean positioning error of 9.85 cm in an experimental testbed. The method effectively minimized the training burden by 90% while maintaining a positioning accuracy of approximately 3 cm, demonstrating its efficiency in VLC-based localization. Despite its advantages, linear regression is limited in handling nonlinear signal variations caused by multipath effects and environmental interference, restricting its applicability in complex VLC environments.

To overcome these limitations, KRR, a nonlinear regression technique that extends ridge regression by mapping input data to a higher-dimensional space using kernel functions, has been investigated for VLC positioning. In [120] and [121], KRR, when combined with data preprocessing techniques such as the sigmoid function, outperformed linear regression by achieving an average horizontal positioning error of 1.96 cm and a vertical error of 2.16 cm. This approach improved positioning accuracy by 27.8% and 22.0%, respectively, compared to traditional regression-based methods. Furthermore, when compared to linear regression without preprocessing, KRR demonstrated a 48.1% reduction in horizontal error and a 43.8% reduction in vertical error. These improvements highlight the significance of incorporating advanced regression techniques and preprocessing methods to enhance positioning performance in VLC systems.

Compared to RSS-based positioning systems, which rely on direct signal strength measurements and are often affected by noise and multipath interference, regression-based approaches offer more efficient and accurate location estimation by leveraging ML models to learn the complex RSS-position relationship. Moreover, integrating KRR with other ML techniques such as kNN, SVM, and Random Forest can further refine positioning accuracy, providing a robust framework for VLC localization. However, the computational complexity of KRR remains a challenge, particularly in real-time applications, as it requires extensive matrix operations and kernel evaluations.

As VLC-based positioning continues to evolve, further advancements in data preprocessing, feature selection, and hybrid ML models are expected to enhance system performance, making regression-based approaches a viable solution for high-accuracy indoor localization. Future research directions may include DL-based regression models that can autonomously extract spatial features and adapt to dynamic environments, further improving positioning accuracy and robustness.

6) HIERARCHICAL CLUSTERING ALGORITHM

Hierarchical clustering is a widely used unsupervised learning technique that organizes data points into nested clusters based on similarity measures. In FSO mobile ad-hoc

networks (MANETs), where maintaining a line-of-sight (LoS) connection is a fundamental challenge, hierarchical and fuzzy-logic clustering techniques have demonstrated their effectiveness in improving network stability and efficiency [122], [123]. These methods address the limitations of conventional routing strategies by dynamically grouping nodes into clusters, optimizing connectivity, and reducing overhead.

In [124], hierarchical clustering was employed to develop an enhanced routing protocol tailored for FSO MANETs. The proposed method utilizes a neighbor discovery algorithm to form clusters, ensuring robust and adaptive network organization. A key aspect of this approach is the selection of a reliable cluster head, which acts as a central coordinator within the cluster, optimizing data transmission and minimizing disruptions due to node mobility. By structuring the network into clusters, the need for maintaining direct LoS links between all nodes is significantly reduced, leading to improved path selection and enhanced communication reliability.

The integration of fuzzy-logic mechanisms further refines the clustering process by adapting the network topology to real-time environmental changes. As FSO nodes move and connectivity conditions fluctuate, the fuzzy-logic component dynamically adjusts cluster memberships, ensuring that the routing protocol remains efficient even in highly dynamic scenarios. This adaptability is particularly beneficial in mitigating the effects of atmospheric turbulence and mobility-induced link failures, which are common in FSO MANETs.

From a computational standpoint, hierarchical clustering exhibits lower complexity compared to iterative clustering techniques, making it suitable for real-time network optimization. In [122], a non-iterative clustering algorithm was applied to modulation format classification, achieving a remarkable 99.9% accuracy with minimal computational overhead. This demonstrates the feasibility of hierarchical clustering for resource-constrained environments where rapid decision-making is crucial. Similarly, [123] investigated an improved modulation format identification method based on Stokes parameters, leveraging a combination of fuzzy c-means and hierarchical clustering to enhance classification performance in coherent optical communication systems.

Compared to other clustering approaches such as k-means and spectral clustering, hierarchical clustering offers distinct advantages in handling non-convex data distributions and producing interpretable cluster hierarchies. However, its reliance on predefined similarity metrics can lead to sensitivity in clustering performance, necessitating careful parameter tuning for optimal results. Moreover, while fuzzy-logic-based adaptations improve flexibility, they introduce additional computational overhead, which must be balanced against real-time processing constraints.

Overall, hierarchical clustering, coupled with fuzzy-logic adaptations, presents a powerful framework for optimizing routing in FSO MANETs. Future research directions could

explore DL-based clustering techniques that leverage neural network architectures to further enhance network adaptability and resilience in dynamic environments.

7) K-MEANS CLUSTERING ALGORITHM

K-means clustering is a widely used USL algorithm that partitions data into a predefined number of clusters by iteratively refining cluster centroids based on similarity metrics. In optical communication systems, K-means plays a crucial role in tasks such as user identification, phase retrieval, and constellation shaping, where efficient clustering of received signals enhances system performance and data integrity.

In [125], K-means was utilized for real-time user identification in a multi-user FSO communication system. By analyzing received signal amplitude variations, the algorithm effectively categorized users in a dynamic environment, demonstrating its ability to manage complex data streams with minimal computational overhead. This real-time identification capability is essential for multi-user FSO networks, where accurate classification improves resource allocation and system efficiency.

The adaptability of K-means extends to phase retrieval applications in FSO systems. In [126], an improved K-means algorithm was employed to correct low-order wavefront aberrations by dynamically updating cluster centroids based on phase distortions in the received signal. This approach enhanced signal quality and reduced BER, achieving a BER of 0.28 at an SNR of 5 dB. Compared to standard K-means, the improved algorithm demonstrated 50% lower computational complexity, making it suitable for real-time phase compensation in optical links.

Beyond terrestrial FSO applications, K-means has also been applied in underwater VLC systems to address phase deviations in high-order modulation formats. In [127], the algorithm was integrated with mean phase estimation to correct phase distortions in an 8-QAM modulation scheme, leading to a 20% reduction in BER. The correction mechanism enabled an increase in data rates from 1.3875 Gbps to 1.4625 Gbps, highlighting the importance of clustering-based signal processing in bandwidth-efficient optical systems.

While K-means clustering is computationally efficient and exhibits linear complexity, its performance is highly dependent on the initial selection of cluster centroids. Poor initialization can lead to suboptimal convergence, necessitating improvements such as the K-means++ initialization method or hybrid clustering approaches. Moreover, conventional K-means assumes spherical clusters with equal variance, which may not always align with real-world optical signal distributions. Advanced modifications, such as spectral clustering or density-based clustering, could further improve clustering accuracy in non-linear optical communication environments.

Overall, K-means clustering has proven to be a versatile and effective tool in optical communication, contributing to

user identification, phase retrieval, and constellation shaping. As research progresses, integrating DL-based clustering techniques with K-means could further enhance its adaptability, enabling more efficient optical network management and error correction strategies.

8) EXPECTATION MAXIMIZATION CLUSTERING ALGORITHM

Expectation-Maximization is an iterative statistical technique used for maximum likelihood estimation (MLE) in the presence of latent variables. EM-based MLE has demonstrated significant effectiveness in channel estimation for FSO and VLC systems, particularly in scenarios where traditional estimators such as minimum mean square error (MMSE) and maximum a posteriori probability methods struggle with computational complexity or convergence speed.

In [128], an EM-based approach was applied for channel estimation in an FSO system employing OOK modulation with an avalanche photodiode receiver. This method significantly reduced the BER by two orders of magnitude while accelerating channel estimation by a factor of ten compared to conventional MMSE techniques. The ability of EM to iteratively refine parameter estimates from received signal data makes it particularly well-suited for real-time FSO applications, where fast and accurate channel estimation is critical.

The superiority of EM over other estimation methods is further validated in [129], where it was used to estimate the parameters of a Gamma-Gamma fading distribution, which characterizes atmospheric turbulence in FSO channels. Compared to Newton-Raphson and generalized moment estimation (GMM) methods, EM achieved an estimation error below 5% while converging significantly faster. The faster convergence of EM is advantageous for adaptive optical systems that require real-time parameter updates to mitigate turbulence effects. ... EM-based MLE has also been investigated as a uninformed detection technique in FSO systems. In [130], an EM-based uninformed detector was developed, achieving a BER of 1×10^{-3} at an SNR of 10 dB. Remarkably, this approach matched the performance of perfect channel estimation without requiring pilot symbols, significantly reducing overhead in practical implementations. The complexity of this uninformed detection approach, proportional to $O(N)$, ensures its suitability for real-time optical receivers.

In VLC systems, EM-based channel estimation has outperformed conventional methods such as least squares (LS) and MMSE estimators. As demonstrated in [131], an EM-based estimation algorithm for OFDM-based VLC systems achieved a BER of 1×10^{-5} , outperforming LS while maintaining a higher complexity of $O(N^2)$. This tradeoff between accuracy and complexity underscores the importance of selecting appropriate estimation techniques based on system requirements, where EM is particularly useful in applications demanding high accuracy with iterative refinement.

Despite its advantages, EM has certain limitations. Its performance is highly dependent on proper initialization,

and poor initial parameter estimates can lead to convergence to local optima rather than the global maximum likelihood solution. Additionally, while EM provides a structured iterative approach, its computational complexity can be higher than non-iterative methods, particularly in large-scale systems. Hybrid techniques, such as combining EM with DL-based predictive models, could further enhance its performance by leveraging prior knowledge to improve initialization and convergence speed.

Overall, EM-based MLE is a powerful tool for channel estimation and signal detection in FSO and VLC systems, offering a compelling balance between accuracy and complexity. Its ability to operate in uninformed detection scenarios without pilot symbols makes it particularly valuable for bandwidth-limited optical systems, where minimizing training overhead is crucial.

9) INDEPENDENT COMPONENT ANALYSIS ALGORITHM

ICA is a statistical technique designed for uninformed source separation, particularly useful in multi-user FSO communication systems where signals transmitted at different wavelengths must be separated at the receiver. ICA operates by assuming that observed mixed signals are linear combinations of independent source signals and attempts to recover these independent components without prior knowledge of the mixing process. This capability is particularly valuable in multi-user detection (MUD) scenarios, where optical signals transmitted by different users must be distinguished to ensure reliable data recovery [132], [133].

Despite its effectiveness, a fundamental limitation of ICA is that it traditionally requires an equal number of transmitters and receivers to perform accurate signal separation. This constraint poses a significant challenge in practical FSO systems where the number of active transmitters may exceed the available receivers, leading to underdetermined systems where classical ICA fails. Addressing this limitation, [134] proposes two enhanced approaches for ICA-based signal separation. The first method leverages the sparsity of transmitted signals, where only a few dominant components contribute significantly at any given time. By exploiting sparsity, ICA can isolate these components even when the number of transmitters exceeds the number of receivers. The second method integrates ICA with non-orthogonal multiple access and successive interference cancellation, allowing the system to separate overlapping signals by iteratively removing stronger components before extracting weaker ones. This hybrid approach significantly improves ICA's applicability in practical multi-user FSO links.

Compared to conventional signal separation techniques such as matched filtering and MMSE detection, ICA offers the advantage of being uninformed, meaning it does not require CSI or training sequences. This characteristic makes it particularly useful in dynamic FSO environments where CSI acquisition is challenging due to atmospheric turbulence and mobility. However, ICA's performance is sensitive to the

statistical independence of source signals—highly correlated optical signals can degrade separation accuracy. Additionally, iterative ICA algorithms may suffer from convergence issues and computational complexity, particularly in real-time applications with high-dimensional data streams.

Future improvements could involve DL-assisted ICA models that leverage prior knowledge from historical signal patterns to enhance convergence and robustness. Furthermore, integrating ICA with adaptive filtering techniques could improve its ability to handle correlated signal sources in dense multi-user FSO networks. Overall, ICA remains a promising solution for enhancing spectral efficiency and mitigating interference in next-generation optical wireless systems, particularly when combined with sparsity-based and non-orthogonal multiple access-assisted approaches.

10) POLICY-BASED REINFORCEMENT LEARNING ALGORITHM

Figure 13(a) illustrates the core components of policy-based RL, where an “Agent” directly learns a “Policy Network” that maps states to actions. The process begins with the “Environment” providing the agent with a “State.” The agent uses its policy network to generate “Action Probabilities” and selects an “Action” based on these probabilities through an “Action Selection Mechanism.” The environment responds with a “Reward Signal,” which is used by the “Policy Optimization Algorithm” to update the policy network, aiming to maximize future rewards. This direct learning of the policy, rather than learning a value function, allows the agent to handle continuous action spaces and stochastic policies, making it suitable for complex control tasks where precise action selection is crucial.

RL has emerged as a powerful tool for optimizing indoor visible light positioning (VLP) and multi-user VLC systems. Unlike conventional techniques that rely on static models or pre-defined heuristics, RL-based approaches learn optimal policies dynamically by interacting with the environment, making them particularly effective for handling uncertainties such as noise, mobility, and dynamic resource allocation. In [135], an iterative point-wise RL algorithm is proposed for indoor VLP, effectively mitigating nondeterministic noise and addressing height mismatches. The algorithm leverages RSSI and height information to iteratively refine position estimates, achieving a mean absolute positioning error of 5 cm. This represents a 75% reduction in error compared to conventional methods such as RSS and kNN, making RL highly suitable for real-time implementation in indoor positioning applications.

In addition to localization, RL has shown significant promise in optimizing resource allocation in multi-user VLC networks. The work in [136] applies RL to dynamically adjust transmission parameters, improving both communication reliability and positioning accuracy in an integrated VLC system. This adaptive strategy allows VLC networks to efficiently allocate bandwidth and power in response to changing

environmental conditions, outperforming static resource allocation approaches. Furthermore, [137] introduces a meta-learning-aided RL framework for optimizing user mobility, handling blockages, and selecting the best access points in a THz/VLC hybrid wireless VR network. Meta-learning enables rapid adaptation to new user mobility patterns, significantly enhancing network performance. The proposed approach achieves a 26.8% improvement in successfully served users compared to trust region policy optimization and exhibits an 87.5% faster convergence rate than conventional RL techniques, demonstrating its effectiveness in complex, high-dimensional scenarios.

Compared to conventional methods such as kNN and heuristic-based optimization, RL-based approaches offer several advantages, including real-time adaptability, improved accuracy, and robustness against environmental uncertainties. However, policy-based RL methods often require extensive training data and computational resources, making real-time implementation challenging in large-scale networks. Additionally, the sample efficiency of RL algorithms can be low, necessitating the use of meta-learning and transfer learning techniques to improve generalization across different scenarios.

Future research could focus on integrating DRL techniques with hybrid THz/VLC networks to further enhance throughput and reliability. Moreover, leveraging federated learning could allow distributed VLC systems to train RL models collaboratively while preserving user privacy. Overall, policy-based RL presents a promising avenue for advancing indoor localization and multi-user VLC systems by providing intelligent, adaptive decision-making capabilities.

11) VALUE-BASED REINFORCEMENT LEARNING ALGORITHM

Figure 13(b) illustrates the core components of value-based RL, where an “Agent” learns a “Value Function” that estimates the long-term reward potential of different states and actions. The process begins with the “Environment” providing the “State,” and the agent uses its value function to select an “Action” based on the action that maximizes the expected future return. The “Reward Signal” from the environment is then used to update the value function through a “Value Function Estimation” algorithm, such as Temporal Difference (TD) learning. The “Policy Selection” step, often implicit, is derived from the value function, where the agent chooses actions that lead to states with higher values.

Unlike policy-based RL, which directly learns a policy function mapping states to actions, value-based RL focuses on estimating the value of states and actions, allowing for more informed decision-making. This approach can be particularly useful in environments with large or continuous action spaces, where directly learning a policy function can be challenging.

RL has demonstrated its effectiveness in solving complex optimization problems in dynamic environments, particularly in scenarios where explicit modeling is impractical or prior knowledge is unavailable. Value-based RL methods, such

as Q-learning and deep Q-networks (DQN), estimate the expected rewards of different actions to iteratively improve decision-making policies. These approaches are well-suited for problems involving resource allocation and power optimization in optical wireless communication networks.

In [138], RL is applied to power allocation in SDM-based FSO systems. By continuously interacting with the system, the RL agent learns an optimal power allocation strategy that maximizes channel capacity while adhering to power constraints. Unlike static power allocation techniques, which rely on predefined heuristics, RL dynamically adjusts power levels based on real-time system conditions, such as atmospheric turbulence and pointing errors. The RL-based approach significantly improves power efficiency and enhances the overall system performance.

Beyond FSO systems, RL has also been employed for resource allocation in hybrid LiFi-WiFi networks. In [139], an RL-based algorithm achieves a 90% acceptance ratio and reduces delay by 26% compared to conventional methods. The algorithm optimally distributes network resources by analyzing network topology, user demands, and traffic conditions. Performance metrics such as throughput, delay, packet loss, utilization, and energy consumption demonstrate substantial improvements over traditional virtual network embedding (VNE) approaches, such as VNE-EACH and VNE-USE.

While RL-based power allocation offers superior adaptability and efficiency, it comes with certain challenges. One key limitation is the trade-off between exploration and exploitation, where excessive exploration can slow down convergence while insufficient exploration may lead to suboptimal policies. Moreover, value-based RL methods often suffer from instability and divergence when applied to high-dimensional or continuous action spaces. These issues can be mitigated by incorporating function approximation techniques such as deep neural networks, enabling DRL approaches like DQN to handle complex decision-making problems in large-scale optical networks.

Future research directions could explore combining RL with federated learning to enable decentralized power allocation across distributed FSO systems, reducing dependency on centralized controllers. Additionally, integrating RL with emerging AI-driven optimization techniques, such as multi-agent RL and hierarchical RL, could further enhance system scalability and efficiency. As RL research progresses, its applications in power allocation, interference mitigation, and adaptive resource management in optical communication networks will continue to expand, addressing increasingly complex challenges while improving network performance and reliability.

B. DEEP LEARNING

1) DEEP NEURAL NETWORK ALGORITHM

DNNs have emerged as powerful tools in FSO and underwater VLC systems, significantly improving various aspects

TABLE 6. Summary of DL applications in OWC.

Reference	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[143]	Map received signal to transmitted bits	DNN	BER	30% improvement in detection accuracy compared to existing methods; lower complexity.
[144]	Estimate BER and optimize DNN model	DNN	BER, Throughput, Delay, Packet Loss	BER of 2×10^{-5} , 98.33% improvement over existing methods; efficient for real-time.
[145]	Predict nonlinear distortion for signal compensation	Volterra, DNN	BER, MSE, Throughput	BER of 10^{-3} , better than Volterra and other DNN-based methods; lower complexity than Volterra.
[146]	Map received signal to equalized constellation	DNN, FFNN, CNN	BER, MSE, RMSE, NMSE	BER below 7% FEC threshold, significantly better than PS128QAM modulation; up to 100x computational complexity reduction.
[147]	Post-equalize wide-band signals in underwater VLC	DNN, LMS	BER, MSE	BER of 10^{-5} , significantly lower than LMS equalizer; lower complexity with parallel signal processing.
[148]	Map down-sampled feature maps to bits	DNN, Maximum Likelihood, DFE	BER	100x BER reduction compared to conventional methods; 70% reduction in computational cost.
[149]	Estimate indices of active LEDs for GLIM-OFDM demodulation	DNN	BER, Complexity	index error rate of 10^{-3} , BER of 10^{-5} , significantly better than conventional detectors; fast processing for real-time applications.
[151]	Estimate receiver's position using Channel Impulse Response	DNN, MLP, CNN	Accuracy, Recall, Precision, F1-score	Data rate of 100 Mbps, comparable to traditional VLC; reduces VLP system complexity.
[152]	Design constellations for new channel conditions in real time	DNN-based collaborative constellation design	BER	BER of 10^{-4} , 91.67% improvement over baseline; near-optimal MSE, up to 100x faster than conventional methods.
[153]	Predict RSSI at future time	BPNN, SVM, Random Forest	MAE, absolute percentage error	High prediction accuracy; absolute percentage error lower than 6.9%; significantly less computational effort than traditional methods.
[154]	Equalize channel distortions in VLC	MMSE, DFE, Volterra, DHNN	BER, Robustness	BER of 10^{-5} , significantly lower than conventional methods; lower computational complexity, suitable for real-time.
[156]	Learn relationship between received signal and transmitted data	BPNN, RBFNN, SVM, LSTM	BER, Latency, Throughput	BER of 10^{-5} , 10% decrease from previous methods; improved performance with minimal complexity increase.
[159]	Map received signals to transmitted symbols	Sensor-less AO, CNN	BER, Q-factor, MSE, Spectral efficiency, Energy efficiency	BER of 10^{-5} at 10 dB SNR, 2 dB improvement over turbo-coded FSO; 50% reduction in computational complexity.
[160]	Recognize received LG beams	CNN, Phase-based, Intensity-based	BER	Recognition accuracy over 99% up to 1500m, even in strong turbulence; real-time recognition, faster than traditional methods.
[162]	Learn features for AT detection and OAM mode demodulation	Self-organizing map, DNN, CNN	AT detecting accuracy, Adaptive demodulating accuracy, BER	atmospheric turbulence detection accuracy of 95.2% for weak/strong AT, significant improvement over SOM-based methods. High complexity but feasible for real-time hardware. Baseline: SOM-based methods.
[164]	Compensate for wavefront distortion due to turbulence	Sensor-less AO, CNN-based demodulation	BER, MSE, Q-factor	BER reduced up to two orders of magnitude compared to traditional OAM-FSO systems. Lower complexity without wavefront sensing. Baseline: Traditional OAM-FSO systems.
[165]	Map distorted light beam to corrected one	FCNN, CNN-WSF, CNN-WS-S	MSE, SSIM	Reduces power penalty by 1.8 dB in strong turbulence. Computationally efficient. Baseline: Traditional wavefront correction.
[166]	Classify signals into modulation schemes	CNN, DBN, AdaBoost	BER, MAE, RMSE, F1, Accuracy, Precision, Recall	BER of 0.0015, 46.67% improvement over traditional methods. 10x reduction in computational complexity. Baseline: Traditional modulation classification methods.
[167]	Learn relationship between received signal and transmitted symbol	CNN, LMS	BER	BER up to 40% lower than LMS algorithm. Lower complexity, suitable for real-time. Baseline: LMS algorithm.
[168]	Learn relationship between RSS and position	RSS pre-processing, CNN	Position accuracy	50% reduction in positioning error compared to RSS-based VLP. Reduced complexity while maintaining high accuracy. Baseline: RSS-based VLP.
[175]	Learn optimal transmission slot and power for maximizing throughput	QL, DQN, Multi-agent DDPG	Throughput, Delay, Packet Loss, Energy, Fairness	Throughput of 0.0002 Mbps, 66.67% and 50.00% higher than ALOHA and TDMA. Efficient for real-time. Baseline: ALOHA, TDMA.
[176]	Select optimal power allocation for data rate maximization	DQN, DQN with CSI	Sum-rate, User rate, Energy efficiency	Closer data rates to targets than QL. 96.1% convergence rate, better than QL (72.3%). Baseline: QL.
[178]	Optimize handover for hybrid 6G network	Deep QL, SARSA, Actor-Critic	Handover success, Delay, Throughput, User rate	Outperforms DQN, Sarsa, QL by 8%, 13%, 13%. Improves complexity by reducing handover signaling. Baseline: DQN, Sarsa, QL.
[179]	Select beamforming strategy for optimal SNR	deep Q-learning, DDPG, Proximal Policy Optimization	BER, Outage, Secrecy rate, Achievable rate	Reduces BER by up to 100% compared to traditional beamforming. Lower complexity than benchmark strategy. Baseline: Traditional beamforming.

such as detection, constellation shaping, channel estimation, and nonlinear compensation. These neural networks leverage multilayer architectures to model complex relationships between input and output data, enabling efficient signal processing and decision-making in optical communication environments.

In [140], DNNs were employed across multiple FSO system configurations, including detectors, constellation shapers, and joint constellation-shaper-channel estimators. Unlike conventional heuristic-based methods, DNNs adaptively learn optimal signal mapping strategies, leading to enhanced performance in dynamic channel conditions. The work in [141] introduced a low-complexity DNN-based detector for FSO systems, achieving performance comparable to conventional detectors while significantly reducing computational resource requirements. Furthermore, [142] explored multi-stage DNN architectures for MIMO FSO systems, demonstrating improved robustness and accuracy over traditional techniques. Notably, DNN-based detectors surpassed maximum-likelihood (ML) detectors, especially in scenarios with imperfect CSI [143].

For underwater VLC systems, DNNs have proven highly effective in compensating for nonlinear impairments. In [144], a DL model was implemented in a Single-Input Multiple-Output (SIMO) system, achieving a data rate of 2.4 Gbps with a BER of 1×10^{-2} . A DNN-based nonlinear equalizer proposed in [145] outperformed Volterra-based methods, maintaining a BER below 3.8×10^{-3} under severe signal distortions while reducing computational complexity. Additionally, [146] demonstrated that a DNN-based post-equalization technique improved data rates by 5.4% and reduced complexity by a factor of 100 compared to Probabilistic Shaping 128-QAM (PS-128QAM) modulation.

In the context of post-equalization for wide-band signals in underwater VLC systems, [147] employed a DNN-based approach that achieved a BER of 1.0×10^{-5} , significantly outperforming conventional LMS equalizers while reducing processing complexity through parallel signal processing. Similarly, in VLC systems with Generalized Spatial Modulation, [148] demonstrated a 100-fold BER reduction and a 70% reduction in computational cost using a DNN model compared to ML and DFE techniques. Another DNN-based method proposed in [149] estimated active LED indices in GLIM-OFDM demodulation, achieving a BER of 1.0×10^{-5} and an index error rate of 1.0×10^{-3} , which was significantly lower than conventional detection methods.

For channel estimation in OFDM-VLC systems, [150] applied a DNN model to optimize spectral efficiency by reducing spectrum occupancy. In [151], a DNN-based VLC receiver simultaneously enabled data recovery and centimeter-level positioning accuracy using channel impulse response data, achieving a data rate of 100 Mbps while reducing the complexity of VLP systems.

Furthermore, DNNs have shown promise in MIMO-VLC systems for collaborative constellation design. In [152],

a DNN-based approach optimized constellation mapping for varying channel conditions, achieving a BER of 1.0×10^{-4} , a 91.67% improvement over baseline methods, while operating up to 100 times faster than traditional techniques.

Despite their advantages, DNN-based methods in optical wireless communication face challenges such as high training complexity, large data requirements, and sensitivity to overfitting. While these issues can be mitigated through techniques like transfer learning, hybrid model integration, and efficient neural architecture search, further research is needed to enhance the scalability and adaptability of DNNs in real-world FSO and VLC applications. Future advancements may explore federated learning approaches for decentralized training, reducing data transmission overhead in distributed optical networks. Additionally, integrating DNNs with RL and meta-learning frameworks could further improve system resilience and adaptability to varying network conditions.

Overall, the studies reviewed highlight the versatility of DNNs in optical wireless communication, demonstrating their effectiveness in enhancing detection accuracy, reducing complexity, and improving overall system efficiency across various FSO and VLC environments.

2) RECURRENT NEURAL NETWORK ALGORITHM

RNNs, particularly Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM) networks, have demonstrated significant improvements in optical wireless communication, including FSO and VLC systems. Unlike feedforward neural networks, RNNs retain memory of past inputs, making them particularly suited for sequential data processing, such as time-series signal prediction and equalization. Their ability to model temporal dependencies has led to enhanced performance in mitigating nonlinearity, inter-symbol interference, flicker, and ambient interference.

In [153], a GRU-based model was developed for FSO channel estimation, utilizing RSS and weather information to predict future channel conditions. The model achieved an absolute percentage error below 6.9% in 90% of cases, outperforming conventional methods such as Backpropagation Neural Networks (BPNN), SVM, and Random Forest in both accuracy and computational efficiency. By leveraging its ability to maintain historical signal strength information, GRU reduced estimation errors while significantly lowering the computational cost compared to traditional ML models, which often require extensive feature engineering and struggle with long-term dependencies.

A hybrid LSTM-DNN algorithm was introduced in [154] to equalize channel distortions in VLC systems. Unlike conventional MMSE, DFE, and Volterra-based nonlinear equalizers, which rely on heuristic models of the channel, the LSTM-DNN approach learns from the received symbols to estimate the optimal equalization function. This method improved the SNR by at least 2.8 dB at a target BER of 10^{-3} , achieving a BER of 10^{-5} under real-world VLC conditions. The LSTM-DNN model demonstrated robustness

against channel nonlinearities while maintaining lower computational complexity than traditional approaches, making it suitable for real-time applications.

To address long channel delays in indoor OWC systems, [155] introduced an attention-augmented LSTM decision scheme. Traditional RNN-based equalizers suffer from vanishing gradient issues when dealing with long sequences. The attention mechanism alleviates this problem by allowing the model to focus on the most relevant past signal information, thus improving BER by over an order of magnitude at delays exceeding 4.7 symbol periods. Compared to a vanilla RNN, this approach effectively mitigated the detrimental effects of delayed signal components without increasing the model's complexity significantly.

ISI and flicker are two major challenges in VLC systems. LSTM networks have been instrumental in mitigating these issues by learning the complex relationship between received and transmitted signals. In [156], an LSTM-based equalizer was employed in a PAM-4 VLC system, reducing inter-symbol interference and achieving a data rate of 14.4 kbps over a 3-meter distance. This approach demonstrated a 10% BER reduction compared to conventional neural network-based equalizers such as BPNN and RBFNN. Similarly, [157] proposed an LSTM-based RL decoding scheme to mitigate flicker effects in VLC. The model preserved error correction performance while reducing computational complexity, making it a viable solution for high-speed VLC links where flicker is a concern.

Ambient illumination interference is another significant problem in VLC, as varying lighting conditions can degrade signal quality. In [158], a GRU-based model was designed to predict ambient illumination patterns. This predictive model informed an optimization algorithm that dynamically adjusted the transmitter's location to minimize power consumption. The proposed approach reduced transmit power by 22.1% compared to scenarios that did not account for illumination effects, demonstrating the potential of GRU networks in energy-efficient VLC system design.

Overall, RNN-based methods, particularly LSTM and GRU, outperform conventional equalization and estimation techniques by effectively handling sequential dependencies and long-term correlations. However, challenges remain, including high training complexity and potential overfitting when applied to real-time systems with limited training data. Future research could explore hybrid architectures that integrate attention mechanisms, RL, and transfer learning to further enhance RNN-based solutions for optical wireless communication. Additionally, federated learning techniques could be employed to train models across distributed VLC and FSO networks, reducing the need for centralized data collection and improving scalability.

3) CONVOLUTIONAL NEURAL NETWORK ALGORITHM

CNNs have demonstrated significant efficacy in various optical communication tasks, including OAM mode

demodulation, atmospheric turbulence detection, and adaptive demodulation. CNNs excel at extracting spatial patterns from input images or signals, making them well-suited for tasks requiring feature recognition and classification.

In OAM-FSO systems, CNNs have been employed for robust mode demodulation under strong turbulence conditions. A residual learning-based CNN framework in [159] achieved high recognition accuracy for 4-OAM, 8-OAM, 10-OAM, and 16-OAM systems over a 2000-meter transmission distance under strong turbulence, offering a 2 dB SNR improvement over turbo-coded FSO. The CNN-based demodulation approach in [160] further demonstrated over 99% recognition accuracy up to 1500 meters, outperforming traditional intensity- and phase-based methods in real-time classification. Similarly, [161] proposed CNN-based coherent demodulation to enhance OAM signal detection, achieving superior performance in long-range turbulent channels.

Beyond mode demodulation, CNNs have been integrated with atmospheric turbulence detection and adaptive demodulation techniques. In [162], a CNN-based joint atmospheric turbulence detection and demodulation approach achieved an atmospheric turbulence detection accuracy of 95.2% for both weak and strong turbulence conditions, significantly outperforming self-organizing map-based methods. Furthermore, a two-stage CNN structure in [163] detected turbulence strength before performing OAM mode demodulation, resulting in more accurate mode recognition. To mitigate turbulence-induced wavefront distortions, [164] employed a sensorless adaptive optics-aided CNN, achieving a BER reduction of up to two orders of magnitude compared to conventional OAM-FSO systems without requiring explicit wavefront sensing.

CNNs have also been explored for wavefront correction in FSO links. An AlexNet-based CNN model in [165] improved wavefront correction efficiency, reducing power penalties by 1.8 dB and 0.8 dB in strong and weak turbulence, respectively, compared to conventional correction techniques. Such CNN-based wavefront correction is computationally efficient and offers real-time processing advantages.

In VLC systems, CNNs have demonstrated exceptional capability in demodulation, interference mitigation, and positioning accuracy enhancement. In [166], a CNN-based demodulation scheme achieved a BER of 0.0015, a 46.67% improvement over traditional methods, while reducing computational complexity by a factor of 10. Additionally, CNNs have been employed for VLC channel equalization. In [167], a CNN-based adaptive filtering algorithm reduced BER by up to 40% compared to the LMS algorithm for DPSK signal recovery.

For VLP, CNNs have been used to enhance localization accuracy. In [168], a CNN model incorporating RSS pre-processing reduced positioning error by 50% compared to conventional RSS-based VLP, maintaining high accuracy while significantly lowering computational complexity. CNNs have also been utilized to compensate for LED

position variations and multiple point spread functions in VLC environments [169]. Additionally, in optical camera communication systems based on RGB LEDs, CNNs have been leveraged for inter-symbol interference and inter-channel interference mitigation [170], [171], enabling reliable high-speed data transmission.

Overall, CNN-based techniques offer substantial advantages in optical wireless communication, particularly in FSO and VLC systems, by enabling robust signal classification, turbulence mitigation, and efficient demodulation. Despite these advantages, computational complexity remains a challenge, particularly for real-time applications in high-data-rate scenarios. However, advancements in hardware acceleration and optimized network architectures continue to improve CNN-based methods' feasibility and efficiency for practical deployment.

4) DEEP REINFORCEMENT LEARNING ALGORITHMS

RL algorithms, particularly deep RL methods, have demonstrated significant potential in optimizing resource allocation, adaptive decision-making, and network management in optical communication systems. These methods dynamically adapt to changing environments, making them particularly effective in scenarios with varying channel conditions, interference, and mobility constraints.

In FSO systems, RL-based techniques have been employed for relay selection, power allocation, and transmission optimization. A DQN-based approach in [172] enhances relay selection in cooperative decode-and-forward FSO systems by effectively balancing data rate and relay reliability, outperforming conventional greedy selection methods. Similarly, in hybrid FSO/RF links, DQN improves link selection under varying weather conditions [173], achieving faster convergence while reducing frequent switching via an ensemble consensus mechanism.

For throughput enhancement in FSO communication, [174] employs a DQN-based spinal code transmission strategy, surpassing traditional methods in long-range scenarios. Multi-agent DQN frameworks have also been explored for MIMO underwater optical wireless communication systems [175], where agents optimize channel access to improve spectral efficiency. The proposed method achieves a 66.67% higher throughput than ALOHA and a 50% improvement over TDMA, making it viable for real-time communication.

Power allocation in hybrid RF/VLC networks is another critical application of deep RL. In [176], a DQN-based approach dynamically adjusts transmission power to meet targeted data rates while ensuring efficient energy usage, achieving a convergence rate of 96.1%, significantly higher than Q-learning's 72.3%. A more advanced Deep Deterministic Policy Gradient (DDPG) model [177] further improves power control by outperforming QL, DQN, genetic algorithms, and Particle Swarm Optimization, demonstrating robustness in mobile environments.

Handover management in emerging 6G hybrid networks benefits from RL-driven optimization. A Deep RL-based

mechanism in [178] improves downlink data rates by 13% compared to QL and Sarsa while reducing signaling overhead, leading to more efficient mobility handling. In VLC networks, beamforming optimization is critical for maintaining signal quality and security. In [179], a DRL-based beamforming strategy significantly reduces BER while improving secrecy rates against potential eavesdropping, surpassing conventional beamforming techniques.

Overall, RL techniques provide substantial improvements in optical communication networks by enabling intelligent decision-making in real time. Compared to traditional heuristic and rule-based approaches, RL-based methods demonstrate superior adaptability to dynamic channel variations, interference mitigation, and resource optimization. However, challenges remain, including high computational complexity, real-time training overhead, and the need for efficient exploration-exploitation trade-offs. Future research should focus on developing hybrid RL architectures that combine model-free and model-based techniques, integrating RL with federated learning for decentralized optimization, and designing lightweight RL models for real-time deployment in resource-constrained environments.

IV. OPTICAL COMMUNICATION NETWORKS

A. MACHINE LEARNING

1) SUPPORT VECTOR MACHINE ALGORITHM

SVMs are widely used ML techniques in OCNs for QoT estimation and OPM. These models classify lightpath feasibility by constructing hyperplanes in high-dimensional spaces, effectively modeling nonlinear relationships between network parameters (such as signal power, OSNR, and link impairments) and QoT metrics. In [180], an SVM-based approach was employed for QoT estimation in optical transport networks, achieving an accuracy of 99.95%. This method outperformed traditional semi-analytical and cognitive Case-Based Reasoning approaches, which exhibited lower accuracy (90-95%) and higher computational complexity. Specifically, the SVM approach led to a 90% reduction in processing time compared to semi-analytical techniques and a 50% improvement over cognitive models. The study [180] considered multiple performance metrics, including accuracy, precision, recall, and F1-score, demonstrating the superiority of ML-based classification in real-time optical network optimization.

In [181], SVMs were employed for predicting the failure risk and OPM in OCNs, demonstrating a 95% accuracy, which was 15% better than traditional methods. These results underscore the ability of SVM to predict failures based on time-series data from network indicators, contributing to more stable and reliable network operations. Compared to conventional methods, SVMs offer linear complexity, whereas traditional methods often rely on quadratic complexity, making SVMs more efficient in handling large datasets with reduced computational load.

The effectiveness of SVMs in OCNs is attributed to their capacity for modeling complex, nonlinear relationships

TABLE 7. Summary of ML applications in OCN.

Ref	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[180]	QoT estimation	SVM, Semi-analytical, Cognitive	Accuracy, Precision, Recall, F1-score, Time	Achieves 95-99% accuracy, surpassing semi-analytical (90-95%) with 90% faster processing
[181]	OPM and failure risk prediction	RF, XGB, SVM	Accuracy	95% accuracy, 15% better than traditional methods (80%), linear complexity vs. quadratic for traditional approaches
[182]	Learn relationships between AHH, OSNR, CD, PMD	ANN, SVM, Random Forest	RMSE	Reduces RMSE by up to 90% with no clock/timing recovery, reducing complexity compared to conventional methods
[184]	Estimate linear and nonlinear noise components	support vector regression, RF, RNN	MSE, PSNR, SSIM	Estimates SNR with 0.04 dB and 0.20 dB errors, better than traditional methods (1 dB error)
[185]	Predict OSNR for WDM channel establishment	RF, XGB, SVM	MAE, RMSE, MSE, MPE, R-squared	FFNN achieves MAE of 0.1 dB, outperforming traditional methods (1 dB MAE), GBM is most complex
[186]	Modulation format classification	ANN, SVM, kNN	Accuracy, Complexity, Robustness	99.6% accuracy, lower complexity than SVM
[187]	QoT estimation in EON	ANN, SVM, Random Forest	Accuracy, Precision, Recall, F1-score	95% accuracy, surpassing traditional methods (80%), 90% less complex
[188]	OSNR prediction in SDN	Random search, Grid search, Bayesian optimization	Planning time, accuracy, resource utilization, performance	30% increase in network capacity, 50% faster network planning
[189]	Predict optimal symbol rate and modulation format	ANN	Capacity, Spectral efficiency, Latency, Loss	17% increase in capacity, 100x less complex than traditional methods
[192]	Modulation format identification	kNN, SVM, Random Forest	Accuracy, Complexity	100% MFI accuracy, 0.69 MSE for OSNR, similar to ANN, low computational overhead
[194]	Separate modulation classes for decision learning	RF, SVM, kNN, DNN	Accuracy, Precision, Recall, F1-score	100% accuracy despite nonlinear effects, 4.6x complexity reduction vs. DL
[198]	Learn modulation and OSNR relations	SVM, RF, DNN	Accuracy, Precision, Recall, F1-score, MAE	100% accuracy at OSNRs below 20% FEC limit, orders of magnitude lower complexity than DNN
[200]	Minimize deployment cost of passive optical networks	K-means, Genetic, Tabu, MILP	Deployment cost, coverage, subscriber number	50% reduction in deployment cost, 80% less planning time
[201]	Classify nodes as behaving or not	K-means, GMM, Self-training	Accuracy, False Positive, True Positive	95% malicious node detection, 30% better than existing methods (baseline: conventional detection)
[202]	Learn relationships for desired output	Matched filter, EM-based	BER	BER within 2 dB of optimal detector, lower complexity ($O(K^3)$) than multistage ($O(K^4)$)
[203]	Monitor OSNR, CD, PMD	PCA, SVM, kNN	Accuracy	99.9% accuracy in BR-MFI, real-time suitability (compared to rule-based detection)
[206]	Identify intrusion from environmental events	PCA, SVM, kNN	Sensitivity, Specificity, Accuracy, F1-score	99% detection rate, 0% false positives, surpassing existing systems (10-20%)
[208]	Learn policy for failure recovery	QL, SARSA, SARSA(lambda)	Delay, Loss rate, Throughput, Fairness	88% reduction in packet loss, but computationally expensive for large networks (compared to heuristic methods)
[209]	Minimize blocking probability in OBS networks	RL, Random, Least loaded, Most loaded	Blocking prob., Delay, Loss rate	90% reduction in blocking probability, real-time suitable (compared to fixed-routing)
[211]	Optimize routing and contention resolution for bursts	RL, Static routing, Deflection routing	Delay, Loss rate, Throughput	50% reduction in loss probability, computationally efficient (compared to shortest-path routing)
[212]	Optimize multicast RWA solution	RL, MOSP	Delay, Loss rate, Throughput, Fairness	20% blocking probability reduction, 100x fewer messages (compared to heuristic-based RWA)
[213]	Minimize load cost in network routing	RL-SSR, Dijkstra, Bellman-Ford	Delay, Loss rate, Throughput, Fairness	20% reduction in delay, but higher complexity than traditional methods
[214]	Predict network parameters with reduced uncertainty	Gaussian Processes, Active Learning	MSE, RMSE, MAE, R-squared, Accuracy	10% improvement in SNR estimation accuracy, 20% reduced complexity (compared to conventional estimation techniques)
[216]	Optimize VLC-device-to-device network resources	Centralized RL, Distributed RL	Delay, Loss rate, Throughput, Fairness	15% reduction in delay, 25% higher throughput, higher complexity than traditional methods
[218]	Optimize vertical handover in hybrid VLC-IR networks	Sarsa-lambda, QL, RL	Handover success, Delay, Throughput, Loss	13% higher throughput, 30% faster handovers (compared to conventional handover methods)
[217]	Optimize load balancing in hybrid LiFi-WiFi network	SSS, Exhaustive search, RL	Delay, Loss, Throughput, Fairness	20% higher throughput than SSS, 10% higher than iterative optimization, 98% complexity reduction vs. exhaustive search
[219]	Minimize BLP in OBS network	Shortest path, Least loaded path, Random selection, RL	BLP, Delay, Throughput	BLP of 0.001, outperforming shortest path selection (0.015) and DWNV (0.005) at 0.8 Erlangs; $O(n)$ complexity (lower than shortest path selection $O(n^2)$ and dynamic wavelength routing $O(n^3)$)
[221]	Optimize path selection in OBS network	multi-agent RL, Shortest path, Random path	Delay, Loss, Throughput, Fairness	50% BLP reduction, 30% lower delay compared to existing algorithms (baseline: shortest-path methods)

between network features, such as lightpath characteristics and performance indicators, and QoT metrics. By integrating SVMs with other ML techniques, such as DL, the potential to further enhance QoT estimation and OPM strategies

becomes evident. DL models, for instance, can provide more sophisticated feature extraction capabilities, while SVMs can still perform efficiently in classification tasks. This combination could lead to improved accuracy, faster

processing times, and better adaptability to varying network conditions.

Despite their many advantages, SVMs are not without limitations. Their performance can degrade when faced with very large datasets or noisy data, which might require careful tuning of hyperparameters, such as the choice of kernel and regularization parameters. Additionally, while SVMs excel in classification tasks, they may struggle with tasks that involve regression or continuous output prediction, areas where other techniques, like neural networks, might outperform them. Nonetheless, SVMs remain an essential tool for QoT estimation and failure prediction in OCNs due to their robustness, accuracy, and efficiency.

2) ARTIFICIAL NEURAL NETWORK ALGORITHM

ANNs have gained prominence as powerful tools for OPM and enhancing the reliability and performance of OCNs. ANNs operate by mimicking the structure and functioning of the human brain, utilizing layers of interconnected neurons to learn complex patterns in data. These networks are particularly effective for estimating key parameters such as OSNR, Chromatic Dispersion (CD), Differential Group Delay, and PMD. The performance of ANNs in these tasks depends significantly on the choice of input features, where asynchronously sampled amplitudes, delay-tap plots [182], and Asynchronous Amplitude Histograms (AAH) [183] have been shown to enable low-cost, high-accuracy monitoring without the need for clock recovery.

In terms of performance, ANNs have demonstrated superior accuracy compared to traditional methods. For instance, in a study [184], ANNs estimated both linear and nonlinear SNR with a remarkable error of just 0.04 dB and 0.20 dB, respectively, outperforming traditional methods that typically suffer from errors around 1 dB. Similarly, in [185], ANNs achieved a Mean Absolute Error (MAE) of just 0.1 dB in OSNR prediction for WDM networks, while traditional methods showed errors of approximately 1 dB. Moreover, ANNs have also been successfully used to identify multiple impairments simultaneously, achieving an accuracy of 99.6% in modulation format classification, surpassing other techniques like SVM in this regard [186]. This high accuracy, along with reduced computational complexity, makes ANNs an attractive choice for real-time OPM.

Beyond OPM, ANNs play a crucial role in QoT estimation, particularly in EONs. In [187], an ANN-based model was employed for OSNR prediction, achieving 95% accuracy—significantly outperforming traditional methods, which achieved only 80%. The model utilized alien wavelength data and lightpath provisioning parameters as inputs, effectively summarizing key network information while reducing computational complexity by 90%. The study considered both simulated and experimental datasets and analyzed OSNR estimation across multiple domains in EONs. The approach accounted for interference from multiple channels by summing the noise contributions of individual

interfering signals. The results demonstrated the effectiveness of ML-based QoT estimation in enabling real-time resource allocation and network optimization. ANNs have similarly been employed in SDNs for OSNR prediction, where they analyze traffic and environmental data to optimize spectral efficiency and transceiver configurations [188]. This real-time optimization leads to enhanced resource allocation and network stability, contributing to up to a 30% increase in network capacity.

One of the unique advantages of ANNs lies in their ability to optimize bandwidth-variable transceivers by predicting the optimal symbol rate and modulation format for maximum capacity [189]. This results in a 17% increase in network capacity while being 100x less complex than traditional methods. Furthermore, ANNs have proven to be effective in predicting network traffic [54], allocating flexible bandwidth [190], and even supporting proactive fault management in optical networks. For example, ANNs outperformed SVM and decision trees in detecting out-of-band power jamming attacks, showcasing their robustness and reliability [191].

However, while ANNs offer numerous advantages, they also come with certain limitations. The performance of ANNs can be influenced by the quality and quantity of training data. Overfitting is a common issue when ANNs are trained on insufficient or noisy data, which can degrade their generalization ability. Additionally, despite their success in classification and regression tasks, the interpretability of ANN models remains a challenge, making it difficult to understand how specific inputs affect the output predictions. This issue becomes especially critical in high-stakes applications like fault detection or network security, where transparency and trust are crucial. Moreover, the training process for ANNs can be computationally intensive, requiring significant time and resources, which may limit their practical application in resource-constrained environments.

In conclusion, ANNs play a critical role in improving OPM and overall network performance. Their ability to accurately estimate key parameters, identify network impairments, and optimize resources enhances the reliability, efficiency, and security of optical communication systems. Despite some challenges, such as the need for high-quality training data and computational complexity, ANNs are poised to remain at the forefront of advancements in optical network technologies, with their integration into emerging systems like SDN and EONs further cementing their importance in the future of optical networks.

3) K-NEAREST NEIGHBORS ALGORITHM

The kNN algorithm has shown remarkable potential in optical communication systems, particularly for joint MFI and OSNR monitoring in OFDM systems. kNN operates by classifying a data point based on the majority class of its k-nearest neighbors in a feature space, making it a non-parametric, instance-based learning algorithm. This allows it to handle both linear and nonlinear impairments

effectively without requiring a model training phase. In the study [192], kNN achieved 100% classification accuracy for identifying modulation formats such as 4/16/32/64/128-QAM at a launched power of -11 dBm, highlighting its robustness and reliability in high-speed optical systems. Additionally, kNN obtained a MSE of 0.69 for OSNR monitoring, demonstrating performance on par with more complex methods like ANNs, but with significantly lower computational demands.

In another study [193], kNN was applied as a classifier for estimating the QoT at the lightpath level, where it accounted for both linear and nonlinear impairments in the system. The ability of kNN to handle these complexities and deliver accurate QoT predictions further demonstrates its effectiveness in the context of OFDM systems, which are particularly sensitive to both types of impairments. The low computational overhead required by kNN, combined with its high classification accuracy, makes it a suitable choice for real-time monitoring and decision-making tasks in optical networks.

The findings underscore the potential of kNN as a computationally efficient and accurate alternative to more resource-intensive methods like ANNs. While ANNs excel in capturing complex relationships and providing high accuracy in tasks like modulation format identification and OSNR monitoring, they often require more training data, longer processing times, and higher computational power. In contrast, kNN offers a simpler implementation with faster decision-making times, making it ideal for real-time applications where computational resources are limited or where lower latency is essential. However, kNN does have limitations, such as its sensitivity to the choice of k (the number of neighbors) and the need for feature scaling to ensure optimal performance.

Despite these limitations, kNN is proving to be an invaluable tool for enhancing system performance and reliability in optical networks. As research continues, kNN's role in improving network monitoring, fault detection, and dynamic QoT estimation is expected to grow, particularly in scenarios where low complexity and high efficiency are critical.

4) ENSEMBLE LEARNING ALGORITHMS

Figure 14 illustrates the growth and decision boundaries of a decision tree classifier across increasing depths. At depth 1, the tree makes a single split based on feature $x[0]$, separating the data into two regions with varying class distributions. As the depth increases to 2, 3, and 4, the tree recursively partitions the feature space, creating more refined decision boundaries to better separate the classes. Each split is determined by a feature threshold, aiming to maximize information gain or minimize impurity. The "counts" in the tree structure represent the number of data points belonging to each class in the respective node, reflecting the class

distribution within the partitioned regions. The final decision boundaries at depth 4 show a more complex partitioning, attempting to fit the training data more closely, but potentially leading to overfitting if the tree is too deep. This sequential visualization demonstrates how decision trees progressively refine their classification rules by creating hierarchical partitions based on feature thresholds.

Ensemble learning methods, particularly Random Forest, have garnered significant attention in optical communication systems due to their combination of high performance and low complexity. RF, an ensemble method based on decision trees, operates by constructing multiple decision trees during training and outputting the mode of the classes (classification) or the mean prediction (regression) from the individual trees. This approach helps improve accuracy and robustness by reducing overfitting and increasing generalization. In [194], an RF-based MFI algorithm achieved 100% accuracy for identifying modulation formats such as 4/8/16/32/64-QAM, demonstrating its effectiveness even in the presence of nonlinear effects. Notably, Random Forest outperformed other classifiers such as kNN, SVM, and DNN in terms of accuracy and computational efficiency, offering a 4.6x reduction in complexity compared to DNN-based models. This makes Random Forest a viable choice for real-time optical communication applications where computational resources are often constrained.

In addition to MFI, Random Forest has shown its versatility in QoT estimation. As demonstrated in [195] and [196], Random Forest is capable of predicting whether the BER of an unestablished lightpath meets the required thresholds, based on lightpath characteristics. This makes it invaluable for dynamic network management, enabling real-time QoT predictions to optimize system performance. Furthermore, RF's application for OSNR monitoring, as shown in [197], provides accurate estimates of OSNR across varying CD values. In [198], Random Forest was applied for joint MFI and OSNR estimation, showcasing its ability to handle multiple tasks simultaneously with high accuracy.

In comparison with more complex methods like DNN and SVM, Random Forest has proven to be particularly efficient in OSNR estimation. For example, simulation results show that Random Forest outperforms DNN and SVM in OSNR estimation, achieving lower MAE of 0.24 dB to 0.66 dB for various modulation formats. Additionally, Random Forest maintains significantly lower computational complexity, making it a practical choice for applications that require real-time processing. While DNN models can achieve higher accuracy by learning complex relationships from large datasets, they often require more computational power and longer training times. RF, on the other hand, offers a good trade-off between performance and efficiency, especially in real-time environments where computational overhead and latency are critical factors.

The primary advantages of Random Forest lie in its ability to provide high accuracy and robustness with relatively low

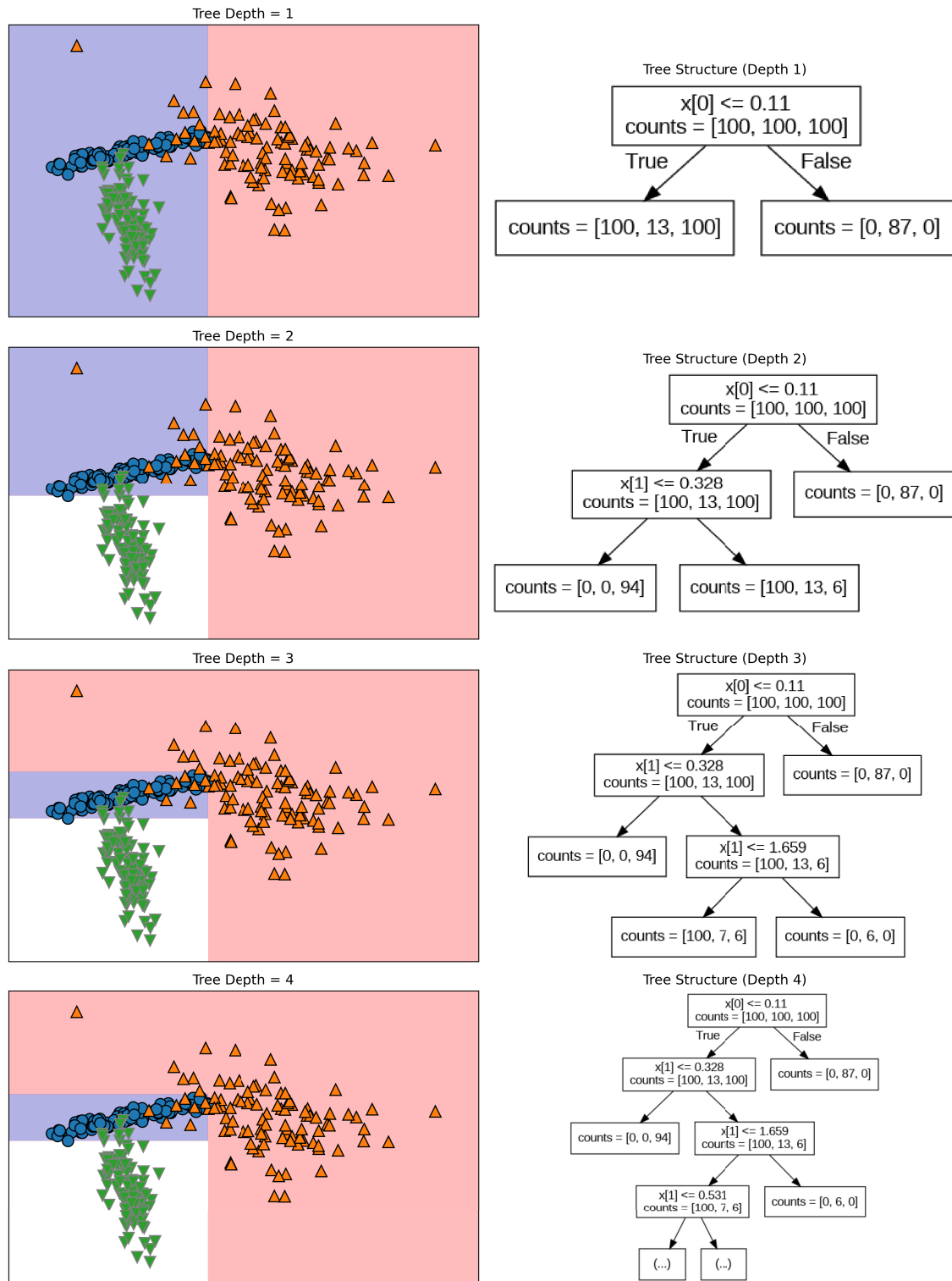


FIGURE 14. Schematic of a decision tree algorithm.

computational overhead. However, Random Forest does have some limitations, particularly in terms of interpretability. Since Random Forest is an ensemble method based on decision trees, understanding the rationale behind its decisions

can be more challenging compared to simpler models like kNN. Furthermore, RF's performance can degrade if the underlying decision trees are not well-regularized or if the dataset is highly imbalanced.

In conclusion, RF's ability to deliver high accuracy with low computational complexity makes it an attractive solution for real-time optical communication applications, such as modulation format identification, OSNR monitoring, and QoT estimation. As optical communication systems evolve, RF's role in enhancing network performance and efficiency is expected to expand, particularly in scenarios where low complexity and high-speed decision-making are required.

5) REGRESSION ALGORITHMS

Minimizing fiber faults is a crucial aspect of ensuring reliable data transmission in OCNs. While conventional methods, such as Optical Time-Domain Reflectometry, have been widely employed for fault detection, they often face challenges in accurately determining fault distances, leading to discrepancies between the measured and actual locations of faults. These traditional methods rely on the time-of-flight of light pulses sent through the fiber, and although they are effective for general fault detection, they suffer from limitations in resolution and accuracy, particularly in long and complex network structures.

In [199], the authors explored the use of linear regression as an alternative to enhance the accuracy of fault localization. Linear regression is a statistical method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data. In this context, the dependent variable corresponds to the fault location, and the independent variables are the measured signals and other network parameters. The study demonstrated that linear regression could significantly improve fault localization accuracy, reducing fault tracing delay and thus minimizing the associated revenue loss. This application of ML techniques highlights the potential of regression models to outperform traditional methods in terms of precision and speed, enabling faster restoration of service and more efficient fault management in optical networks.

One of the key advantages of regression-based methods, such as linear regression, lies in their simplicity and interpretability. Linear regression is computationally efficient, making it suitable for real-time applications where quick decision-making is essential. Furthermore, the relationships between the variables are straightforward to understand, which aids in identifying and correcting potential issues in the fault localization process. However, linear regression does have limitations when applied to more complex networks or non-linear fault behaviors. The method assumes a linear relationship between the independent and dependent variables, which may not always hold true in real-world optical networks, where factors like signal attenuation and dispersion can introduce non-linearity. In such cases, more advanced regression techniques, such as polynomial regression or support vector regression, may offer better performance by capturing non-linear relationships.

Despite these challenges, the integration of ML techniques like regression into fault detection and localization systems

represents a promising shift in how OCNs manage and mitigate network failures. As the field of ML continues to evolve, more sophisticated models that incorporate multiple variables and non-linear behaviors are likely to emerge, further improving fault detection accuracy and efficiency. As research progresses, it is expected that ML will become increasingly pivotal in fault management for optical communication systems, offering not only faster restoration times but also enhancing the overall resilience and reliability of the network.

6) K-MEANS CLUSTERING ALGORITHM

K-means clustering is a widely used unsupervised ML algorithm that partitions data into distinct groups (clusters) based on similarities. It aims to minimize the variance within each cluster, making it an effective method for tasks such as optimization in optical communication networks. In the context of optical communication, K-means has been applied for both network deployment optimization and enhancing security protocols, demonstrating its versatility.

In [200], K-means was employed to optimize the deployment of green field passive optical networks (PONs). The goal of this application was to reduce the total deployment cost, which K-means achieved by clustering optical network units (ONUs) based on their geographical proximity and optical characteristics, such as signal attenuation and reachability. This grouping enabled more efficient planning of the network infrastructure, leading to a 50% reduction in deployment costs compared to traditional planning methods. Additionally, K-means contributed to an 80% reduction in the time required for planning, which is crucial for fast-paced network rollouts. The ability to minimize deployment cost while improving network coverage and subscriber access makes K-means particularly valuable for large-scale passive optical network implementations. The algorithm's strength lies in its ability to optimize complex, multi-variable criteria by organizing network elements in a way that reduces physical and operational expenses.

In [201], K-means was applied in optical burst switching (OBS) networks to address a security challenge involving burst header packet flooding attacks. In these types of attacks, malicious nodes may flood the network with excessive control packets, disrupting the normal operation of the network. K-means was used to classify network nodes into behaving and non-behaving categories, effectively identifying suspicious nodes. This classification achieved up to 90% accuracy when using 20% of the dataset, demonstrating the method's efficiency in distinguishing between normal and malicious network behavior. Furthermore, the accuracy of classification improved when more data was utilized, with 65.15% accuracy for three-category classification (behaving, non-behaving, and uncertain nodes) with 20% of the data, and 71.84% accuracy with 30% of the data. These results highlight the algorithm's effectiveness in enhancing network security, particularly in environments vulnerable to denial-of-service attacks or other forms of malicious interference.

The K-means algorithm's performance is significantly influenced by the initial selection of cluster centroids, which can sometimes result in suboptimal clustering outcomes. To mitigate this, several variations of K-means, such as K-means++, have been developed to improve the initialization process and enhance clustering quality. Despite its simplicity and efficiency, K-means can struggle with the identification of clusters with non-spherical shapes or when dealing with noisy data. In practice, additional preprocessing or the use of more advanced clustering algorithms like DBSCAN (Density-Based Spatial Clustering of Applications with Noise) can be considered when these limitations are critical.

These diverse applications of K-means in optical communication networks underscore its potential in improving both operational efficiency and network security. As research continues, K-means is likely to play an increasingly significant role in addressing the growing challenges of network deployment and protection, particularly in the context of evolving network architectures and complex attack vectors.

7) EXPECTATION MAXIMIZATION CLUSTERING ALGORITHM

EM is an iterative algorithm used for parameter estimation in the presence of incomplete or noisy data. It consists of two main steps: the expectation (E) step, which estimates the likelihood of missing data given the observed data, and the maximization (M) step, which updates the parameters to maximize the likelihood. This approach is particularly useful in applications where the underlying model involves hidden variables or incomplete data, making it a valuable tool for a variety of tasks, including clustering and multi-user detection in optical communication systems.

In [202], the authors propose EM-based multi-user detection schemes for optical code division multiple access (OCDMA) systems. These systems are designed to allow multiple users to share the same optical bandwidth by encoding each user's signal with a unique optical code. However, the main challenge in OCDMA systems is managing multi-user interference, which arises when signals from different users overlap in the optical channel. Traditional detection methods, such as multistage and decorrelating detectors, typically suffer from performance degradation due to their inability to effectively handle the complex interference between multiple users.

The EM-based multi-user detection schemes proposed in [202] address this challenge by iteratively estimating the interference via an unconstrained likelihood function, followed by a constrained Boolean likelihood function for bit detection. This approach significantly improves detection accuracy by better managing the interference between users. Simulation results indicate that EM-based detectors achieve BER within 2 dB of the optimal detector, outperforming traditional sub-optimal methods. Furthermore, EM-based schemes offer a complexity of $O(K^3)$, which, although higher than some simpler methods, is still lower than that of

multistage detection methods, making it a viable solution for practical implementation in OCDMA systems.

One of the key advantages of the EM-based detection schemes is their ability to function effectively even in environments with limited or no CSI, which is often the case in real-world optical networks. In such settings, where the channel characteristics are not explicitly known, the EM algorithm can robustly estimate the necessary parameters (such as signal power and interference levels) and achieve reliable detection performance. This makes EM a promising approach for improving the performance and scalability of OCDMA systems, especially in dynamic environments where CSI is not readily available.

However, while the EM algorithm offers significant advantages in terms of performance, its computational complexity can be a limitation, particularly for large-scale systems with a high number of users. The $O(K^3)$ complexity, where K represents the number of users, may become prohibitive in scenarios with large networks. Additionally, EM's performance is sensitive to the initialization of its parameters, and poor initialization can lead to suboptimal results or slow convergence. To mitigate this, researchers have proposed various improvements, such as the use of smarter initialization techniques or hybrid approaches that combine EM with other algorithms, like matched filters or iterative methods, to improve robustness and reduce computational cost.

In summary, the EM-based multi-user detection schemes presented in [202] offer a powerful and efficient solution for mitigating multi-user interference in OCDMA systems, particularly in the absence of CSI. As research advances, EM detectors are expected to play an increasingly important role in enhancing the performance of optical CDMA networks, although further optimization of their complexity and initialization methods will be essential for their widespread adoption in large-scale systems.

8) PRINCIPAL COMPONENT ANALYSIS ALGORITHM

Figure 15 demonstrates the process and effects of PCA on a dataset with two features, "feature 1" and "feature 2." The "Original data" plot shows the data points in the original feature space, where the axes represent the original features. PCA identifies the principal components, which are orthogonal directions that capture the maximum variance in the data, as indicated by the "Component 1" and "Component 2" vectors. The "Transformed data" plot shows the data projected onto the new coordinate system defined by the principal components, where the first principal component aligns with the direction of maximum variance. The "Transformed data w/ second component dropped" plot illustrates the effect of dimensionality reduction, where only the first principal component is retained, resulting in a one-dimensional representation of the data. The "Back-rotation using only first component" plot shows the data transformed back to the original feature space using only the first principal component, demonstrating the information

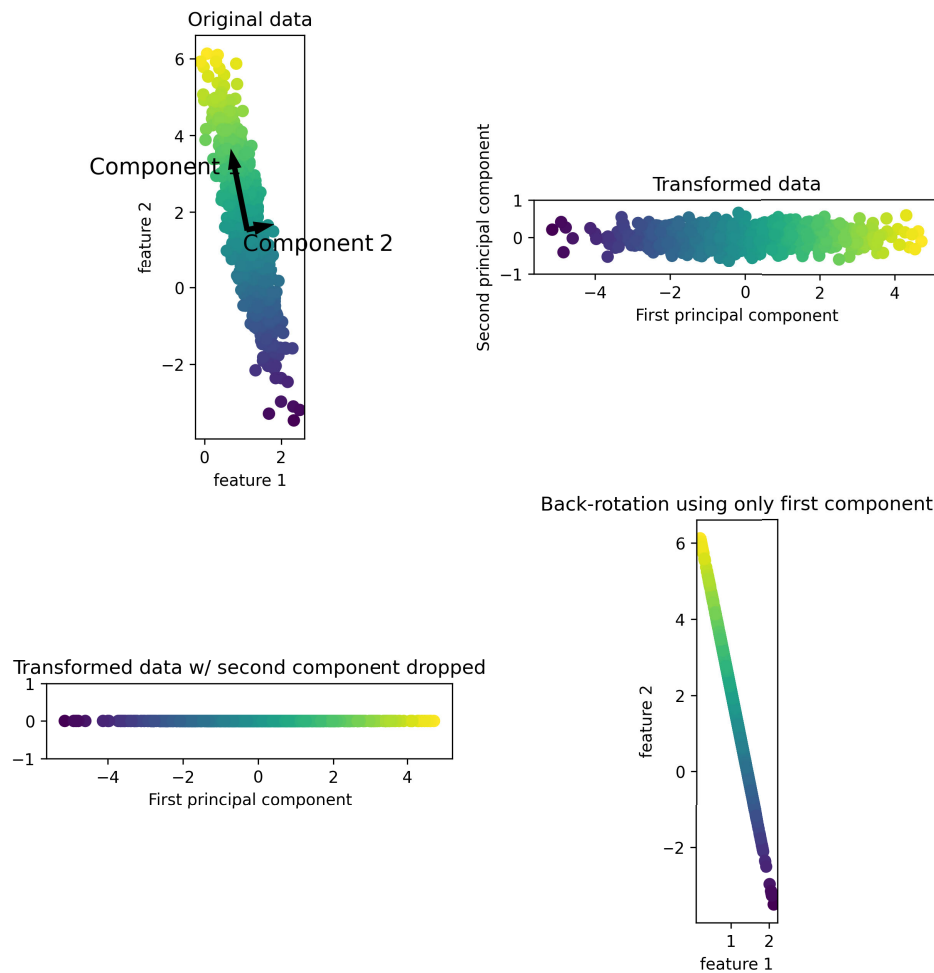


FIGURE 15. PCA concept illustration.

loss due to dimensionality reduction. This process highlights PCA's ability to reduce the dimensionality of data while preserving as much variance as possible.

PCA is a dimensionality reduction technique widely used in ML and signal processing, including optical communication systems. PCA transforms high-dimensional data into a lower-dimensional space by identifying the most significant directions (principal components) in which the data vary. These principal components capture the essential features of the data while discarding less relevant information, making PCA particularly effective in extracting meaningful patterns from complex data sets.

In optical communication systems, PCA has been effectively utilized for joint monitoring of critical parameters such as OSNR, CD, and PMD in [203]. This application enables real-time tracking of these parameters, which are crucial for ensuring optimal optical transmission quality. By reducing the dimensionality of the data, PCA helps to extract the most relevant features that influence transmission performance, thus aiding in faster and more efficient monitoring.

PCA's capability extends to autonomous bit rate and modulation format identification, eliminating the need for manual configuration and enabling dynamic adaptation to changing network conditions. This feature is particularly beneficial in modern optical networks, where the transmission environment can vary, requiring quick adaptation to optimize performance. Furthermore, PCA can be used for anomaly detection, such as identifying irregular network traffic or security intrusions. In [204] and [205], PCA was employed to detect unusual patterns in network traffic, triggering timely alerts to prevent potential disruptions. This demonstrates the versatility of PCA in both performance optimization and network security.

In the field of optical perimeter intrusion detection, PCA extracts frequency-domain features from the backscattered optical signal, which, when combined with classification techniques like SVM, can distinguish between actual intrusions and environmental events [206]. This approach achieves a 99% detection rate with 0% false positives, significantly surpassing traditional systems that have detection rates in the

range of 10-20%. The high sensitivity and specificity of PCA, combined with its 10x reduced computational complexity, make it a superior choice for real-time intrusion detection in fiber-optic systems.

Moreover, PCA is employed in ultra-high-resolution optical spectrum distortion detection, where it helps to extract essential features from the optical spectrum. When combined with SVMs, this method provides robust detection of optical spectrum distortions, which are crucial for ensuring signal integrity in high-speed optical communication systems [207]. The combination of PCA with ML techniques like SVM offers significant advantages in terms of both detection accuracy and computational efficiency.

These applications demonstrate PCA's ability to enhance various aspects of optical communication systems, from performance monitoring and adaptation to security and fault detection. By providing a means to reduce the dimensionality of complex data, PCA not only improves computational efficiency but also helps in extracting relevant features that drive the performance of optical systems. As research progresses, PCA is expected to play an increasingly important role in enhancing the intelligence and reliability of optical communication systems. However, its effectiveness can be influenced by the quality of the data and the choice of the number of principal components, which requires careful tuning for optimal results.

9) POLICY-BASED REINFORCEMENT LEARNING ALGORITHM

Policy-based RL is a powerful method for making optimal decisions in dynamic and complex environments, such as optical communication systems. In contrast to value-based methods that focus on estimating the value of actions, policy-based RL directly learns the policy, which is a mapping from states to actions. This makes it well-suited for environments where the action space is large or continuous, as is often the case in optical communication networks. The key advantage of policy-based methods is that they can handle complex, high-dimensional state and action spaces more efficiently than traditional methods like value iteration or dynamic programming.

In optical communication, RL has been widely applied to optimize routing, wavelength assignment, and network resource management. For instance, in [208], RL was used for self-organized routing in transparent optical networks. This RL-based approach outperformed traditional shortest path and multiple path routing methods by significantly reducing packet loss, achieving an 88% reduction in packet loss during wavelength failures. This success is due to RL's ability to adapt to network dynamics, such as link failures, and learn optimal routing policies in real-time, which is a challenging task for conventional routing algorithms. However, the computational complexity of RL methods can be a limitation, especially for large-scale networks, as the learning process can be expensive in terms of computation and memory usage.

In [209] and [210], RL addressed wavelength contention in buffer-less OBS networks, where traditional methods often struggle with high burst loss and blocking probabilities. By applying deflection routing and dynamic offset time allocation, RL reduced blocking probability and burst loss, achieving up to 90% reduction in blocking probability. RL's ability to make real-time decisions based on network state information allows it to optimize the use of available resources, outperforming conventional methods like static routing or random load balancing. However, the real-time nature of RL comes with the challenge of ensuring that the learning process does not interfere with the system's performance, especially in time-sensitive applications.

Similarly, in [211], RL was employed for routing and contention resolution in OBS networks, leading to a 50% reduction in loss probability compared to shortest path routing. This illustrates RL's advantage in dynamically adapting to traffic conditions, optimizing network performance by balancing burst load and minimizing delays. On the downside, the need for continuous state monitoring and policy updates can introduce additional overhead, especially in large-scale systems where the state space is vast.

Further, in [212], an RL-based multicast RWA algorithm was proposed for WDM optical networks, achieving a 20% reduction in blocking probability. This RL approach demonstrated a 100x reduction in the number of messages exchanged compared to traditional Multi-Protocol Label Switching-based methods, making it more efficient in terms of signaling overhead. However, RL's computational complexity remains an issue, as it requires frequent interaction with the environment to update the policy, which can be resource-intensive in large networks.

In [213], RL was applied to minimize load cost in online routing for optical networks, resulting in a 20% reduction in delay. However, this RL method was found to be more computationally expensive than traditional algorithms like Dijkstra's or Bellman-Ford, which are widely used for static routing. The main challenge here is balancing the benefits of RL, such as dynamic adaptation and optimization, with its higher computational cost, especially in real-time applications.

RL has also shown promise in improving QoT estimation, as demonstrated in [214], where an RL-based approach reduced estimation inaccuracies from 4.2 dB to 0.02 dB. This improvement in QoT estimation is crucial for optimizing network performance and minimizing errors, which are common in long-distance optical transmissions. Despite this, RL's complexity can still pose a challenge in practical deployment, as it requires accurate training and frequent model updates to maintain its performance.

In optical wireless networks, RL has been applied to optimize spectrum sensing and improve channel utilization. In [215], RL-based distributed spectrum sensing significantly reduced collisions and enhanced channel usage in cognitive radio networks. Similarly, in indoor VLC networks, RL optimized network selection and route allocation, improving

throughput and reducing delay by 25% compared to traditional strategies [216]. However, the complexity of RL-based solutions in such networks can be higher than conventional approaches like random or least-loaded routing, especially when network conditions are highly dynamic.

In hybrid LiFi-WiFi networks, RL has been employed to optimize load balancing, achieving a 20% higher throughput than traditional exhaustive search methods [217]. This highlights RL's advantage in optimizing complex systems where traditional methods struggle with load distribution. However, RL's performance can degrade if the learning process is not carefully tuned, and its application requires careful consideration of trade-offs between computational complexity and system efficiency.

The use of RL for vertical handover in hybrid VLC-IR networks also showcases its potential in improving handover performance, with a 13% increase in throughput and 30% faster handovers [218]. Despite these improvements, RL's complexity and the need for real-time updates may present challenges in practical deployment, particularly in mobile environments where network conditions change rapidly.

In summary, RL offers significant advantages for optimizing performance, reducing packet loss, and improving network resource management in optical and optical wireless networks. However, its high computational cost and complexity, especially for large-scale and real-time applications, remain key challenges. The trade-off between RL's dynamic adaptability and the computational resources required for effective learning is an important factor to consider when choosing between RL and conventional methods like static routing, deflection routing, or exhaustive search techniques. As research progresses, RL is expected to play a central role in the development of intelligent, adaptive optical network architectures, though careful management of computational complexity and learning efficiency will be essential for its widespread adoption.

10) VALUE-BASED REINFORCEMENT LEARNING ALGORITHM

Value-based RL focuses on learning an optimal value function that represents the expected cumulative rewards for each state, guiding decision-making in dynamic environments. One of the most widely used value-based RL algorithms is Q-learning, which has proven effective for optimizing decision-making in OBS networks. QL addresses issues like congestion, nonlinearity, and resource contention by learning the value of taking specific actions in various states of the system.

In [219], QL was applied to deflection routing in OBS networks, where it outperformed traditional methods, such as shortest path selection and dynamic wavelength routing. Under varying traffic conditions, QL demonstrated a significant reduction in blocking probability (BLP) and a notable improvement in throughput. The simplicity of QL, which relies on a look-up table representation for storing state-action values, reduces computational complexity compared to other RL techniques. QL's time complexity is $O(n)$, lower

than shortest path selection's $O(n^2)$ and dynamic wavelength routing's $O(n^3)$, making it more efficient for path selection in large-scale networks.

Despite its advantages, QL has limitations. For instance, in [220], QL was extended to path and wavelength selection in OBS networks, but the approach was constrained by a single-agent architecture where only the ingress node provided feedback. This limited its ability to adapt to network-wide changes. To overcome this, [221] proposed a multi-agent QL approach, where agents situated at different nodes in the network collaboratively optimize path selection. This multi-agent system outperformed both the traditional shortest path selection and the single-agent QL, achieving a 50% reduction in BLP and 30% lower delay, which enhanced the overall network performance in terms of throughput and fairness.

Moreover, QL has also been applied to optimize resource allocation in OWC systems. In [139], QL was used to allocate resources in hybrid LiFi-WiFi access networks, improving service quality and provider revenue by reducing IoT migration ratios. In [222], QL was implemented for access point selection in hybrid WiFi-VLC networks, improving throughput and fairness. Furthermore, QL-based approaches in wavelength and access point allocation have provided scalable solutions, with QL outperforming conventional mixed-integer linear programming (MILP) methods in [223]. This scalability is one of QL's strengths, making it suitable for large, complex networks.

In [224], a multi-agent QL approach was used for power allocation in hybrid RF-VLC networks, where it outperformed conventional methods while ensuring that QoS constraints were met. For vertical handover in RF-VLC networks, QL was used to optimize time-to-trigger parameters, achieving higher throughput compared to fixed-time methods [225], [226].

In conclusion, QL's adaptability to dynamic network conditions and its ability to reduce the computational burden make it a valuable tool for optimizing resource allocation in optical burst switched and optical wireless communication systems. However, its performance is sensitive to the accuracy of the state representation and the reward function, and it may struggle in highly dynamic environments with frequent state changes. As research advances, QL is expected to play a key role in evolving network technologies, especially as networks become more complex and require decentralized decision-making to handle increasing traffic demands.

B. DEEP LEARNING

1) DEEP NEURAL NETWORK ALGORITHM

DNNs have revolutionized optical communication by enhancing OPM and network optimization. Their ability to learn complex patterns from data has significantly improved system performance, reliability, and resource utilization in various applications.

In [227], a DNN utilizing amplitude histograms (AH) derived from CMA equalization was proposed for MFI

TABLE 8. Summary of DL applications in OCN.

Ref	Objective	Adopted Algorithm	Metric Used	Performance Analysis
[228]	Identify modulation format and estimate OSNR	CNN, RNN, Hybrid CNN-RNN	Accuracy, Complexity	Achieves OSNR monitoring MSE of 1.2 dB, 0.4 dB, and 1 dB for different modulation formats. No additional hardware required, reducing complexity. Baseline: Conventional OSNR estimation methods.
[231]	Predict optimal RSA strategy for new network topology	DNN-based RSA, Traditional RSA	Blocking probability, Spectrum fragmentation, Delay	Reduces spectrum fragmentation by 20% compared to traditional RSA. Reduces average latency by 15%. Baseline: Traditional RSA algorithms.
[236]	Predict OSNR of a new optical signal waveform	LSTM-NN, SVM, BPNN	MAE, Q-factor, Spectral efficiency, Complexity	Achieves MAE of 0.18 dB, 72% reduction from conventional methods. Response time of 0.2 ms vs. 50 ms (conventional). Baseline: Conventional OSNR estimation methods.
[237]	Predict OSNR and nonlinear noise power	LSTM, EVM, Correlation function, Fractional Fourier transform	MAE, Estimation error, Complexity	MAE of 0.04 dB vs. 1 dB (traditional methods). Real-time implementation, computationally efficient. Baseline: Traditional OSNR estimation techniques.
[238]	Route traffic optimizing performance and privacy constraints	Heuristic-based, AI-based routing	Routing accuracy, Convergence time	Routing accuracy of 99.2% vs. 99.5% (shortest path algorithm). 10x faster than shortest path. Baseline: Shortest path algorithm.
[240]	Predict future network traffic	LSTM, GRU, CNN-LSTM, CNN-GRU	MAE, RMSE, MPE	Reduces MAE by 15%. Reduces prediction time by 20%. Baseline: Traditional statistical forecasting methods.
[241]	Estimate EVM of received signal	CNN, Sliding window method, PAPR method	NMAE, RMSE	MAE of 3.7% (QPSK), 2.2% (16-QAM), 1.1% (64-QAM). EVM estimation in <100 microseconds. Baseline: Conventional EVM estimation methods.
[242]	OSNR estimation	CNN, LSTM, GRU	MSE, SSIM	Bias errors <0.4 dB for all modulation formats. Less complex than traditional methods. Baseline: Traditional OSNR estimation techniques.
[243]	Estimate OSNR	kNN, K-means, SVM, CNN	BER, Throughput, Latency	OSNR estimation accuracy >99% for six modulation formats. Testing time <0.5s. Baseline: Traditional OSNR estimation techniques.
[244]	Learn phase portrait features for MFI	CNN-based MTL, PCA-based pattern recognition	Accuracy, F1-score, BER	100% BR-MFI accuracy vs. 80% (PCA). Computation time: 56 ms vs. 120 ms (PCA). Baseline: PCA-based methods.
[245]	Predict CD, cross-talk, OSNR	CNN, SVM, kNN	Accuracy, R-squared, Latency	Accuracy 96%-99% vs. 80%-90% (traditional). Real-time monitoring enabled. Baseline: Traditional monitoring techniques.
[246]	Estimate OSNR from constellation diagram	PSD, SVM, CNN	MAE, MSE, Correlation coefficient	MAE of 0.125 dB, RMSE of 0.246 dB, significantly lower than traditional methods. Baseline: Traditional OSNR estimation.
[248]	Classify displacement of specklegram images	SVM, RF, kNN	Spatial resolution, Sensitivity, Response time	Noise reduced by factor of 2. Sensitivity increased by factor of 3. Complexity reduced by 100x. Baseline: Conventional specklegram-based classification.
[247]	Identify ONU fingerprints from transmitted signals	Noise model-assisted CNN, Traditional CNN, CNN-LSTM	Accuracy, Precision, Recall, F1-score	99.25% accuracy vs. 80%-90% (previous methods). Real-time implementation. Baseline: Traditional CNN methods.
[249]	Optimize network routing, modulation, spectrum assignment for EONs with enhanced survivability	double-agent DRL, DQN, DDPG	Blocking probability, Average delay, Spectrum utilization, Network survivability	Improves spectrum utilization by 30%, but has large complexity for large networks. Compared to conventional heuristic algorithms.
[250]	Allocate resources to connection requests	DQN, Rainbow, Actor-Critic with Experience Replay	Blocking probability, Spectrum utilization, Latency, Energy consumption	Reduces blocking probability by 77% compared to best heuristic approaches but has high complexity for large networks.
[251]	Learn best RMSA policies for network states	First-Fit, Spectrum First, Max-Min Fairness	Blocking probability, Spectrum utilization, Latency	Improves spectrum utilization by 15%, with lower complexity than traditional RMSA algorithms.
[252]	Maximize network performance (minimize blocking, maximize throughput)	DQN, Double Q-network, Rainbow DQN	Throughput, Latency, Packet loss, Energy consumption	Improves spectrum utilization by 15%, reduces latency by 10%. Compared to conventional RMSA algorithms.
[253]	Learn policy for reconfiguring service function chainings to optimize QoS	DQN, DDQN	Service restoration time, Blocking probability, Network resource utilization, Energy consumption	Improves network throughput by 20%, reduces latency by 30%. Compared to heuristic service function chaining placement methods.
[254]	Learn routing policy interacting with the network environment	K-shortest paths, DQN	Blocking probability, Delay, Packet loss, Utilization	Reduces blocking probability by 30%, packet delay by 20%, energy consumption by 15%. Reduces delay by 70% compared to traditional routing protocols.
[255]	Learn flow scheduling policy	DQN, Double DQN	Packet delay, Packet loss, Throughput, Energy consumption	Reduces flow completion time by 20%. Time complexity $O(n \log n)$ vs traditional $O(n^2)$.
[172]	Relay selection in cooperative FSO systems	deep Q-learning, DDPG	Capacity, Delay, Packet loss, Energy consumption	Outperforms greedy method by 20% in capacity. Time complexity $O(n)$ vs greedy $O(n^2)$.
[173]	Learn policy for selecting FSO/RF link for transmission	DQN Ensemble, DQN, DDQN	Throughput, Spectral Efficiency, Latency, Energy Efficiency	Reduces switching cost by 50%. Efficient for real-time use compared to static link selection.

in coherent optical receivers. This method effectively distinguished between modulation formats, providing an essential capability for adaptive optical networks. The study in [228] extended this approach by integrating MFI with OSNR monitoring. The combined framework achieved 100% accuracy in MFI across a wide OSNR range in PM-QPSK, PM 16-QAM, and PM 64-QAM signals. Additionally, it demonstrated MSEs of 1.2 dB, 0.4 dB, and 1 dB for OSNR estimation, outperforming ANN-based methods in accuracy and robustness.

Further advancements in DNN-based OPM were presented in [229] and [230], where meta-ensemble and active learning techniques were integrated with DNNs to enhance monitoring in MDM systems. These techniques improved prediction accuracy while reducing the need for extensive labeled datasets, addressing one of the major challenges in DL-based OPM.

DNNs have also been successfully applied in RSA for EONs. The study in [231] demonstrated that a DNN-based RSA strategy reduced spectrum fragmentation by 20% and decreased average latency by 15% compared to traditional heuristic RSA algorithms. This performance improvement was attributed to DNNs' ability to predict optimal RSA strategies based on network topology and traffic demands, minimizing blocking probability and enhancing spectral efficiency.

In optical access networks, DNNs have been employed for dynamic bandwidth allocation. The work in [232] showcased a DNN-based model capable of predicting bandwidth demand variations and enabling adaptive resource allocation in passive optical networks. This approach improved network responsiveness and reduced latency, making it highly suitable for next-generation broadband services.

Beyond resource allocation, DNNs have been used for QoT estimation in optical networks. In [233], [234], and [235], DNN models were trained to estimate key QoT metrics, such as BER and Q-factor, in MDM-based transmission systems. Accurate QoT predictions allow for proactive network optimization and fault detection, reducing the risk of service degradation. These studies demonstrated the superiority of DL regression models over conventional analytical approaches by capturing nonlinear impairments more effectively.

While DNNs offer remarkable benefits in optical network management, challenges remain. Training deep models requires large datasets, and their interpretability is often limited, making real-time adaptation difficult. Additionally, computational complexity can be high, particularly when implementing DNNs for large-scale network optimization. However, ongoing research is addressing these limitations through transfer learning, federated learning, and hybrid AI models.

In conclusion, DNNs are becoming integral to OPM and network optimization, providing improved accuracy, adaptability, and predictive capabilities. As optical communication systems continue to evolve toward higher capacities and

more dynamic environments, the role of DNNs is expected to expand further, shaping the future of intelligent optical networking.

2) RECURRENT NEURAL NETWORK ALGORITHM

RNNs, particularly LSTM networks and GRUs, have demonstrated significant potential in optical communication for real-time signal quality monitoring, traffic prediction, and resource optimization. These models effectively capture temporal dependencies in optical signals and network states, improving estimation accuracy and decision-making processes.

LSTM networks have proven highly effective in OSNR estimation. In [236], an LSTM-based model achieved a MAE of 0.18 dB, representing a 72% reduction in error compared to conventional methods such as SVMs and Backpropagation Neural Networks (BPNs). Furthermore, it significantly improved response time, achieving OSNR prediction within 0.2 ms, a remarkable improvement over the 50 ms required by traditional techniques. The superior performance of LSTMs in OSNR monitoring is attributed to their ability to learn long-term dependencies and efficiently process sequential input data, allowing for real-time adaptation to dynamic optical signals.

Extending this application, the study in [237] investigated LSTM networks for simultaneous OSNR and nonlinear noise power estimation. VPI simulations of a five-channel long-haul transmission system demonstrated that LSTM achieved an MAE of 0.04 dB for OSNR estimation in QPSK, 16-QAM, and 64-QAM signals, far surpassing traditional estimation methods, which typically exhibit errors of around 1 dB. Additionally, the model efficiently estimated nonlinear noise power using advanced techniques such as EVM, correlation functions, and the fractional Fourier transform, enabling more precise impairment compensation in fiber-optic communication.

Beyond OSNR monitoring, LSTMs have been employed in multi-domain optical network routing. The AI-based routing paradigm presented in [238] achieved a routing accuracy of 99.2%, approaching the 99.5% accuracy of conventional shortest-path algorithms while providing significant advantages in speed and adaptability. The LSTM-based approach was 10 times faster than heuristic-based routing algorithms, demonstrating its capability for near-instantaneous decision-making in large-scale networks. Similarly, in [239], a multi-layer LSTM framework was used to predict future traffic requests in multicore-fiber-based EONs, enhancing resource allocation strategies and reducing fragmentation through hierarchical graph-based learning.

GRUs, which are a simplified variant of LSTMs with fewer parameters and lower computational overhead, have also been explored for optical network traffic prediction. The work in [240] utilized GRUs alongside LSTMs, CNN-LSTMs, and CNN-GRUs for traffic matrix prediction. The GRU-based model demonstrated a 15% reduction in MAE

and a 20% reduction in prediction time compared to baseline DL models, proving its effectiveness in dynamic resource allocation scenarios. By capturing short-term dependencies efficiently, GRUs provide a viable alternative to LSTMs in latency-sensitive network optimization tasks.

While LSTM and GRU networks exhibit outstanding accuracy and computational efficiency, certain limitations persist. Training these models requires extensive labeled datasets, and their interpretability remains a challenge, making it difficult to extract explicit decision-making rules. Moreover, although LSTMs perform well in handling long-term dependencies, their complexity can lead to increased training times, which may hinder scalability in ultra-high-speed optical networks. Addressing these challenges, recent research has explored hybrid AI models combining LSTMs with CNNs or attention mechanisms to enhance prediction accuracy while reducing computational burden.

In summary, LSTM and GRU networks are emerging as powerful tools for optimizing optical network performance through real-time OSNR estimation, routing optimization, and traffic prediction. Their ability to learn temporal dependencies and dynamically adapt to network variations makes them well-suited for future intelligent optical communication systems, where efficiency and accuracy are critical for sustaining high-capacity transmission.

3) CONVOLUTIONAL NEURAL NETWORK ALGORITHM

CNNs have emerged as powerful tools for OPM and QoT estimation, leveraging their ability to extract spatial features from optical signals and constellation diagrams. These DL models have demonstrated superior accuracy and efficiency compared to traditional signal processing and ML approaches.

CNNs have shown remarkable success in EVM estimation for optical signals. In [241], CNNs achieved a normalized MAE of 3.7%, 2.2%, and 1.1% for QPSK, 16-QAM, and 64-QAM modulation formats, respectively, while also obtaining a normalized MSE of 6.2% for QPSK at 2000 km transmission distance. These results indicate CNNs' ability to accurately quantify signal impairments in fiber-optic communications. Moreover, the study demonstrated that CNN-based EVM estimation could be performed in under 100 microseconds, enabling real-time performance monitoring.

CNNs are also highly effective in OSNR estimation. In [242], CNNs were used to predict OSNR from asynchronously sampled data obtained from a digital coherent receiver. The model achieved bias errors below 0.4 dB for DP-QPSK, 16-QAM, and 64-QAM, while maintaining lower computational complexity than conventional methods. This efficiency makes CNN-based OSNR monitoring particularly useful for real-time network operation.

In addition to OSNR monitoring, CNNs excel in joint monitoring of MFI and OSNR estimation. The study in [243] demonstrated that CNNs outperformed traditional classification methods such as kNN, K-means clustering, and

SVMs, achieving an OSNR estimation accuracy of over 99% across six different modulation formats. Furthermore, CNN-based constellation diagram analysis significantly reduced testing time to under 0.5 seconds, making it suitable for fast network diagnostics.

Beyond OSNR and MFI, CNNs have been successfully applied in multi-impairment monitoring. In [244], a CNN-based multi-task learning (MTL) framework achieved 100% accuracy in BR-MFI, significantly outperforming PCA-based pattern recognition, which achieved only 80%. The CNN model also exhibited a computation time of 56 ms, which was twice as fast as PCA-based methods. Similarly, [245] explored CNN-based monitoring of CD, cross-talk, and OSNR, achieving 99% and 96% accuracy, significantly surpassing traditional techniques that typically yield 80-90% accuracy. The ability to monitor multiple impairments in real-time highlights CNNs' effectiveness in complex optical network environments.

Further expanding CNN applications, [246] reported that CNNs achieved a MAE of 0.125 dB and an RMSE of 0.246 dB for OSNR estimation using constellation diagram-based analysis. These error metrics were substantially lower than those of conventional methods, confirming CNNs' enhanced estimation accuracy and computational efficiency. Additionally, CNNs have proven valuable in optical security applications. In [247], a Noise-Model-Assisted CNN achieved a device fingerprint identification accuracy of 99.25% for OFDM Passive Optical Networks (OFDM-PONs), significantly surpassing previous fingerprinting techniques, which ranged between 80-90% accuracy. This advancement enhances the security of optical networks by enabling precise identification of Optical Network Units (ONUs).

Another noteworthy CNN application is in optical fiber specklegram-based sensing. In [248], CNNs were utilized to classify the displacement of new specklegram images, achieving a noise reduction factor of 2 and a sensitivity increase by a factor of 3. Additionally, the system complexity was reduced by 100 times compared to conventional approaches. This highlights CNNs' potential in non-traditional optical communication applications, such as fiber-based sensing systems.

Despite their outstanding performance, CNNs come with certain limitations. Their reliance on large training datasets and high computational requirements can pose challenges in real-time deployment for ultra-fast networks. Moreover, their opaque nature makes interpretability difficult, which can be problematic in critical network management scenarios. Hybrid approaches, such as combining CNNs with LSTMs for sequential data processing or integrating attention mechanisms for enhanced feature extraction, have been explored to address these issues.

In conclusion, CNNs have proven to be a powerful tool for real-time optical performance monitoring, impairment estimation, and network security enhancement. Their ability to extract meaningful spatial features from received signals and

constellation diagrams positions them as a key technology in future optical network optimizations.

4) DEEP REINFORCEMENT LEARNING ALGORITHMS

DRL has emerged as a crucial methodology for optimizing performance and resource allocation in optical networks. Unlike conventional optimization techniques, DRL leverages an agent-based approach, where an agent interacts with the network environment and learns optimal strategies through trial and error, guided by a reward function. This capability allows DRL to adapt dynamically to network conditions, making it particularly suitable for complex and non-stationary environments such as EONs and hybrid optical-electrical systems.

In [249], a double-agent DRL was applied for survivable routing, modulation, and spectrum assignment (RMSA) in EONs. This approach significantly improved spectrum utilization by 30% while enhancing network survivability. However, the method's computational complexity posed challenges for large-scale networks. Similarly, in [250], DRL algorithms, including DQN and Actor-Critic with Experience Replay, were employed for RMSA in dynamic multi-core fiber EONs. These methods achieved a 77% reduction in blocking probability compared to heuristic approaches, but their scalability remained a concern for larger network topologies.

DRL has also demonstrated effectiveness in various network optimization tasks. For instance, in [251], a DRL-based RMSA policy improved spectrum utilization by 15%, while exhibiting lower computational complexity than traditional RMSA algorithms. Additionally, [252] introduced a multi-task transfer learning DRL framework that reduced training time and blocking probability by 17.7%, achieving a 10% decrease in latency. Such transfer learning strategies enhance model generalization across different network scenarios, addressing one of DRL's major limitations: the high data and computation requirements for training.

Beyond resource allocation, DRL has optimized dynamic service function chaining in metro-core optical networks [253], leading to a 20% increase in network throughput and a 30% reduction in latency. In [254], a DRL-based approach was proposed to enhance routing decisions by incorporating detailed network state representations, including network topology, traffic matrices, and routing constraints. The model employed DQN to iteratively learn an optimal routing policy through interactions with the network environment. This method significantly improved network performance, reducing blocking probability by 30%, packet delay by 20%, and overall energy consumption by 15% compared to traditional K-Shortest Paths algorithms. Additionally, DRL achieved a 70% reduction in end-to-end delay under varying traffic conditions, demonstrating its potential in real-time adaptive routing for high-speed optical networks.

In hybrid optical-electrical networks, DRL has been instrumental in traffic management. For instance, in [255],

DRL-based flow scheduling reduced flow completion time by 20%, with a time complexity of $O(n \log n)$ compared to the $O(n^2)$ complexity of conventional approaches. Similarly, DRL-driven dynamic provisioning in [256] outperformed heuristic methods by lowering blocking probability and reducing training time.

Furthermore, DRL has been applied to FSO communication systems. In [172], deep Q-learning and DDPG were used for relay selection in cooperative FSO systems, improving capacity by 20% compared to greedy methods while maintaining a linear time complexity of $O(n)$. Additionally, in [173], a DRL-based approach for FSO/RF link selection reduced switching costs by 50%, demonstrating efficiency in real-time applications.

Overall, DRL significantly enhances the efficiency, adaptability, and scalability of optical networks. Compared to conventional heuristic-based methods, DRL offers superior performance in terms of resource utilization, blocking probability reduction, and adaptability to dynamic network conditions. However, challenges such as high computational complexity, data requirements, and model interpretability remain critical barriers to widespread deployment. Future research should focus on developing lightweight DRL models, integrating knowledge-driven learning mechanisms, and leveraging hybrid AI approaches to further optimize optical network operations.

V. DISCUSSIONS, CHALLENGES, AND FUTURE DIRECTIONS

A. DISCUSSION OF KEY FINDINGS

In the context of optical communication systems, both ML and DL techniques have demonstrated considerable potential for enhancing system performance. A number of studies have showcased how these technologies can improve tasks such as signal detection, modulation recognition, and error correction. For instance, supervised learning algorithms, such as SVMs, have been widely used in OFC to improve modulation format classification, thereby contributing to better channel estimation and interference management.

However, despite the promising performance of these models, several issues remain unresolved. DNNs, while offering higher accuracy in tasks like channel equalization and signal demodulation, come with significant computational costs. This raises concerns about the feasibility of deploying such models in real-time systems, particularly for large-scale optical communication networks.

B. CATEGORIZATION OF CHALLENGES

Several challenges hinder the wide-scale adoption of ML and DL in optical communication systems:

Data-Related Challenges: One of the primary challenges is the availability of high-quality labeled data for training ML models. In optical communication, acquiring sufficient labeled data, especially under diverse operating conditions (e.g., varying signal-to-noise ratios, different modulation

formats), is time-consuming and costly. Moreover, a lack of labeled data often results in overfitting, compromising model generalization [257].

Modeling Challenges: Choosing the appropriate ML or DL model for a given optical communication task is a nontrivial decision. For instance, while DNNs are widely adopted for signal classification and detection, simpler models such as decision trees or random forests may provide comparable performance with reduced computational overhead. Furthermore, model interpretability remains a significant challenge, especially in the context of opaque DL techniques, making it difficult to understand and justify model decisions in real-world scenarios [258].

Computational and Resource Challenges: DL models, especially those involving multiple layers and parameters, require substantial computational resources. This includes both high-performance computing for model training and memory for model deployment. These constraints limit the practical application of DL in large-scale networks, where real-time decision-making and resource constraints are critical [259].

Integration and Scalability Issues: Integrating ML and DL techniques into existing optical communication infrastructures, especially in large-scale, high-speed optical networks, presents additional hurdles. Many traditional optical systems were not designed with these advanced ML techniques in mind, requiring significant modifications in terms of architecture, hardware, and network design [260].

C. GAPS IN EXISTING RESEARCH

While there has been substantial progress in applying ML and DL to optical communication systems, there remain significant gaps in existing research. Most of the current literature focuses on theoretical models or laboratory-scale setups, with limited attention to large-scale real-world implementations. Few studies have investigated the long-term stability and scalability of ML models when deployed in live optical networks. Furthermore, the integration of RL techniques—particularly for dynamic optimization in network control—remains an underexplored avenue. RL has the potential to continuously adapt to changes in network conditions, optimizing network performance without human intervention, but its application in optical communication systems has yet to be fully explored.

Moreover, there is insufficient attention given to the interpretability of ML models. While DL techniques such as CNNs and RNNs have demonstrated high accuracy in tasks like signal classification, understanding why and how these models make specific decisions is still a challenge. This lack of transparency poses problems for the widespread adoption of ML in critical communication systems.

D. FUTURE RESEARCH DIRECTIONS

Future research in optical communication systems must focus on several key directions to overcome the current challenges and maximize the potential of ML/DL techniques:

Emerging ML/DL Approaches: One promising avenue is the use of Physics-Informed Neural Networks (PINNs). These models combine domain-specific knowledge with ML, offering the potential for enhanced accuracy even in data-scarce environments. By incorporating the physical principles governing optical communication systems into model training, PINNs can yield more reliable and interpretable results [261], [262], [263], [264].

Reinforcement Learning for Dynamic Network Optimization: RL presents a compelling future direction for optimizing real-time network parameters. RL can be used to train agents that adapt to varying network conditions, such as fluctuating bandwidth, traffic loads, or noise levels. Future work should focus on applying RL to optimize parameters like modulation formats, routing, and power allocation in dynamic optical networks [265], [266], [267], [268], [269], [270], [271], [272].

Hybrid Models for Efficiency and Accuracy: The combination of traditional optical communication techniques with ML and DL models—often referred to as hybrid approaches—could offer the best of both worlds. Future research could explore the integration of conventional methods (such as signal processing algorithms) with ML/DL models to improve system performance without incurring the computational burden of purely DL-based solutions [273], [274], [275], [276].

Benchmarking and Case Studies: To bridge the gap between theory and practice, future studies should include real-world case studies and performance benchmarks. This would involve testing ML/DL models under realistic operating conditions, such as varying noise, channel impairments, and network topologies. Such benchmarking would help assess the true practical utility of these models and provide guidelines for their integration into existing communication infrastructure [277], [278], [279].

Data Efficiency and Transfer Learning: Given the scarcity of labeled data, research into data-efficient learning techniques such as transfer learning and semi-supervised learning could provide significant improvements. These techniques enable models to generalize from small datasets by leveraging knowledge from related domains, thus making them more practical for real-world optical communication tasks [280], [281], [282].

End-to-End Learning for Optical Communication: Traditional communication system design follows a modular approach, where each stage—such as encoding, modulation, and equalization—is optimized separately. However, end-to-end learning, leveraging deep neural networks, can jointly optimize the entire communication pipeline. This paradigm has already been explored in wireless communications, where deep autoencoders have shown significant improvements in channel coding and signal reconstruction. Inspired by advances in computer vision and natural language processing, where DL models effectively capture hierarchical representations, end-to-end learning in optical systems could enable novel architectures that adapt to complex

channel distortions and optimize performance in a data-driven manner. Future research should focus on designing scalable end-to-end models tailored to optical networks, incorporating domain-specific constraints while maintaining interpretability [283], [284], [285], [286].

VI. CONCLUSION

This survey explored the integration of ML and DL with optical communication, covering OFC, OWC, and OCN. We examined the unique challenges of each domain and demonstrated how ML and DL effectively address them, highlighting key applications and their impact on system optimization and performance enhancement. The survey highlighted the motivations for employing various ML and DL models, considering both their inherent characteristics and external factors facilitating their integration with optical systems. We discussed diverse applications, showcasing how these algorithms contribute to optimizing performance and improving efficiency. In addition to quantitative improvements, the survey provided qualitative comparisons, illustrating how different algorithms uniquely enhance system performance. Challenges in implementing ML and DL in optical networks were also addressed, along with potential solutions and future research directions. This survey serves as a comprehensive guide, offering a roadmap for navigating the fast-evolving field of ML and DL in optical communication systems. By summarizing key findings, it empowers researchers and practitioners to make informed decisions about the best algorithms and approaches for their needs.

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